ScaDS.AI DRESDEN LEIPZIG
CENTER FOR SCALABLE DATA ANALYTICS AND ARTIFICIAL INTELLIGENCE

NFDI4 BIOIMAGE
NATIONAL RESEARCH DATA MANAGEMENT INFRASTRUCTURE FOR MICROSCOPY AND BIOIMAGE ANALYSIS

GloBIAS Global BioImage Analysts' Society
GLOBAL BIOIMAGE ANALYST'S SOCIETY

Large Language Models: An Introduction

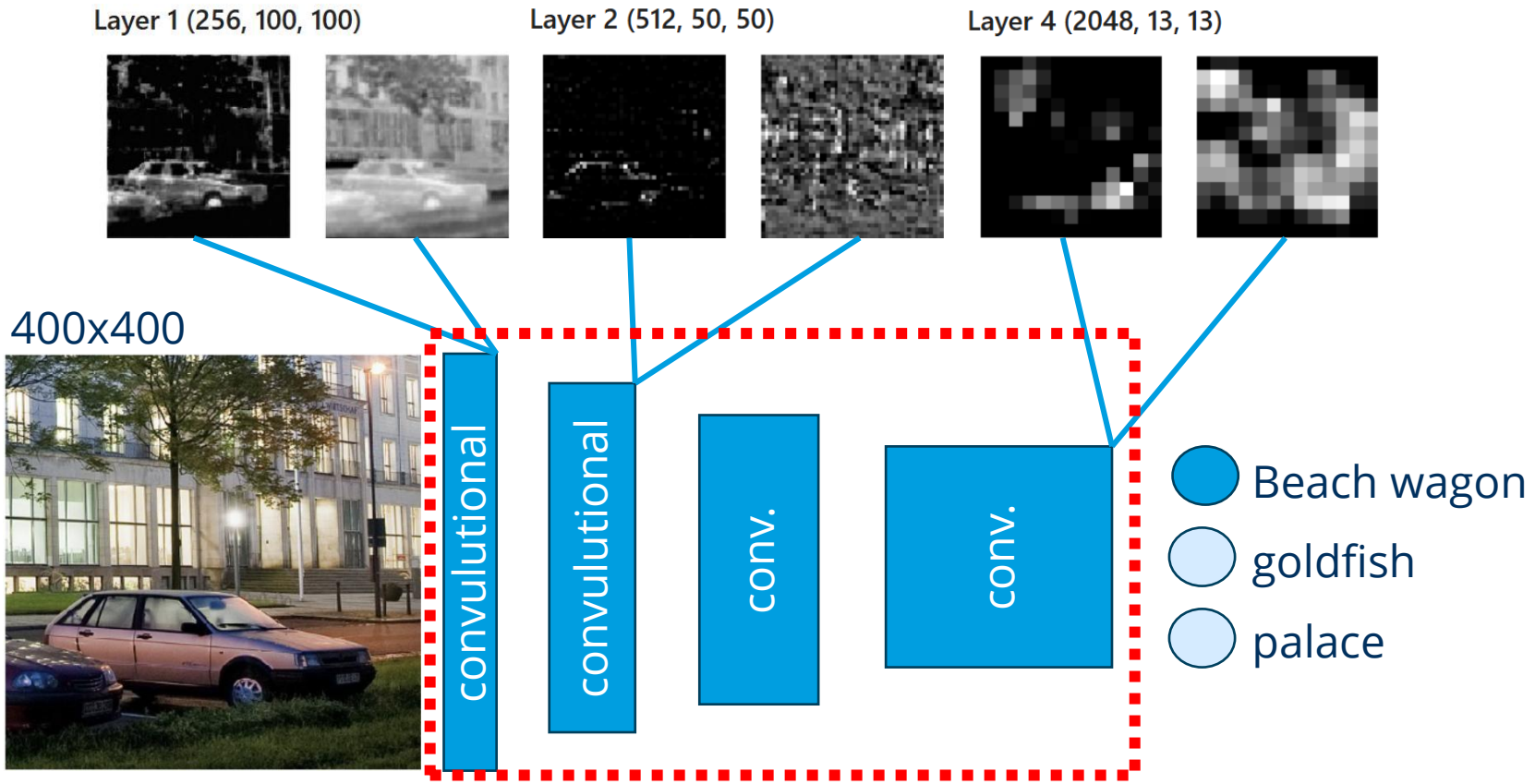
Robert Haase

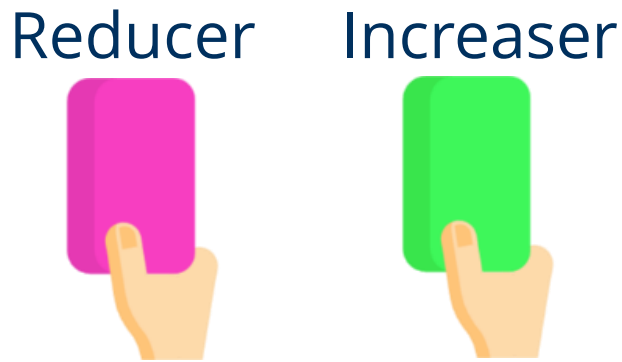
Bundesministerium für Bildung und Forschung
SACHSEN
Diese Maßnahme wird gefördert durch die Bundesregierung aufgrund eines Beschlusses des Deutschen Bundestages.
Diese Maßnahme wird mitfinanziert durch Steuermittel auf der Grundlage des von den Abgeordneten des Sächsischen Landtags beschlossenen Haushaltes.

TECHNISCHE UNIVERSITÄT DRESDEN
UNIVERSITÄT LEIPZIG

Quiz

What is this part of a DNN typically called?





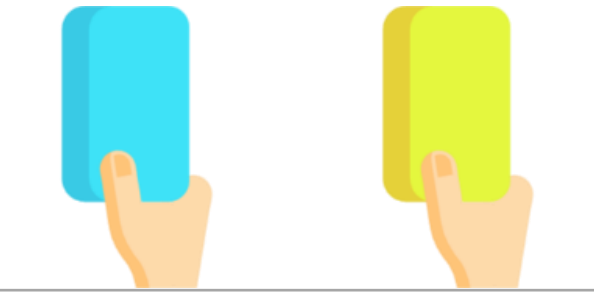
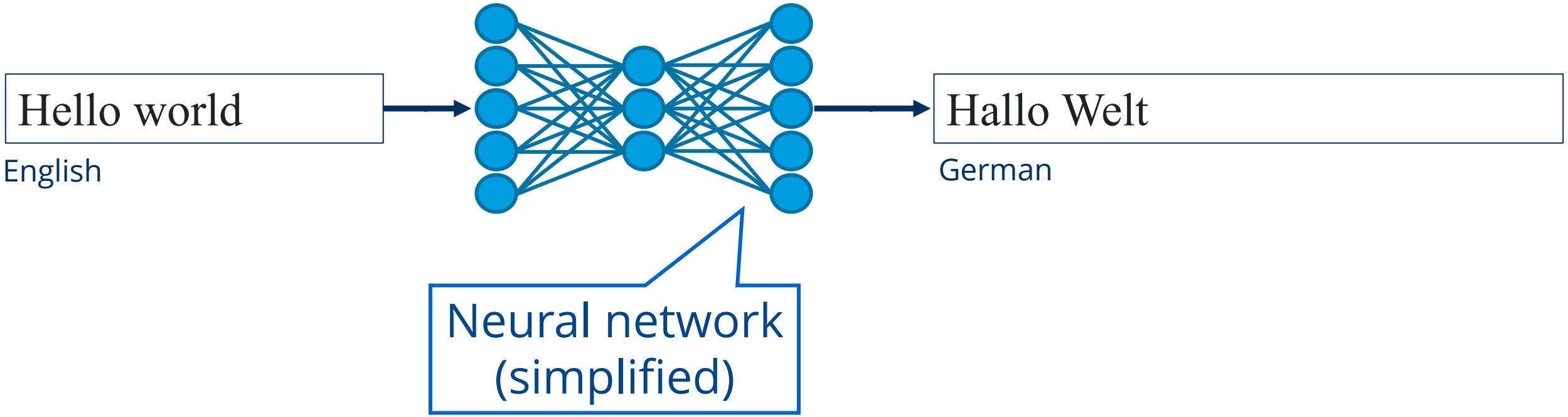


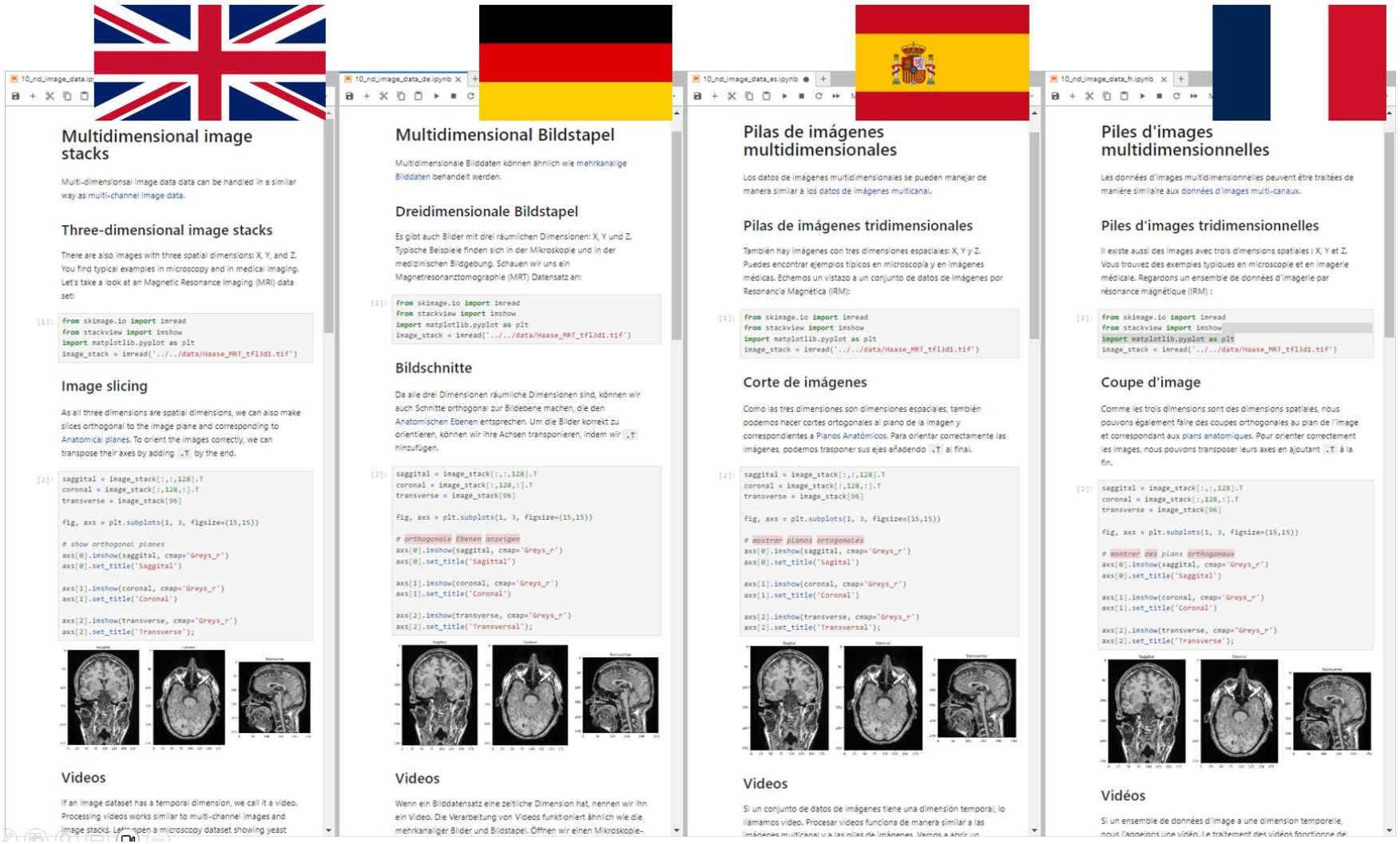
Image source: Cropped from HTW Dresden (Fotograf: Peter Sebb) licensed CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=15652763

Large Language Models (LLMs)

Text-to-text, translation

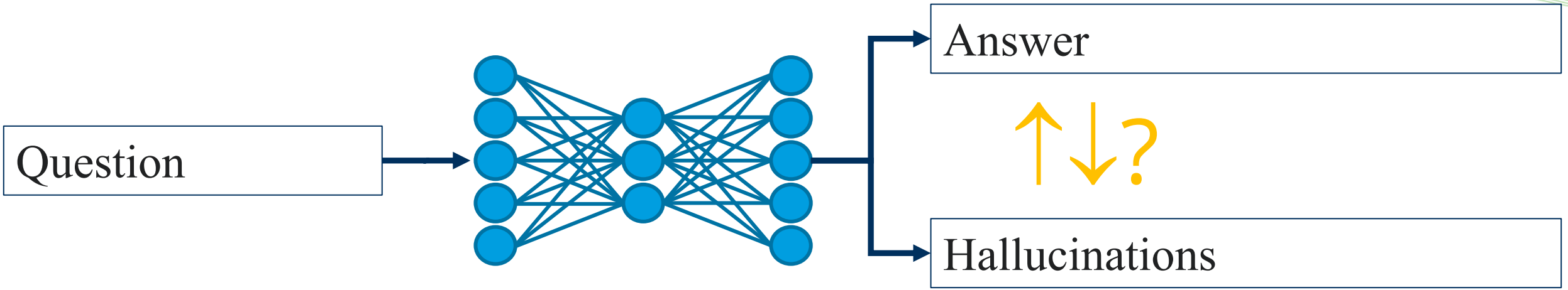


Translating training materials



Large Language Models (LLMs)

Text-to-text, translation, knowledge retrieval



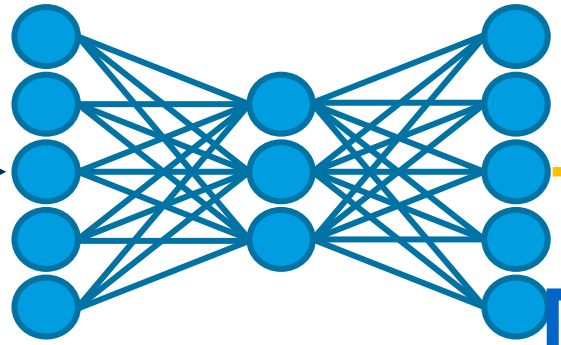
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis[†‡], Ethan Perez[⋆], Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†], Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel[†‡], Sebastian Riedel[†‡], Douwe Kiela[†]

https://arxiv.org/abs/2005.11401

Large Language Models (LLMs)

Text-to-text, translation, reasoning

Hypothesis →  ⇢ Proof + Explanation

2024-8-16

The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery

Chris Lu[1,2,*], **Cong Lu**[3,4,*], **Robert Tjarko Lange**[1,*], **Jakob Foerster**[2,†], **Jeff Clune**[3,4,5,†] **and David Ha**[1,†]
[*]Equal Contribution, [1]Sakana AI, [2]FLAIR, University of Oxford, [3]University of British Columbia, [4]Vector Institute, [5]Canada CIFAR AI Chair, [†]Equal Advising

https://arxiv.org/abs/2408.06292

Large Language Models (LLMs)

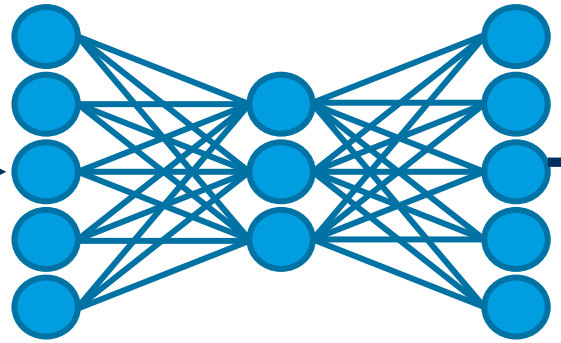
Text-to-text, translation, reasoning

Large Language Models (LLMs)

Text-to-text, translation, code generation

Open hela-cells.tif

English



```
from skimage.io import imread
image = imread("hela-cells.tif")
```

Python

Published as a conference paper at ICLR 2024

SWE-BENCH: CAN LANGUAGE MODELS RESOLVE REAL-WORLD GITHUB ISSUES?

Carlos E. Jimenez[*1,2] John Yang[*1,2] Alexander Wettig[1,2]
Shunyu Yao[1,2] Kexin Pei[3] Ofir Press[1,2] Karthik Narasimhan[1,2]
[1]Princeton University [2]Princeton Language and Intelligence [3]University of Chicago

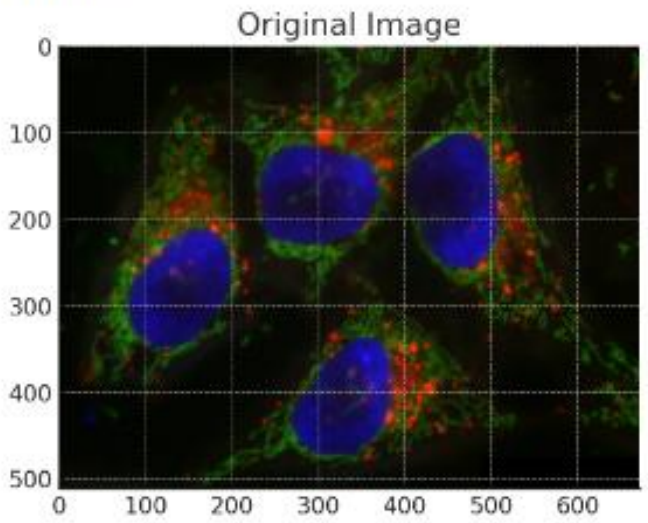
https://arxiv.org/abs/2310.06770

Bio-image Analysis using Large-L

Analysing images in the browser, e.g. using ChatG

Prompt:
“Please segment the nuclei in the blue channel of the image”

ChatGPT



The segmentation of the nuclei has been performed on the blue chan
the figure shows the segmented nuclei in different colors, each corres
the field of view. Each color represents a unique label assigned to the
image.

Analysis

```
from skimage import io, filters, measure, morphology
import matplotlib.pyplot as plt
import numpy as np

# Load the image
image = io.imread('/mnt/data/hela-cells-8bit.png')

# Since it's a color image, we'll just take the blue chann
blue_channel = image[:, :, 2]

# Threshold the image to get the binary image for nuclei
thresh = filters.threshold_otsu(blue_channel)
binary = blue_channel > thresh

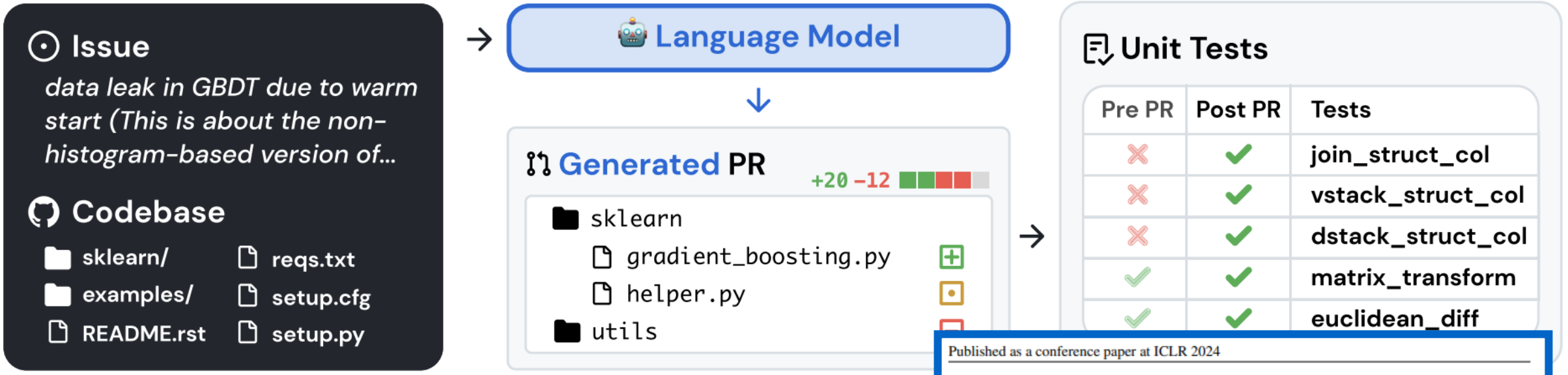
# Remove small objects and perform morphological closing t
cleaned = morphology.remove_small_objects(binary, 50)
closed = morphology.closing(cleaned, morphology.disk(3))

# Label the image
label_image = measure.label(closed)

# Plot the original image and the binary image
fig, ax = plt.subplots(1, 2, figsize=(12, 6))
ax[0].imshow(image)
ax[0].set_title('Original Image')
ax[1].imshow(label_image, cmap='nipy_spectral')
ax[1].set_title('Segmented Nuclei')
plt.show()
```

Large Language Models (LLMs)

Text-to-text, translation, code generation



Published as a conference paper at ICLR 2024

SWE-BENCH: CAN LANGUAGE MODELS RESOLVE REAL-WORLD GITHUB ISSUES?

Carlos E. Jimenez[*1,2] John Yang[*1,2] Alexander Wettig[1,2]
Shunyu Yao[1,2] Kexin Pei[3] Ofir Press[1,2] Karthik Narasimhan[1,2]
[1]Princeton University [2]Princeton Language and Intelligence [3]University of Chicago

Chatting with LLMs on GitHub/Lab

Goal: Transparency + knowledge exchange

haesleinhuepf opened last week

I'd like to have python code for classifying images. I think using the resnet50 model from torchvision. We can use this as example image:

...

Before running the classification, make sure all necessary libraries are installed.

git-bob try to do this.

Create sub-issue 1

github-actions last week Contributor

This message was generated by git-bob (version: 0.19.1, model: claude-3-5-sonnet-20241022, log), an experimental AI-based assistant. It can make mistakes and has limitations. Check its messages carefully.

I worked on implementing image classification using ResNet50 from torchvision for the microscope and cat image. I first saved the input image as microscope_cat.jpg, then created a Jupyter notebook (image_classification.ipynb) that implements the classification logic using the ResNet50 model. The notebook loads the image, preprocesses it, and uses the pre-trained model to predict what's in the image. The classification results were saved to a text file and I made sure to specify the required dependencies in requirements.txt fo anyone who wants to run this code.

...

- classification_results.txt Contains the top predicted categories from the ResNet50 model for the input image
- image_classification.ipynb Jupyter notebook implementing the image classification using ResNet50 model from torchvision
- requirements.txt Lists the Python package dependencies needed to run the classification code

```
Top 5 predictions:
sewing machine: 0.443
toaster: 0.240
iron: 0.132
soap dispenser: 0.047
vacuum: 0.033
```

Image Classification using ResNet50

First, let's install and import the necessary libraries.

In [1]:
```
!pip install torch torchvision pillow requests
```

...

In [2]:
```
import torch
from torchvision import models, transforms
from PIL import Image
import requests
from io import BytesIO
```

Load and preprocess the image

In [3]:
```
# Define image transformations
preprocess = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(224),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])

# Load image
image = Image.open('microscope_cat.jpg')
input_tensor = preprocess(image)
input_batch = input_tensor.unsqueeze(0)
```

Load the pre-trained ResNet50 model and run inference

In [4]:
```
# Load model and weights
weights = models.ResNet50_Weights.IMAGENET1K_V1
model = models.resnet50(weights=weights)
model.eval()

# Run inference
with torch.no_grad():
    output = model(input_batch)

# Get class names
categories = weights.meta['categories']

# Get top 5 predictions
_, indices = torch.sort(output[0], descending=True)
probabilities = torch.nn.functional.softmax(output[0], dim=0)

# Save results
with open('classification_results.txt', 'w') as f:
    f.write('Top 5 predictions:\n')
    for idx in indices[:5]:
        f.write(f'{categories[idx]}: {probabilities[idx].item():.3f}\n')
```

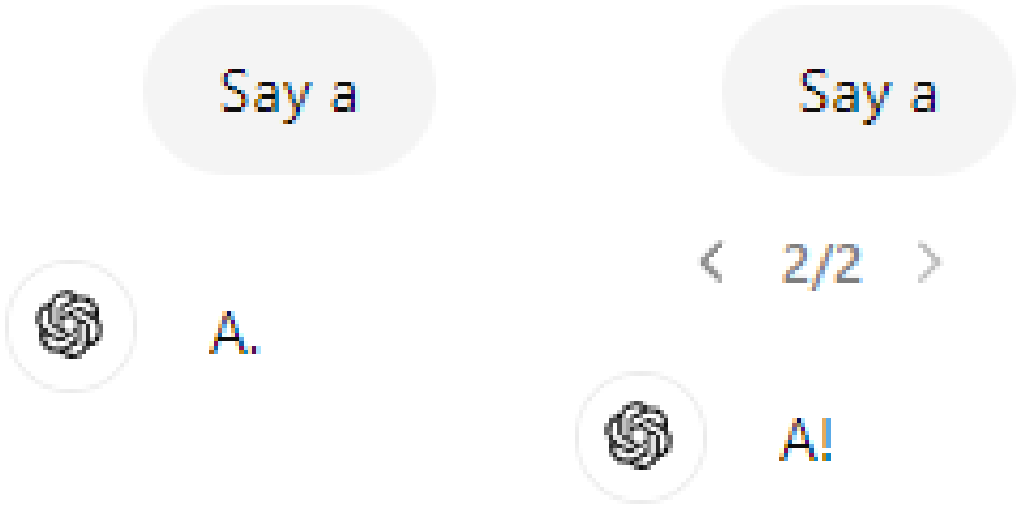
Why code generation using LLMs?

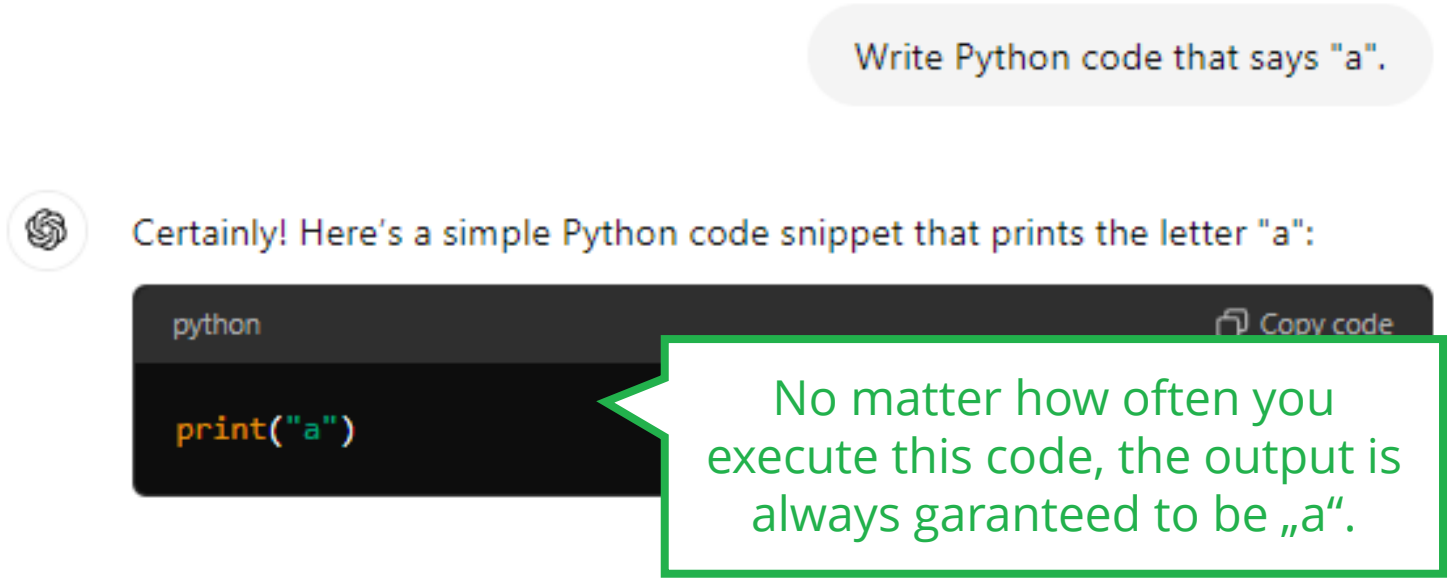
Output of LLMs is often criticized for

- Limited reproducibility
- Hard to evaluate (automatically)

Output of LLM-generated code is

- Reproducible
- Can be executed and evaluted (automatically)





Transformer neural networks

Bundesministerium für Bildung und Forschung
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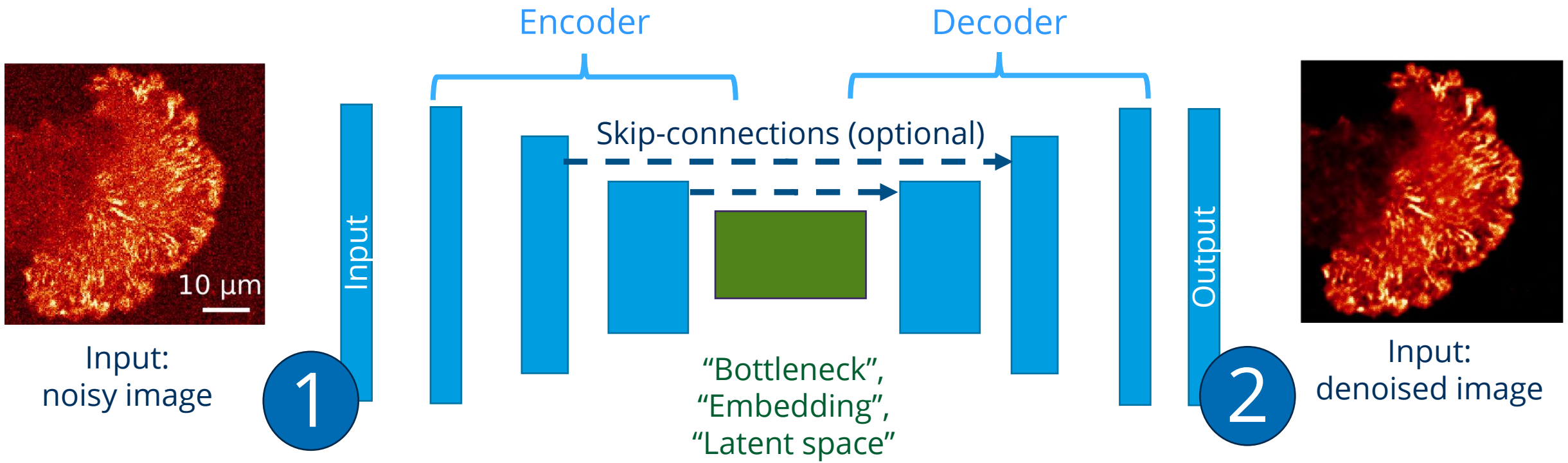
Recap: Encoder-Decoder Networks

Related: „Auto-encoder“, „Variational Auto-Encoder“, „U-Net“



Images demonstrating noise2void, cropped from https://github.com/HenriquesLab/ZeroCostDL4Mic/blob/master/Wiki_files/FigureS2.png license: MIT

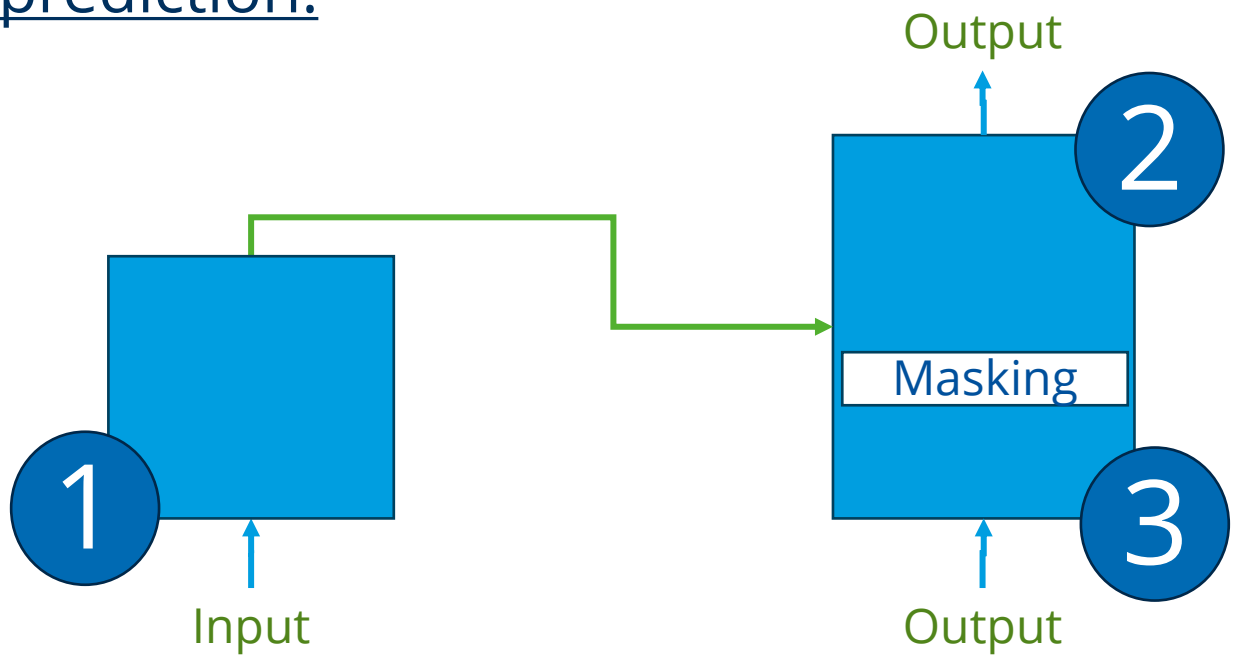
Transformers

LLMs use the transformer neural network architecture

Related: Generative Pretrained Transformer (GPT)

LLMs were originally developed for translation tasks and next-word prediction:

„Die Katze sitzt neben dem Mikroskop.“

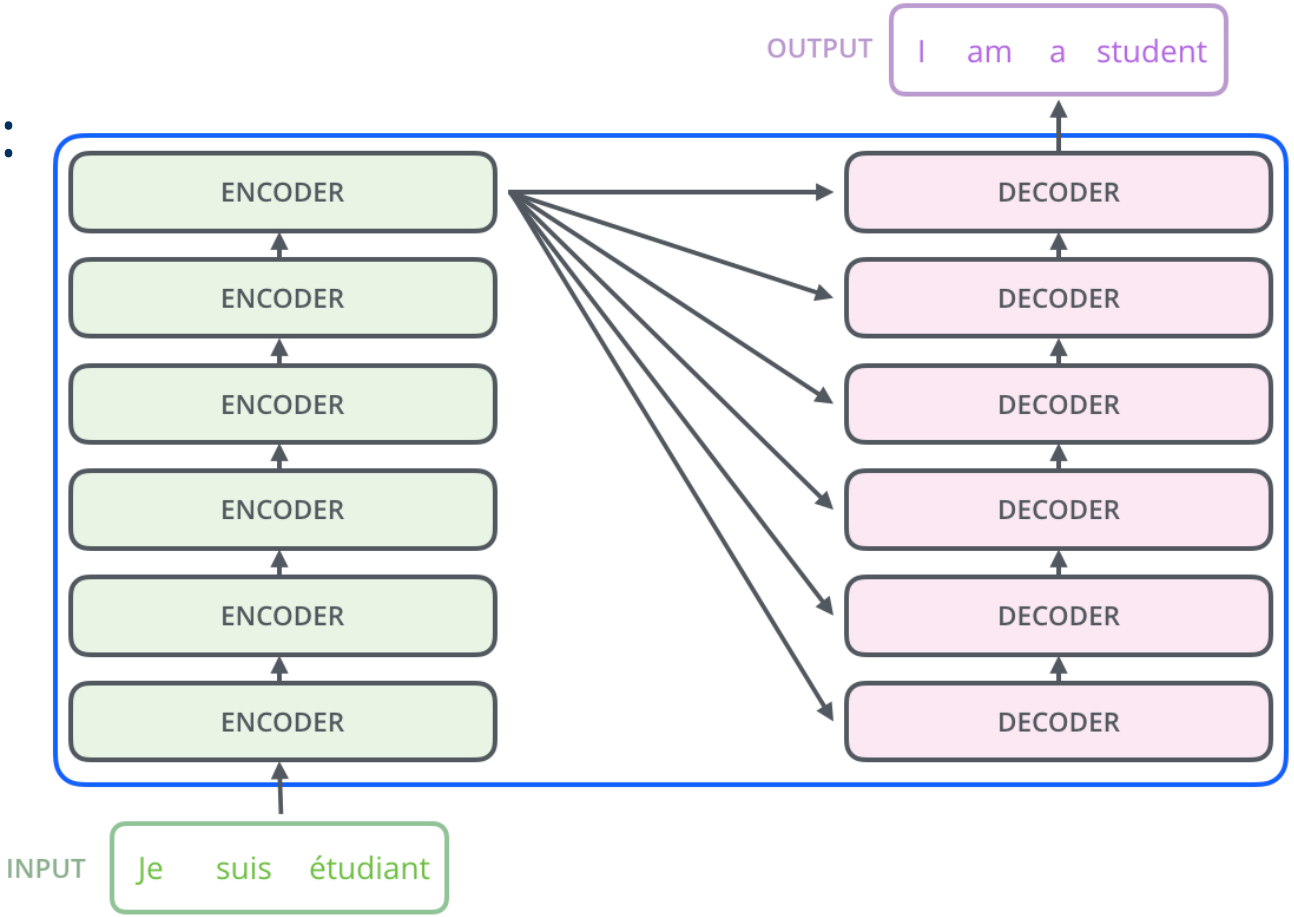


„[...] Microscope“

„The cat sits next to the [...]“

Generative Pretrained Transformer (GPT)

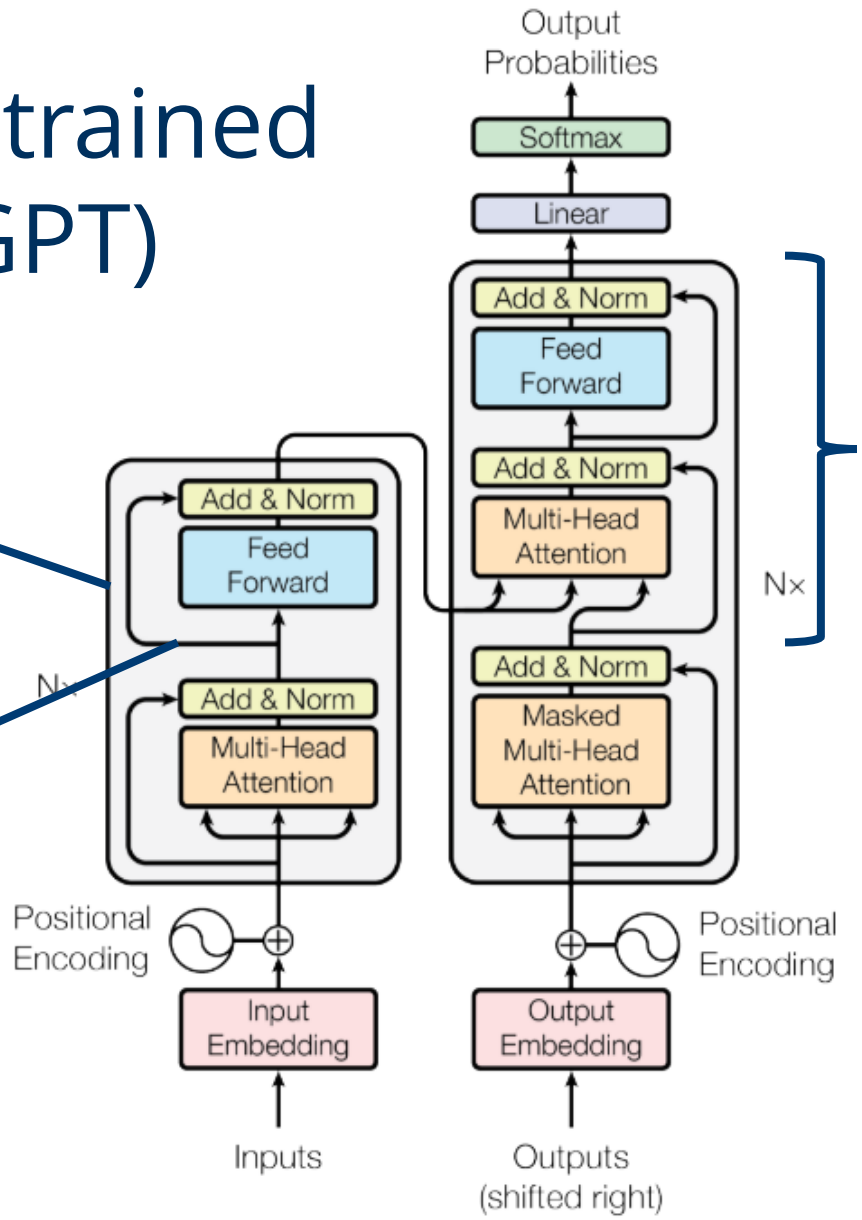
Stacks of encoders and decoders arranged like this:



Generative Pretrained Transformer (GPT)

Encoder

Vectors and matrices

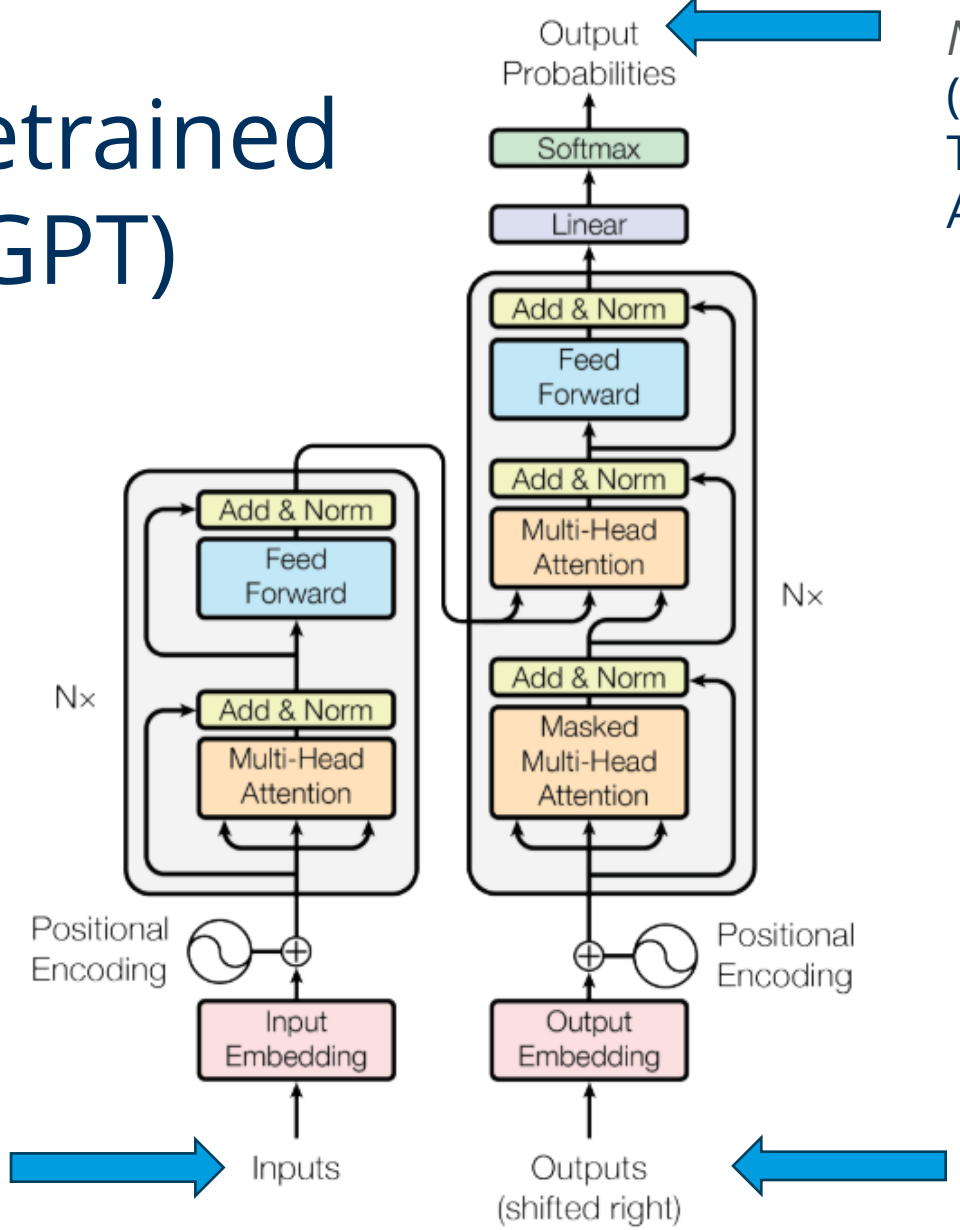


Decoder

Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Generative Pretrained Transformer (GPT)

Task: Translation



Next word probabilities (according to context):
The: 0.9
A: 0.1

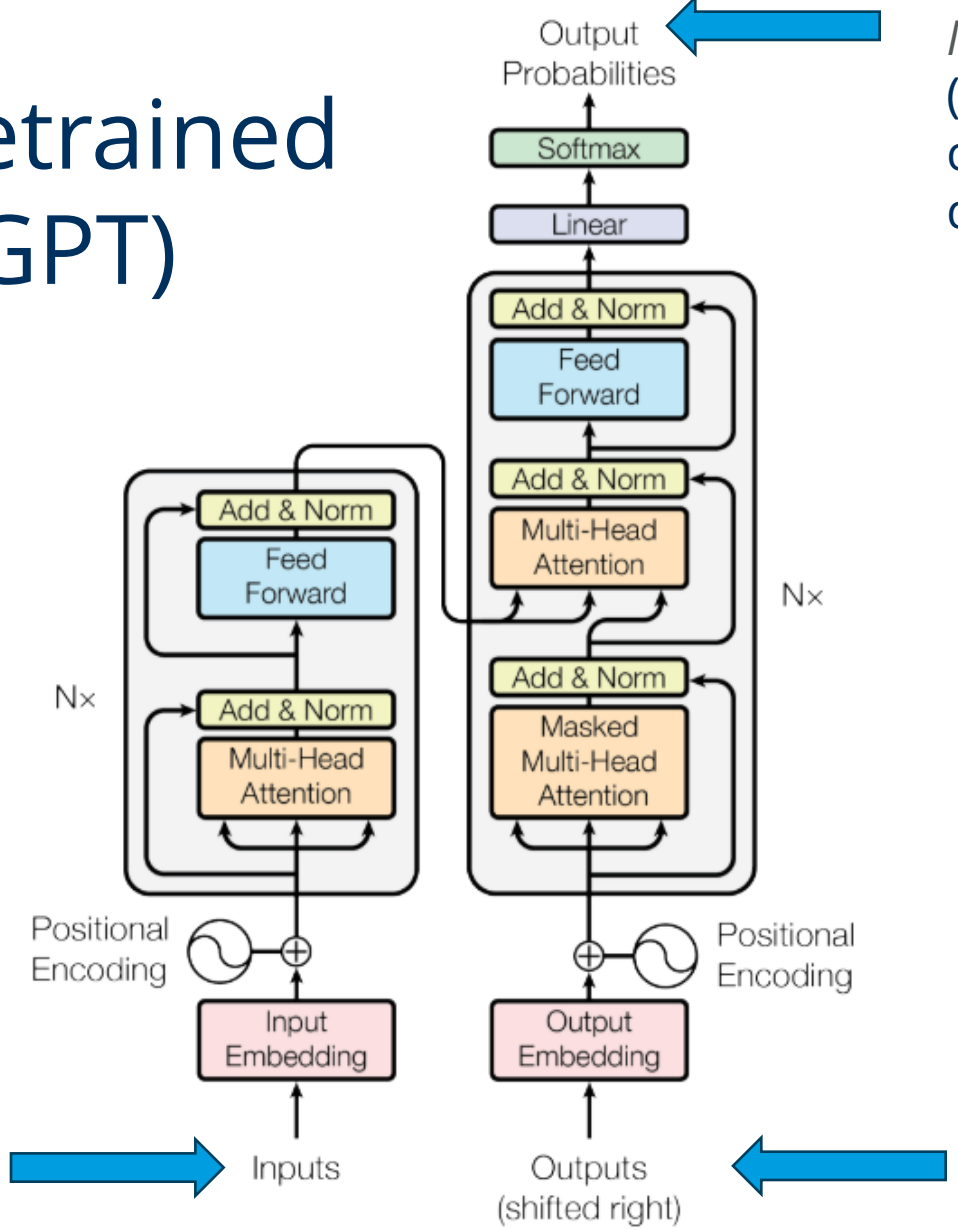
Example input:
Die Katze sitzt neben dem *Mikroskop*

Example output:
The cat sits next to the microscope

Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Generative Pretrained Transformer (GPT)

Task: Translation



Next word probabilities (according to context):
cat: 0.9
dog: 0.0001

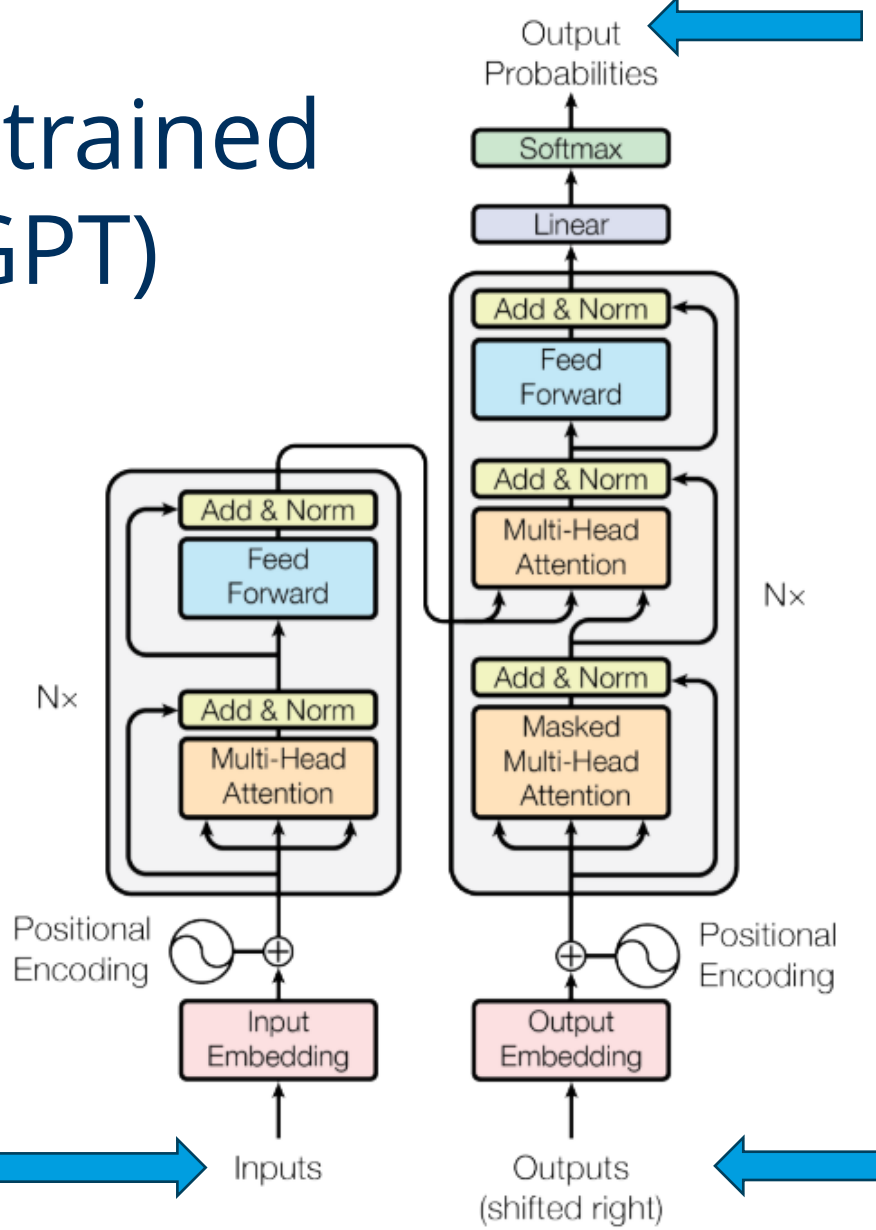
Example input:
Die Katze sitzt neben dem *Mikroskop*

Example output:
The cat sits next to the microscope

Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Generative Pretrained Transformer (GPT)

Task: Translation



Next word probabilities (according to context):
sits: 0.8
takes a seat: 0.2

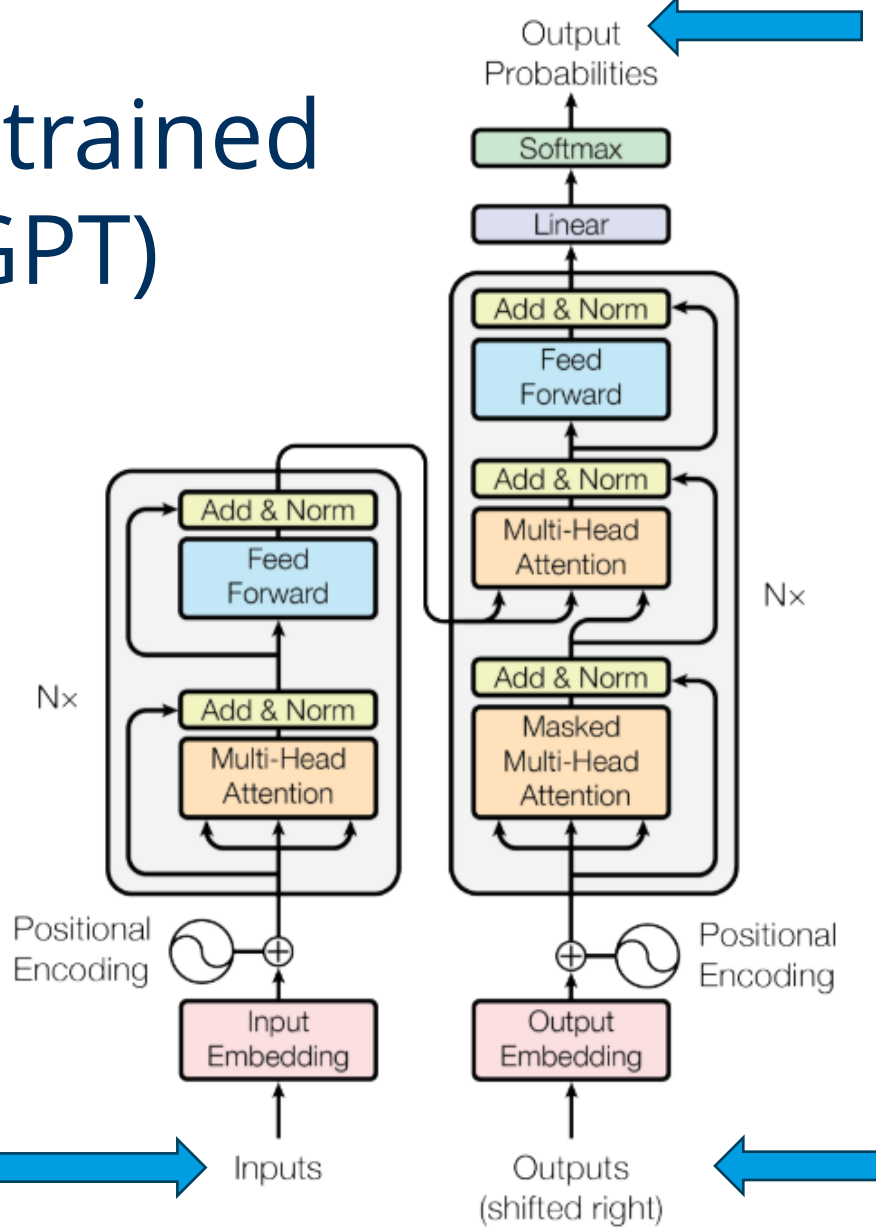
Example input:
Die Katze sitzt neben dem *Mikroskop*

Example output:
The cat sits next to the microscope

Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Generative Pretrained Transformer (GPT)

Task: Translation



Next word probabilities (according to context):
next to: 0.6
near: 0.4

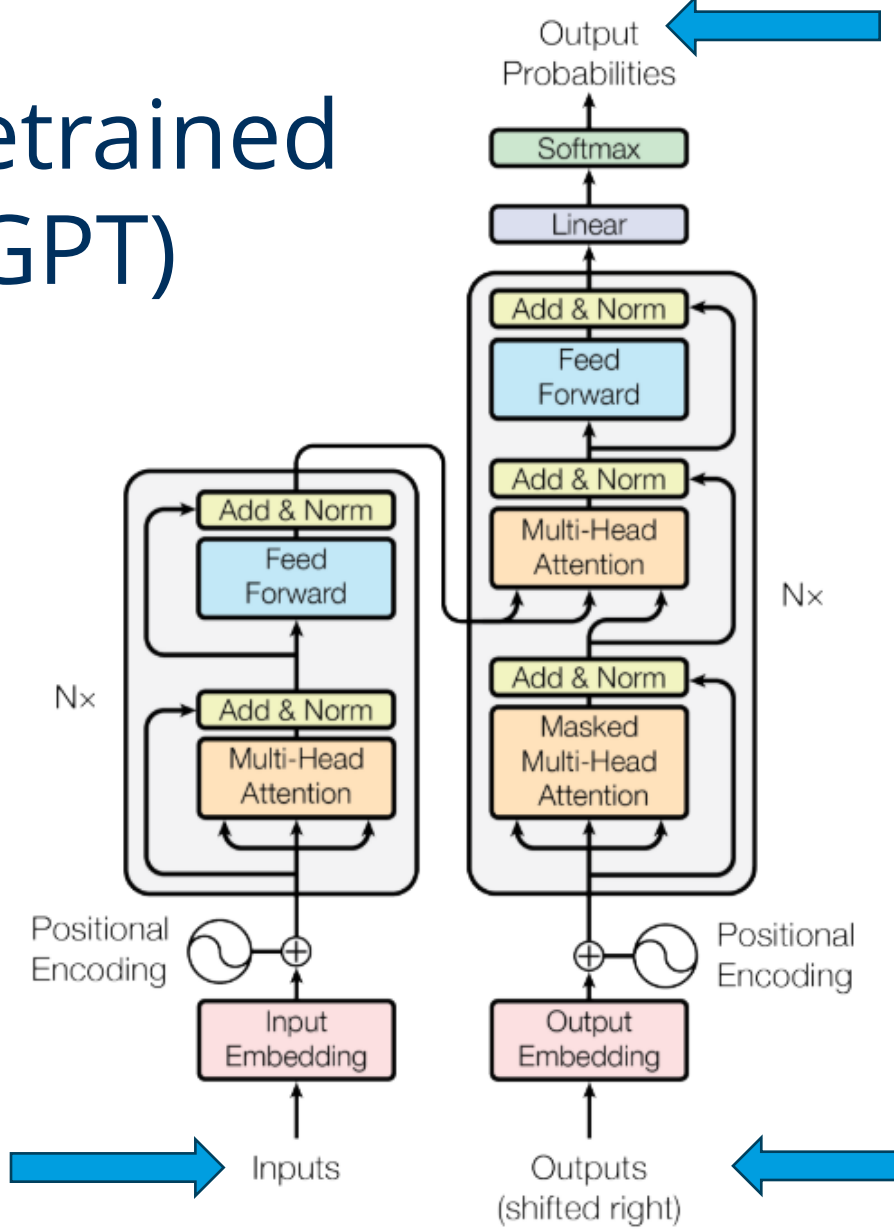
Example input:
Die Katze sitzt neben dem *Mikroskop*

Example output:
The cat sits next to the microscope

Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Generative Pretrained Transformer (GPT)

Task: Translation



Next word probabilities (according to context):
the: 0.9
a: 0.1

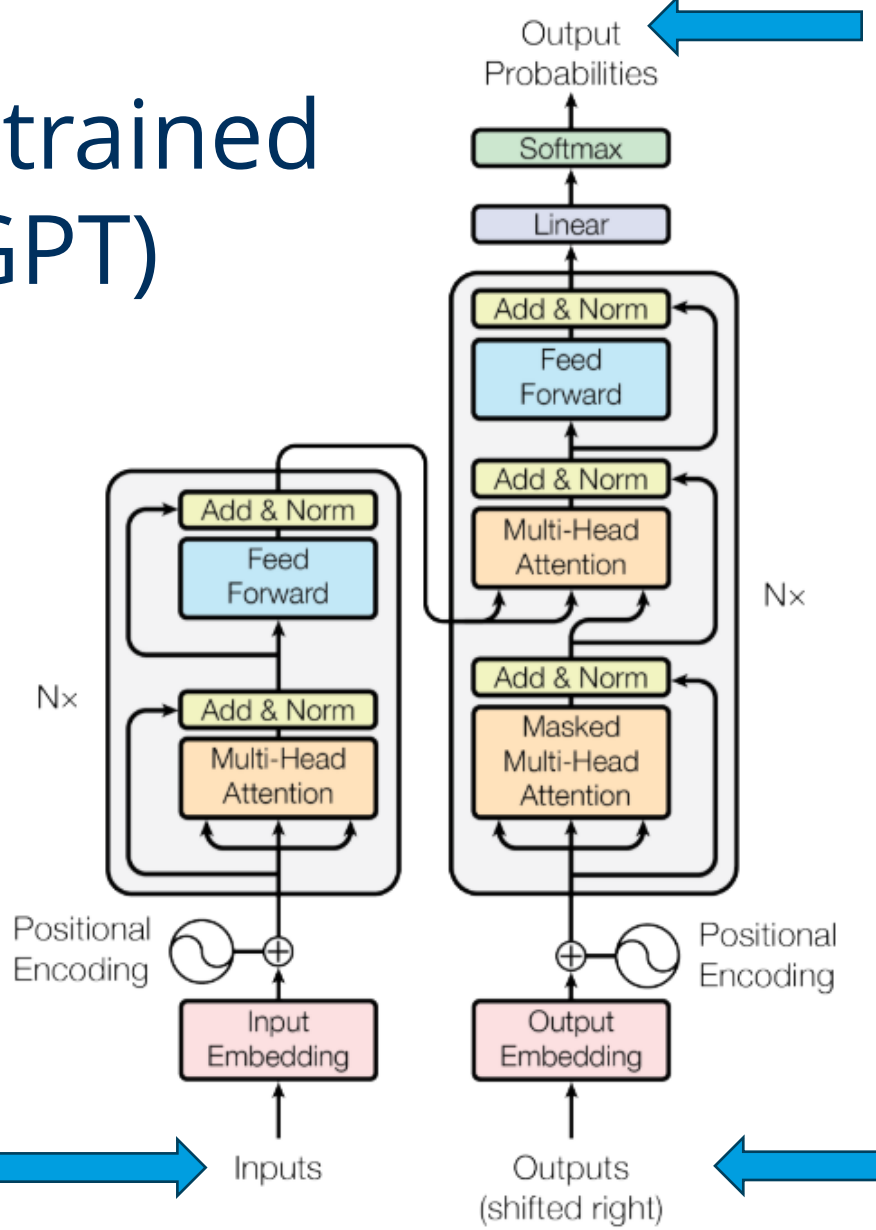
Example input:
Die Katze sitzt neben dem *Mikroskop*

Example output:
The cat sits next to the microscope

Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Generative Pretrained Transformer (GPT)

Task: Translation



Next word probabilities (according to context):
Heating: 0.9
Food: 0.8
Dog: 0.2
Microscope: 0.01

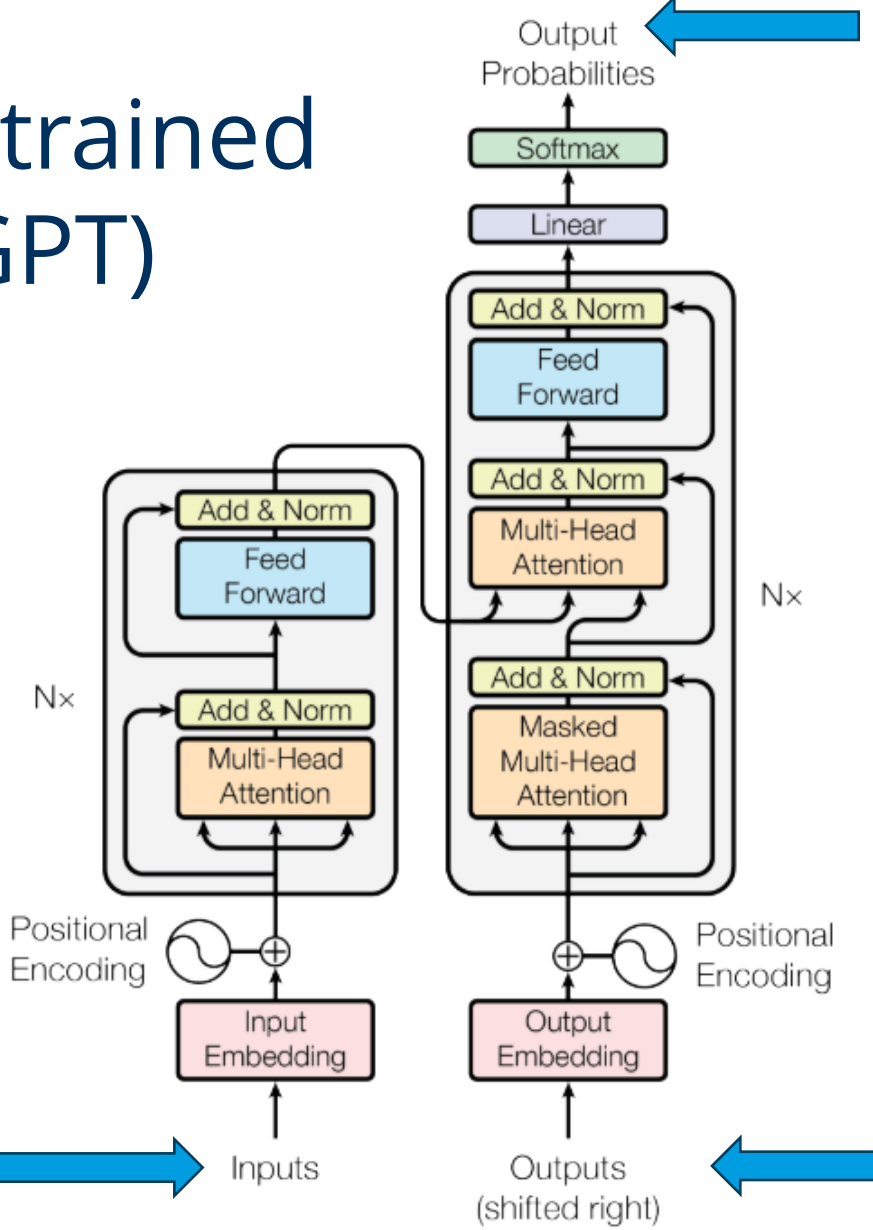
Example input:
Die Katze sitzt neben dem *Mikroskop*

Example output:
The cat sits next to the microscope

Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Generative Pretrained Transformer (GPT)

Task: Translation



Next word probabilities (according to context):
Heating: 0.9
Food: 0.8
Dog: 0.2
Microscope: 0.2

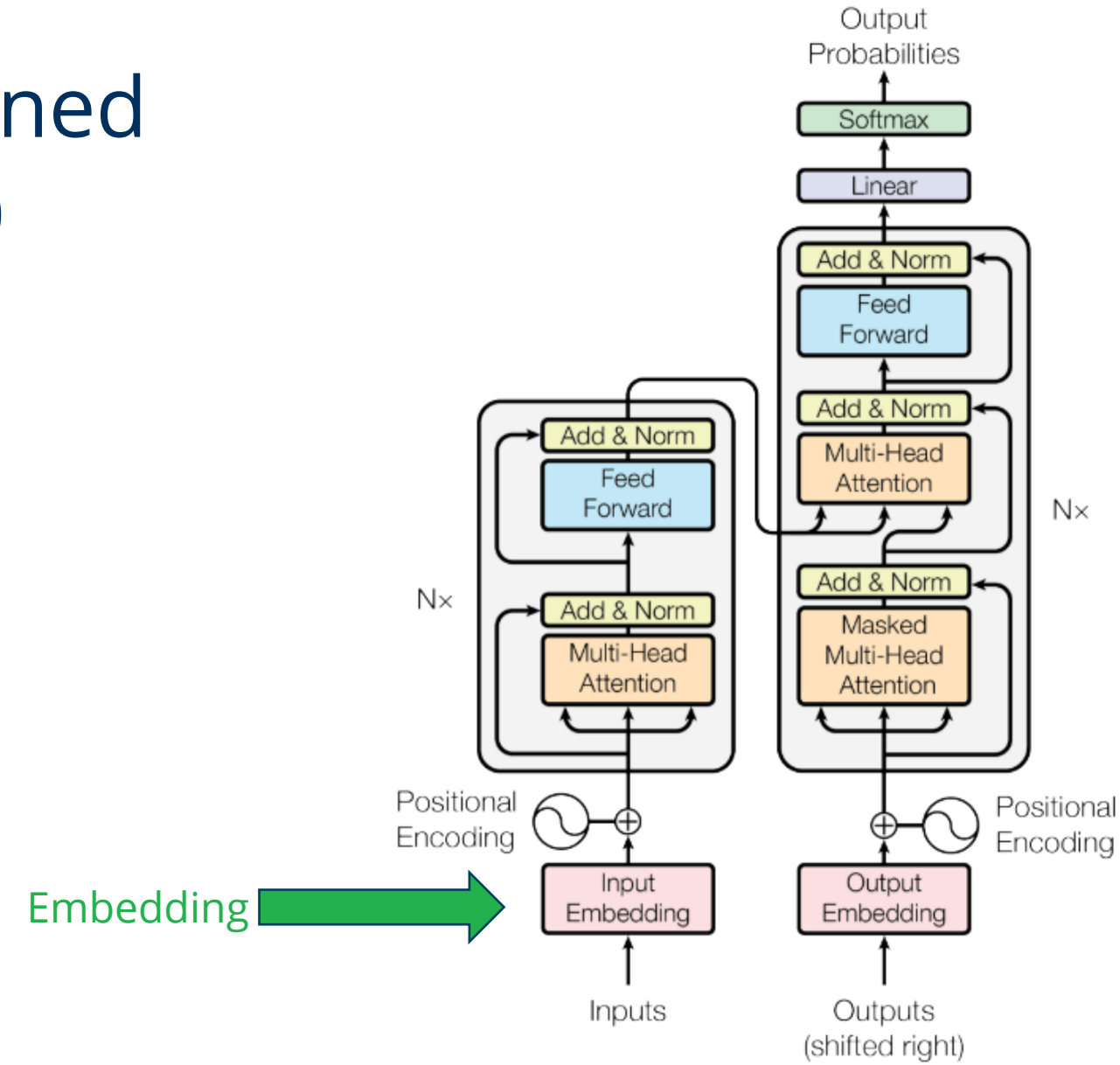
Through fine-tuning the transformer, you can increase the probabilities of words.

Example input:
Die Katze sitzt neben dem *Mikroskop*

Example output:
The cat sits next to the microscope

Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Generative Pretrained Transformer (GPT)



Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Quiz

What is an embedding?

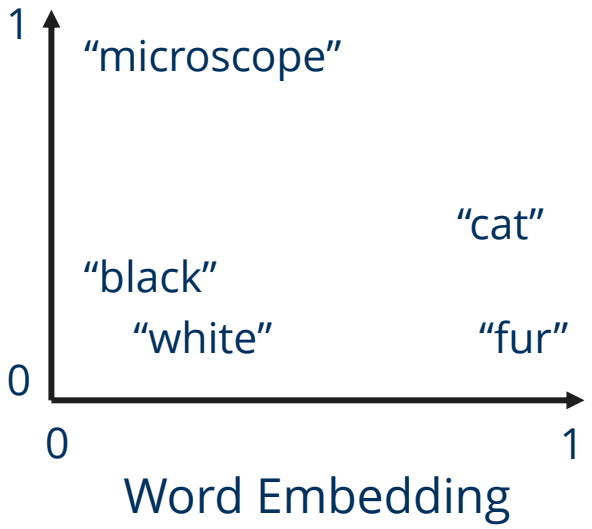
a) in the context of image processing

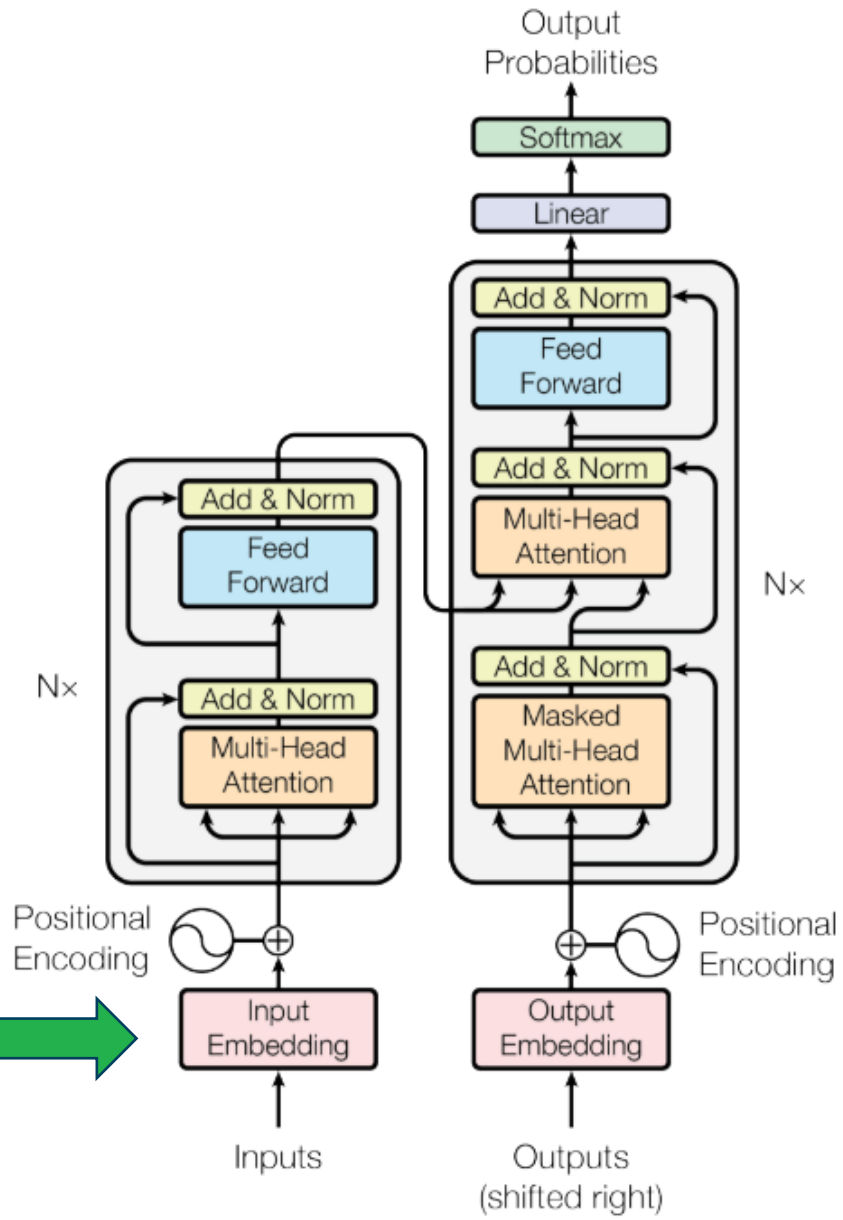
b) in the context of neural networks

c) in general

Generative Pretrained Transformer (GPT)

Words need to be converted into vectors to enable NNs to process them.





Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Attention is all you need

The position of the word in the sentence / context may have influence on its meaning. The position is encoded in a vector as well.

The cat sits next to a microscope.

Next to the microscope there is a cat.

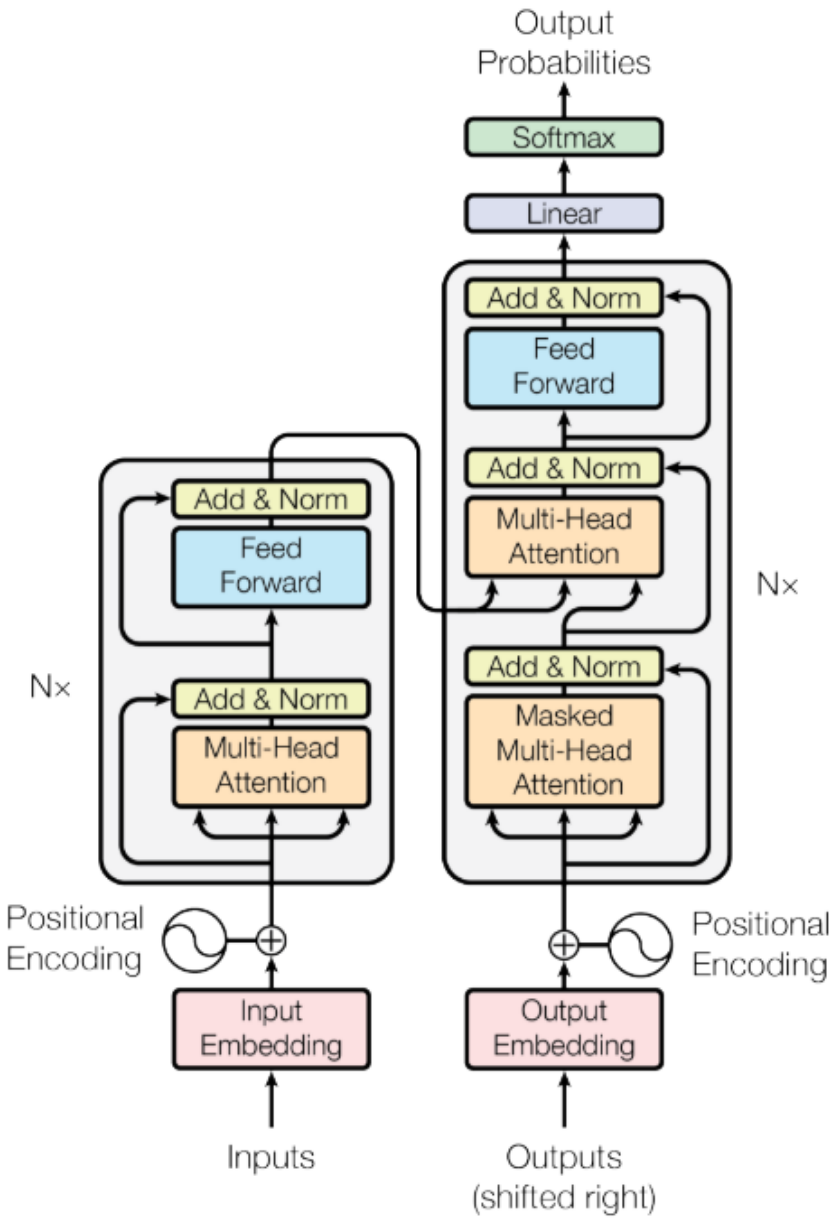
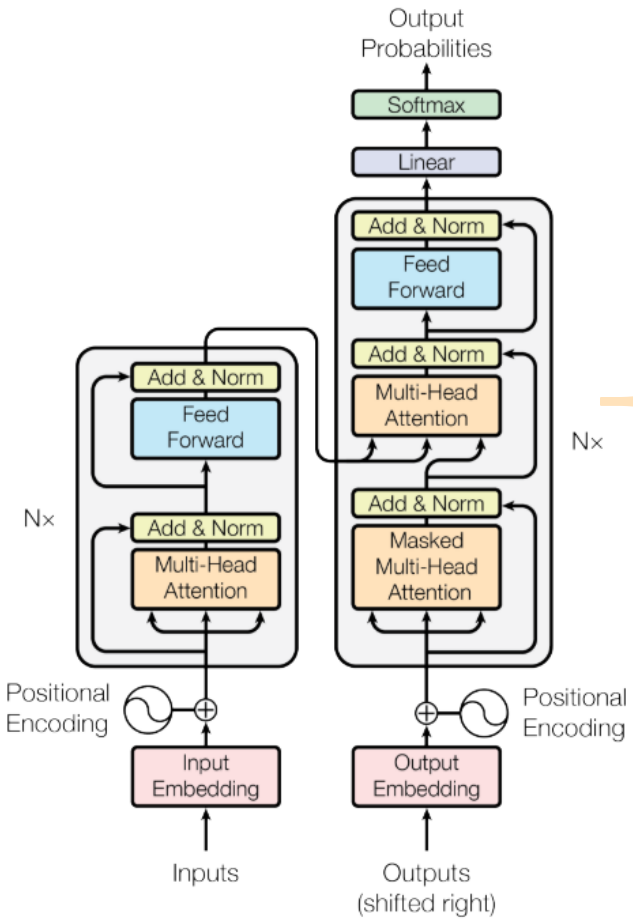
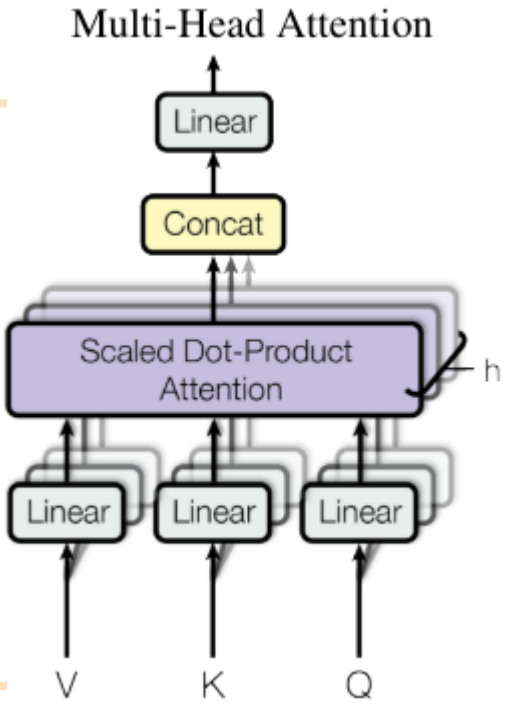
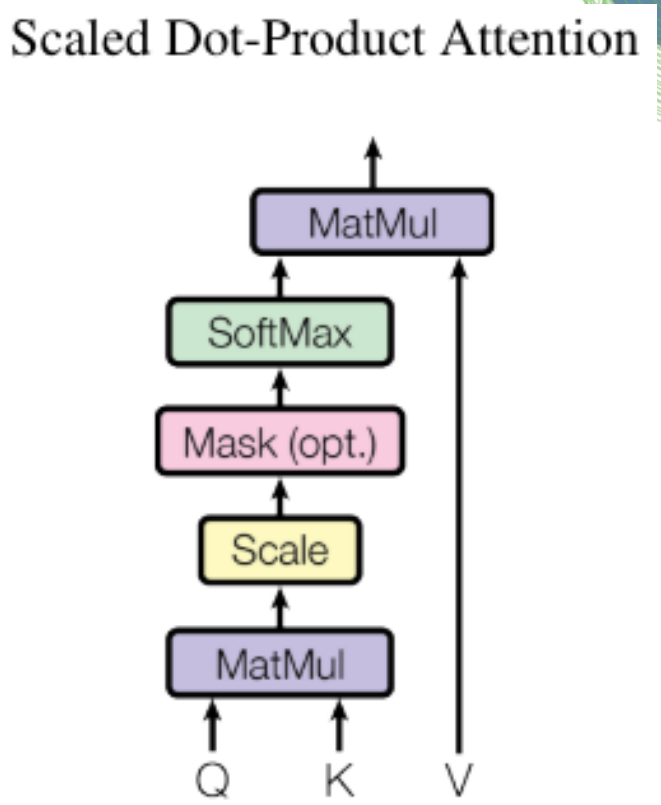


Figure source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

Attention is all you need







Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762

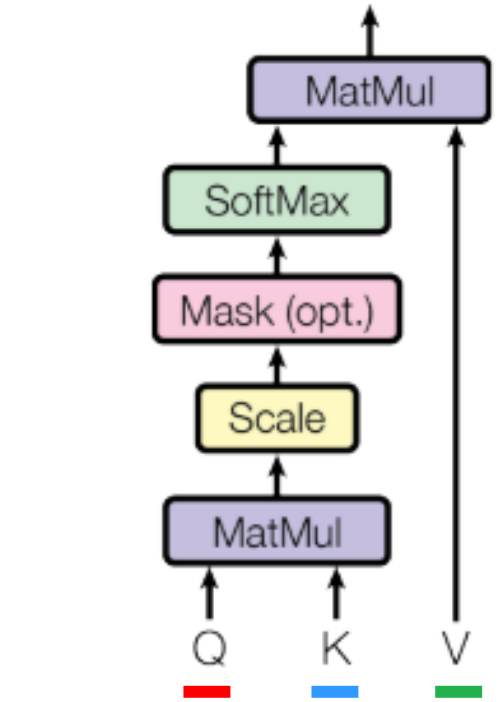
Scaled dot-product attention

Attention score: How much related are two words?

Query: For which word are we calculating attention?

Key: To which word are we calculating attention

Value: Relevance of the query-key relationship



The cat is black and white.

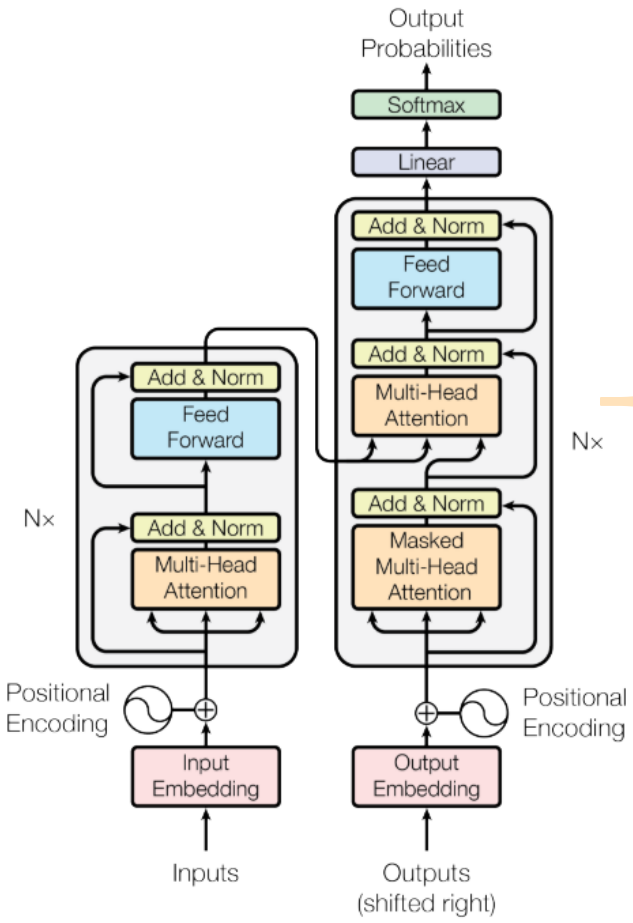
Relevance value: 0.1

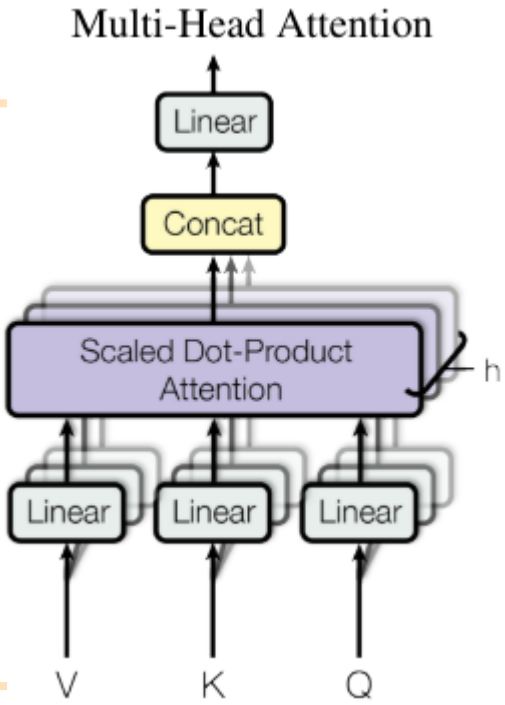
The cat is meowing.

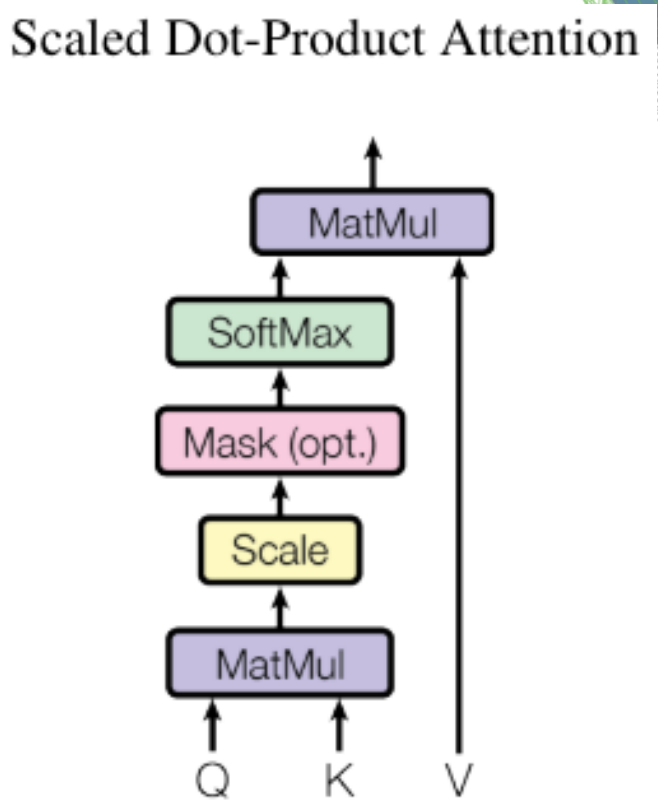
Relevance value: 0.9

Source: Vaswani et al (2017)
https://arxiv.org/abs/1706.03762
See also: https://www.youtube.com/watch?v=sznZ78HquPc

Attention is all you need



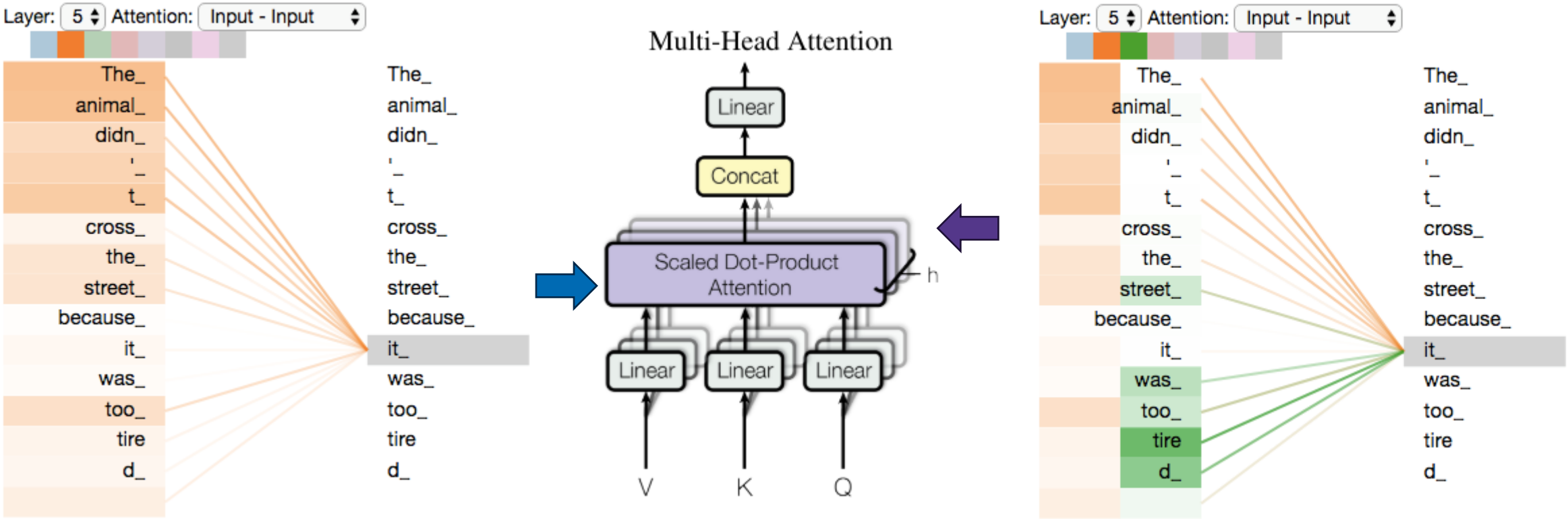




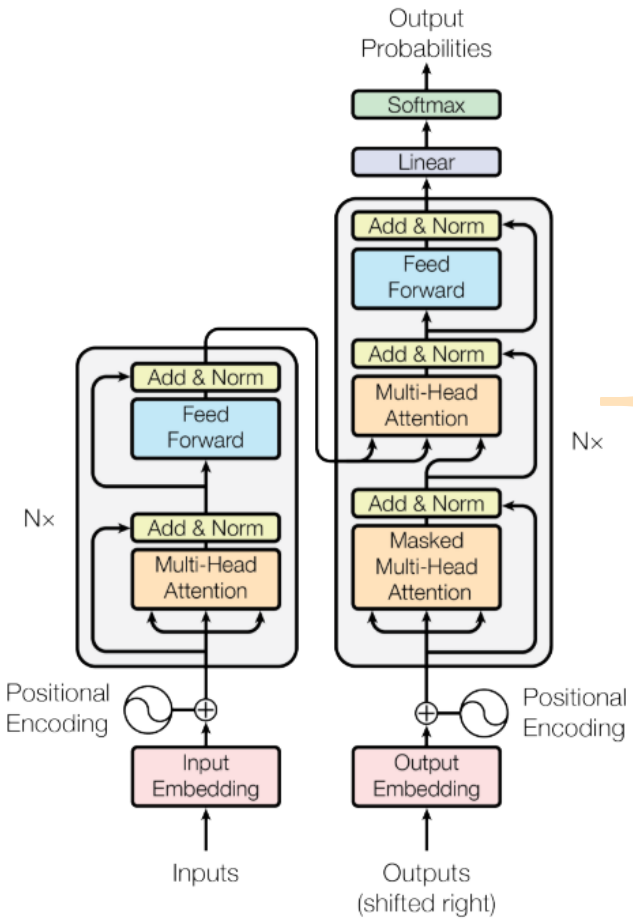
Source: Vaswani et al (2017) https://arxiv.org/abs/1706.03762

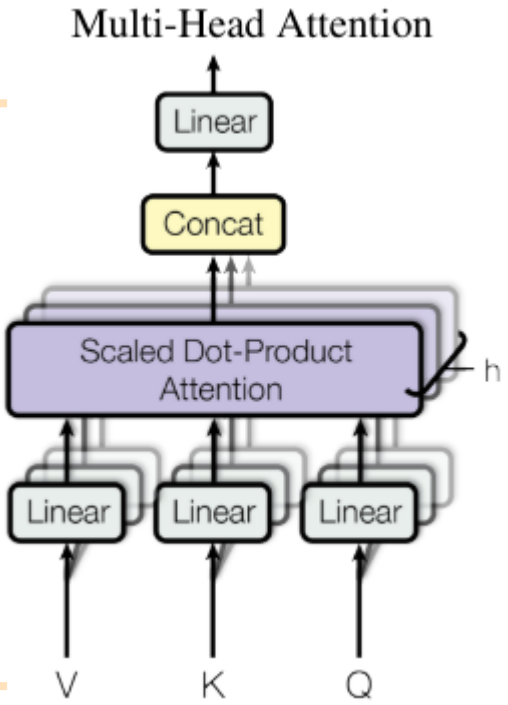
Multi-head attentions

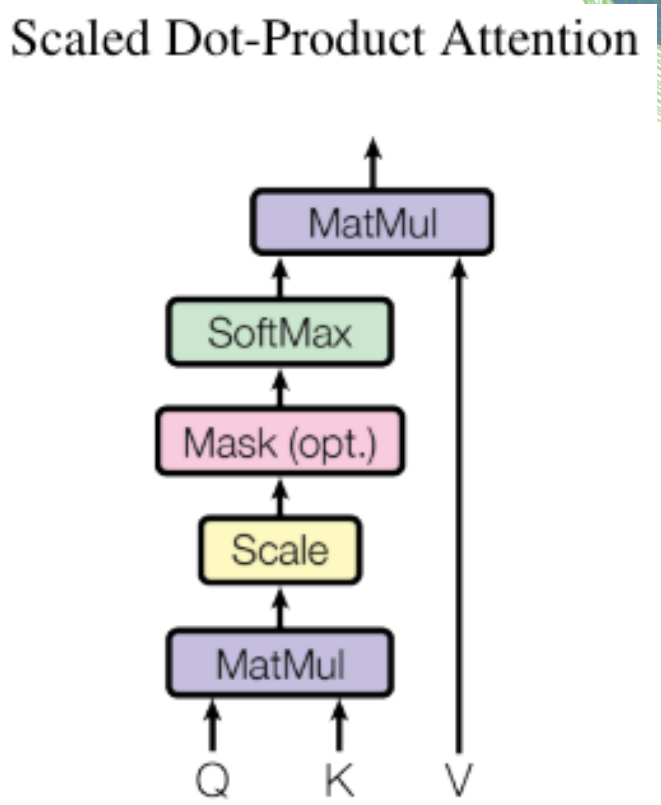
Multiple aspects represented by multiple attention heads



Attention is all you need



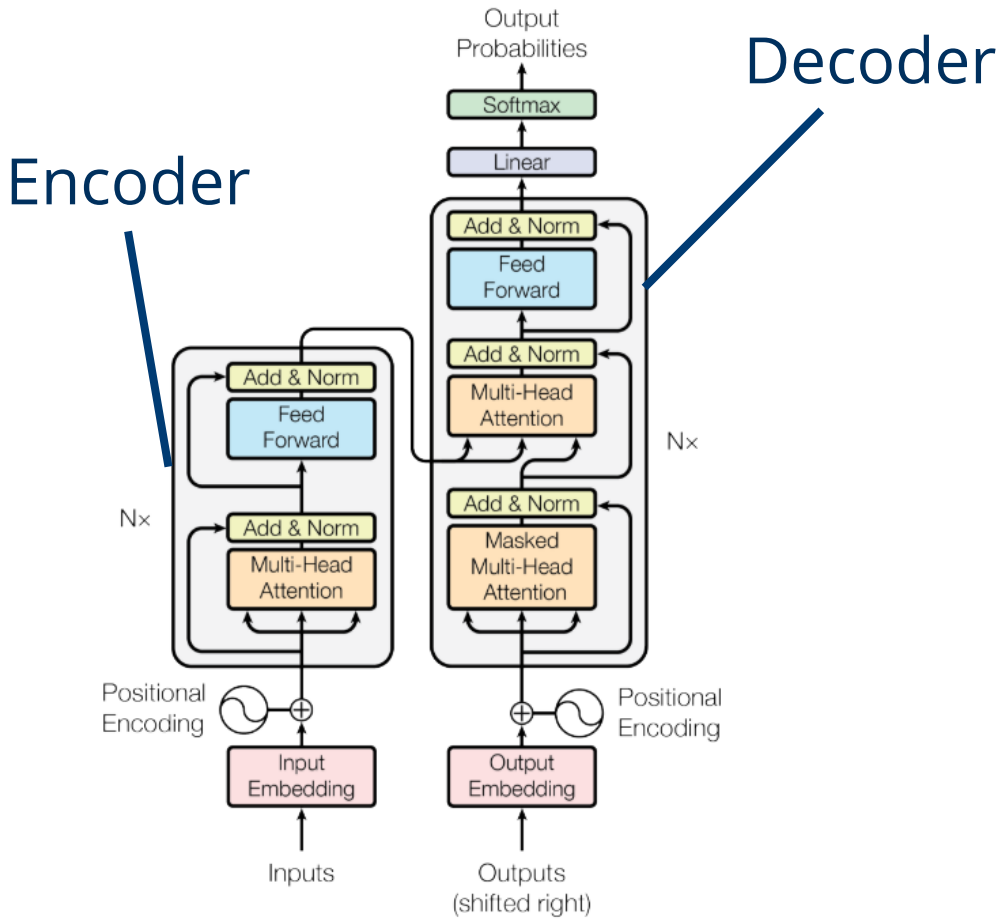


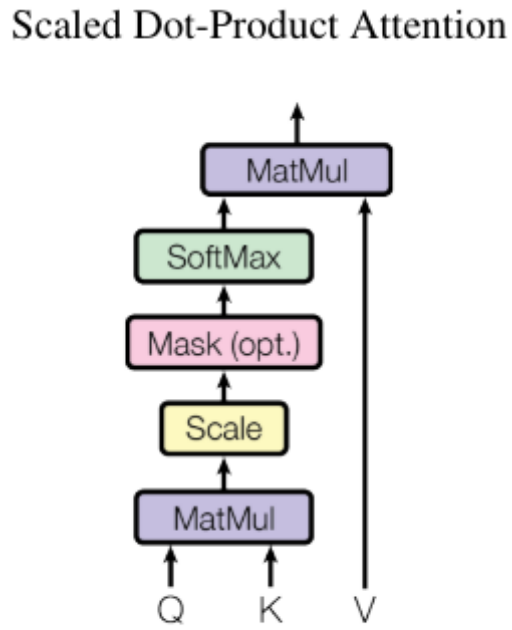


Source: Vaswani et al (2017) https://arxiv.org/abs/1706.03762

Attention is all you need

Summary





The cat is black and white. Relevance

attention score

The cat is meowing. Relevance

attention score

Prompt Engineering

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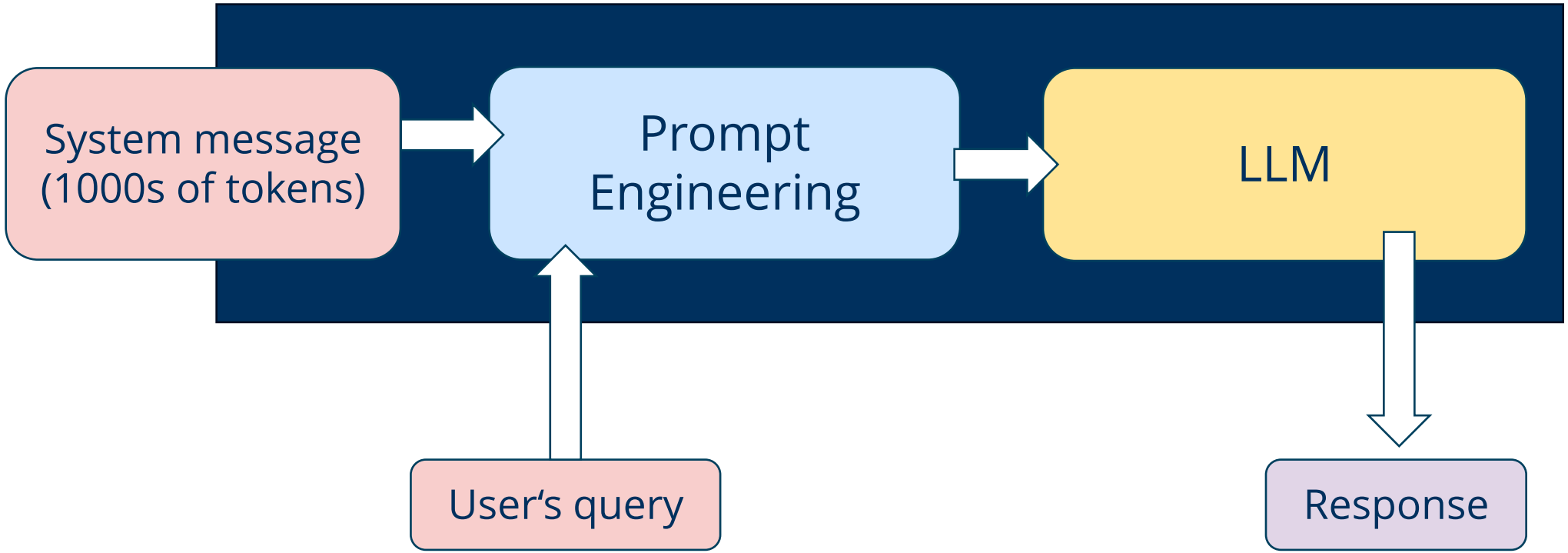
Long-context prompting

Also known as *in-context learning*

Context with plenty of knowledge provided in system message



Prompt Engineering

Modify a prompt until it works.

- Set context / assign role
- Define output style
- Give hints / conditions
- List tasks
- Provide data

```
%%prompt
You are a professional academic consultant who gives advice to students for writing their thesis.



Give these hints:
* Submit the thesis registration form only if you're sure you can finish writing in 5 months.
* Start writing your thesis early, e.g. the fundamentals chapter can be written from day one.
* Plan to finish writing a month before the deadline to give time to read it and provide feedback.
```

Here are some important hints to consider while writing your thesis:

1. Evaluate your readiness: Before submitting the thesis registration form, carefully assess your commitment and availability. Ensure that you are confident in completing your thesis within the given timeframe of five months. This will prevent unnecessary stress and help you maintain a focused mindset throughout the writing process.

2. Begin early: Starting your thesis early is crucial. Begin by working on the fundamentals chapter, as it lays the foundation for your entire thesis. By initiating this chapter from day one, you will establish a good writing routine and make steady progress as you move forward.

3. Plan ahead: Create a realistic schedule to guide your writing process. Break down the work into smaller tasks, setting deadlines for each chapter or section. This will help you maintain a structured approach and make the workload more manageable. Be sure to allocate time for research, data analysis, and revisions.

4. Aim to finish ahead of time: Plan to complete your thesis writing at least one month before the official deadline. This buffer time will allow you to thoroughly review and revise your work before final submission. It will also provide an opportunity to seek feedback from professors or peers, improving the overall quality of your thesis.

Prompt Engineering

Modify a prompt until it works.

- Set context / assign role
- Define output style
- Give hints / conditions
- List tasks
- Provide data

```
%%prompt
You are a professional academic consultant who gives advice to students for writing their thesis.
Keep it short.


Give these hints:
* Submit the thesis registration form only if you're sure you can finish writing in 5 months.
* Start writing your thesis early, e.g. the fundamentals chapter can be written from day one.
* Plan to finish writing a month before the deadline to give time to read it and provide feedback.
```

As a professional academic consultant, here are some helpful hints for writing your thesis:

1. Only submit the thesis registration form if you are confident about completing it within the given timeframe of 5 months.
2. Begin writing your thesis early to ensure a smooth process. For instance, you can start with the fundamentals chapter right from day one.
3. Plan your schedule effectively to finish writing your thesis at least a month before the submission deadline. This will allow ample time for reading, revising, and receiving feedback.

By following these recommendations, you'll have a better chance of successfully completing your thesis on time and with improved quality. Good luck!

Prompt Engineering

Modify a prompt until it works.

- Set context / assign role
- Define output style
- Give hints / conditions
- List tasks
- Provide data

```
%%prompt
You are a professional academic consultant who gives advice to students for writing their thesis.
Keep it short.
Write like Yoda would.

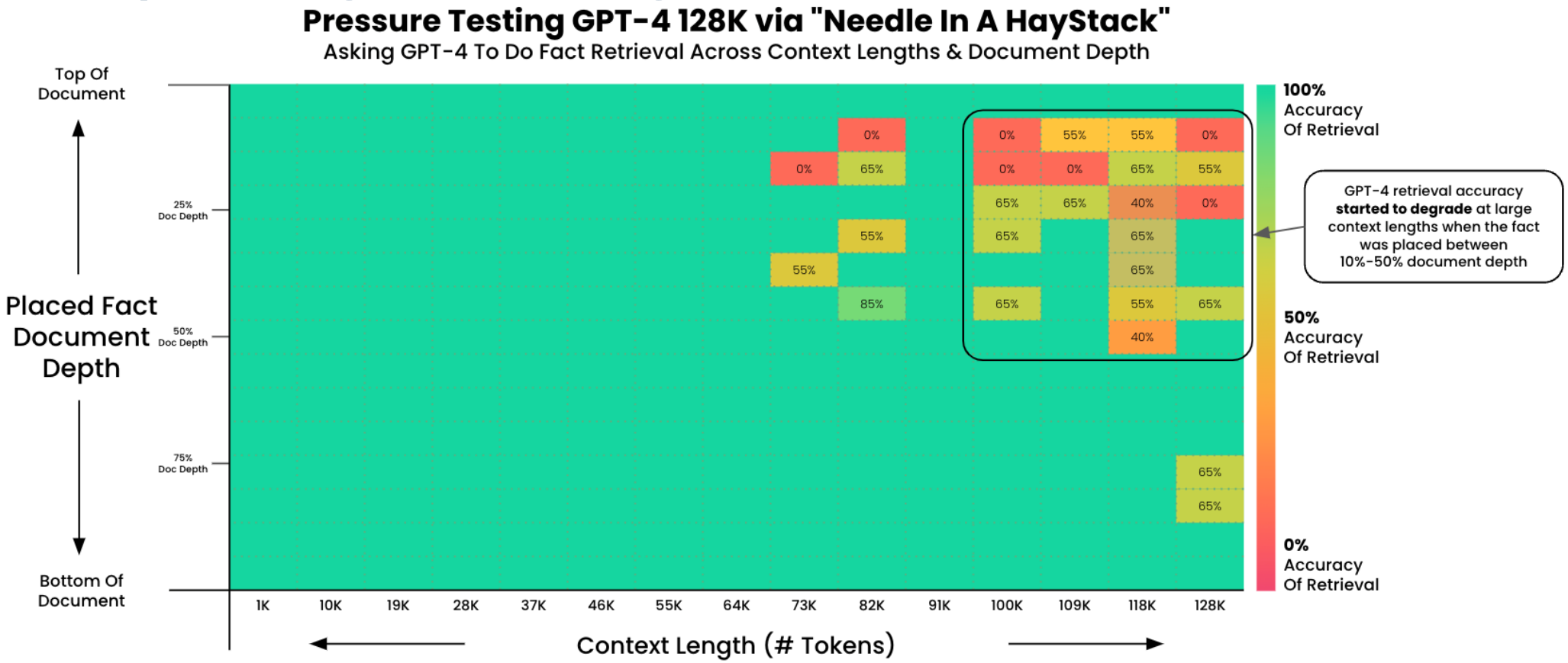
Give these hints:
* Submit the thesis registration form only if you're sure you can finish writing in 5 months.
* Start writing your thesis early, e.g. the fundamentals chapter can be written from day one.
* Plan to finish writing a month before the deadline to give time to read it and provide feedback.
```

Hints for thesis writing, I have. Follow them, you must.

- Only submit the thesis registration form, if in 5 months, complete writing, you are sure.
- From day one, start writing your thesis, you must. Begin with the fundamentals chapter.
- A month before the deadline, finish writing, you should. Time to read and provide feedback, it will give you.

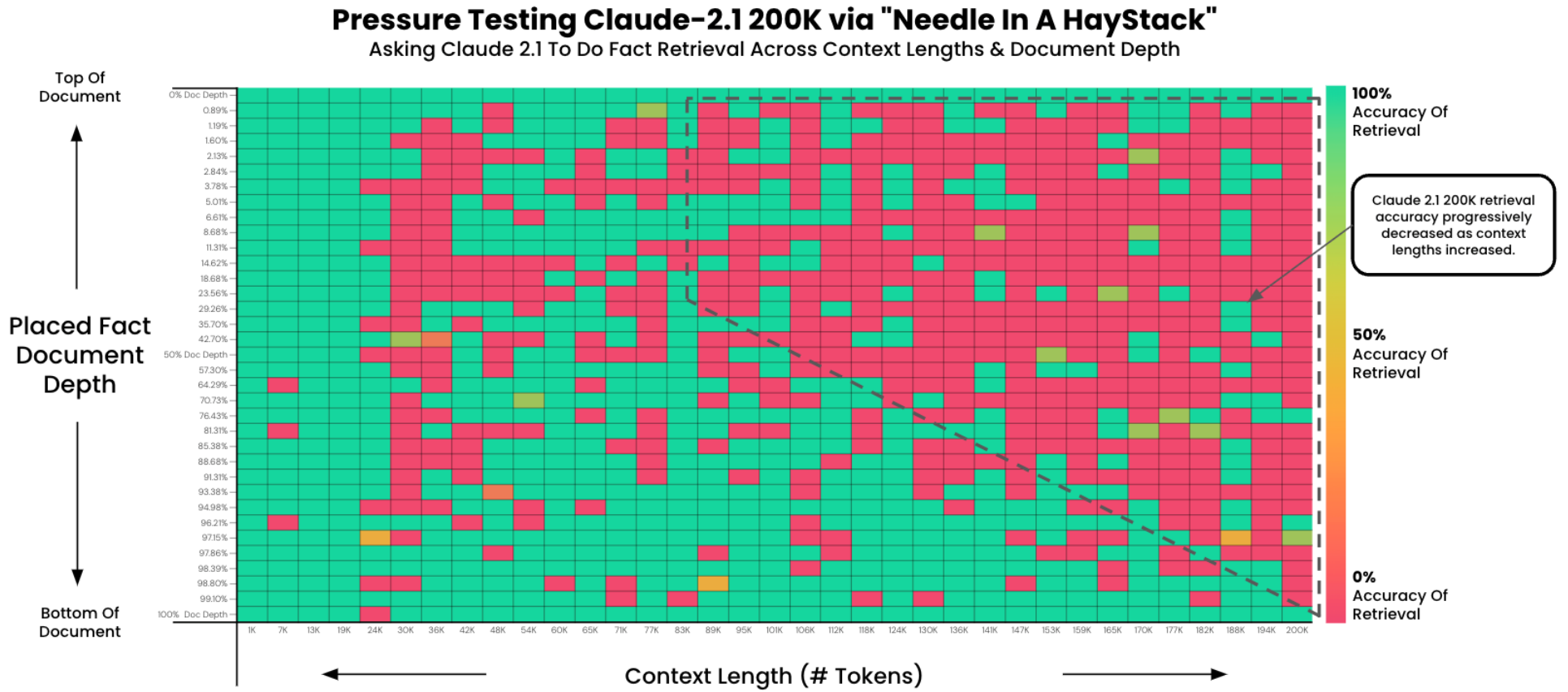
Follow these hints, you must. Successful thesis writing, you will achieve. Hmmmm.

Prompt engineering



Goal: Test GPT-4 Ability To Retrieve Information From Large Context Windows
A fact was placed within a document. GPT-4 (1106-preview) was then asked to retrieve it. The output was evaluated for accuracy.
This test was run at 15 different document depths (top > bottom) and 15 different context lengths (1K >128K tokens).
2x tests were run for larger contexts for a larger sample size.

Prompt engineering



Goal: Test Claude 2.1 Ability To Retrieve Information From Large Context Windows
A fact was placed within a document. Claude 2.1 (200K) was then asked to retrieve it. The output was evaluated (with GPT-4) for accuracy.
This test was run at 35 different document depths (top > bottom) and 35 different context lengths (1K >200K tokens).
Document Depths followed a sigmoid distribution

LLMs are everywhere

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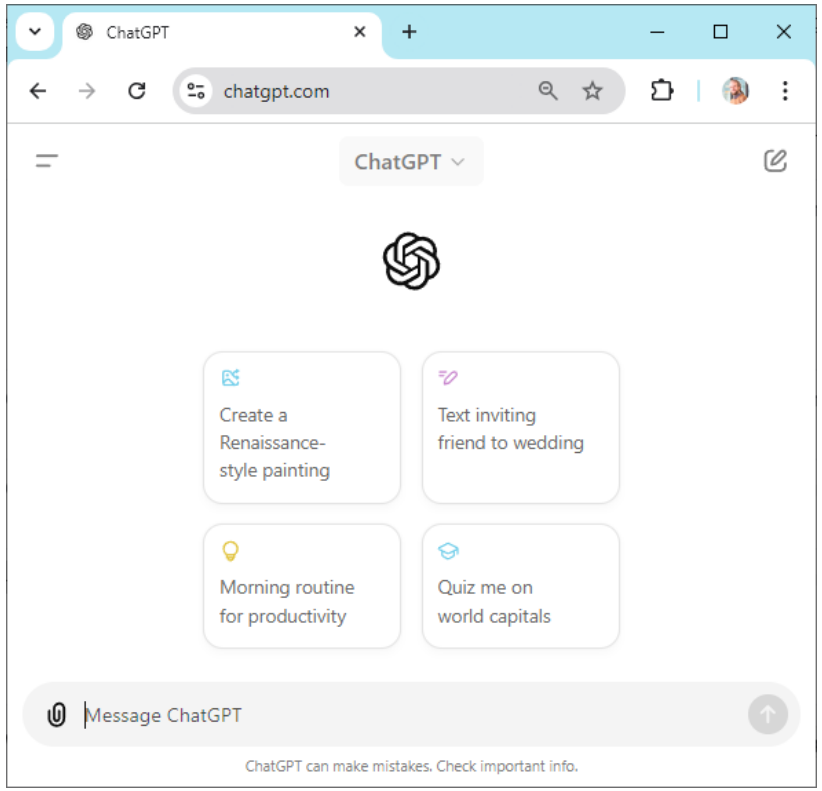
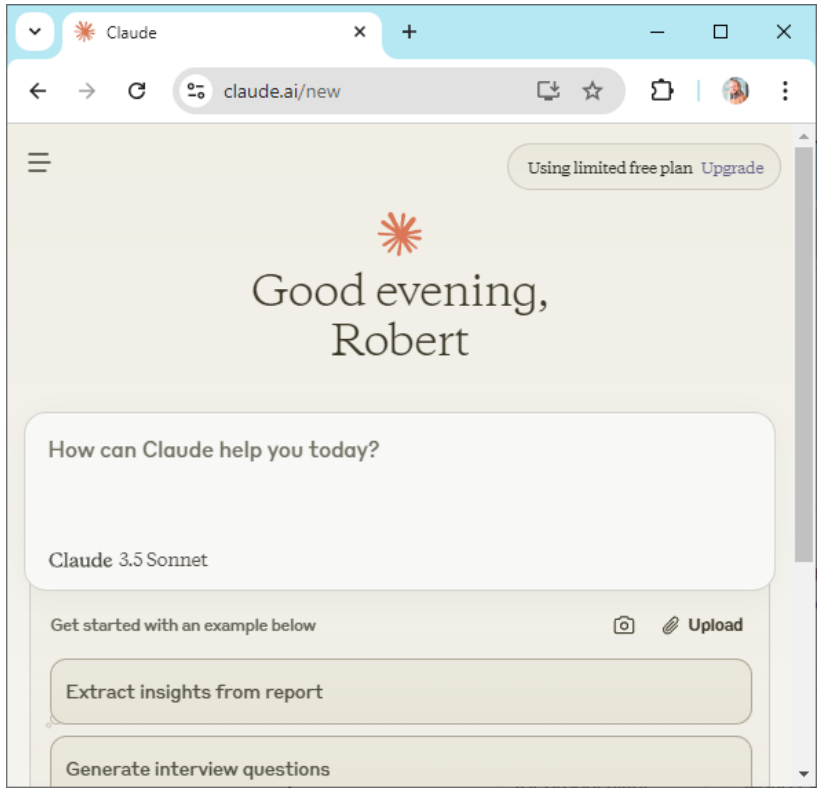
LLMs are everywhere

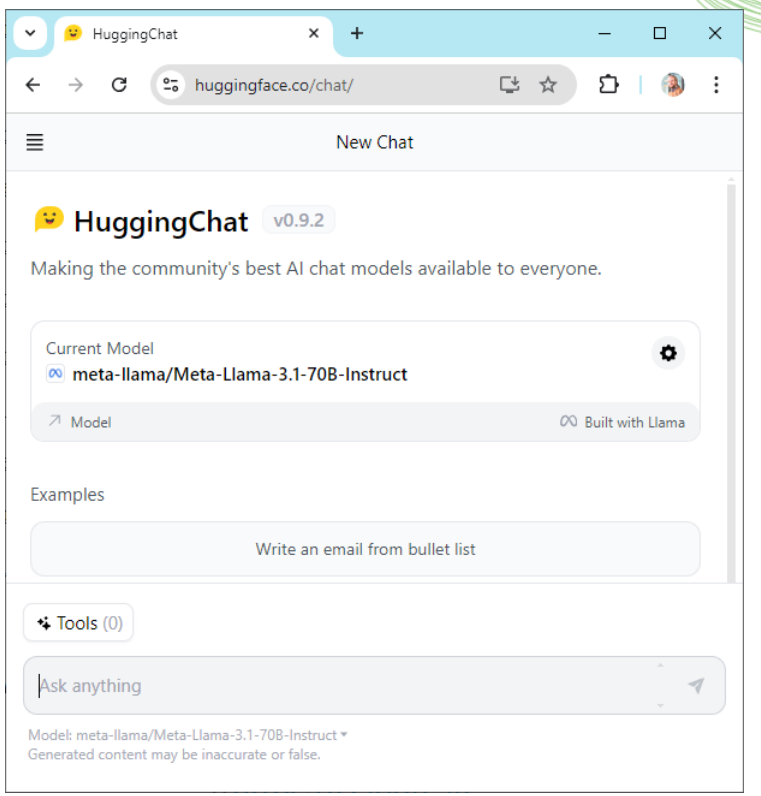
ChatGPT



Claude



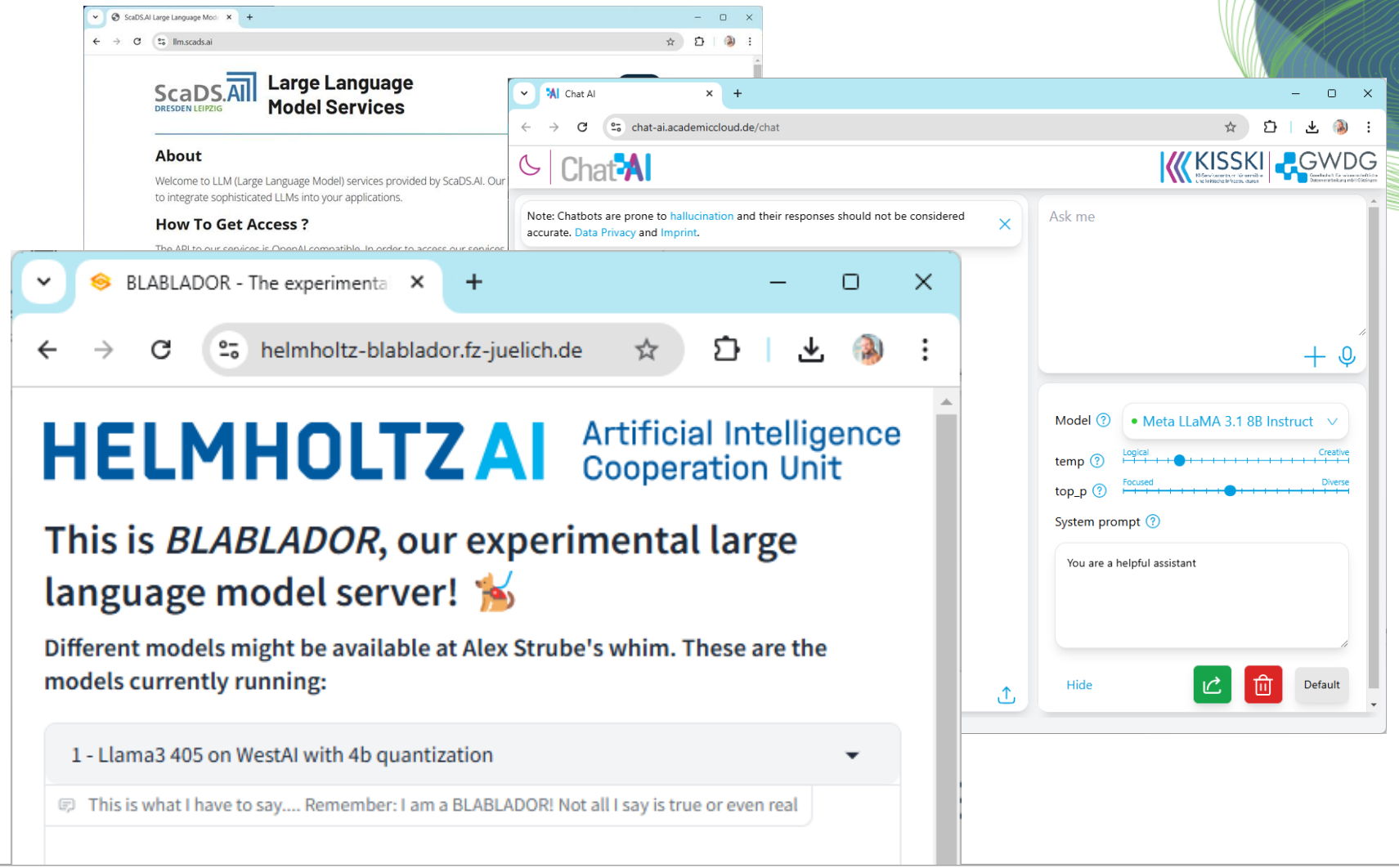
Open models via Huggingface



https://chatgpt.com/
https://claude.ai/
https://huggingface.co/chat/

Institutional AI Infrastructure

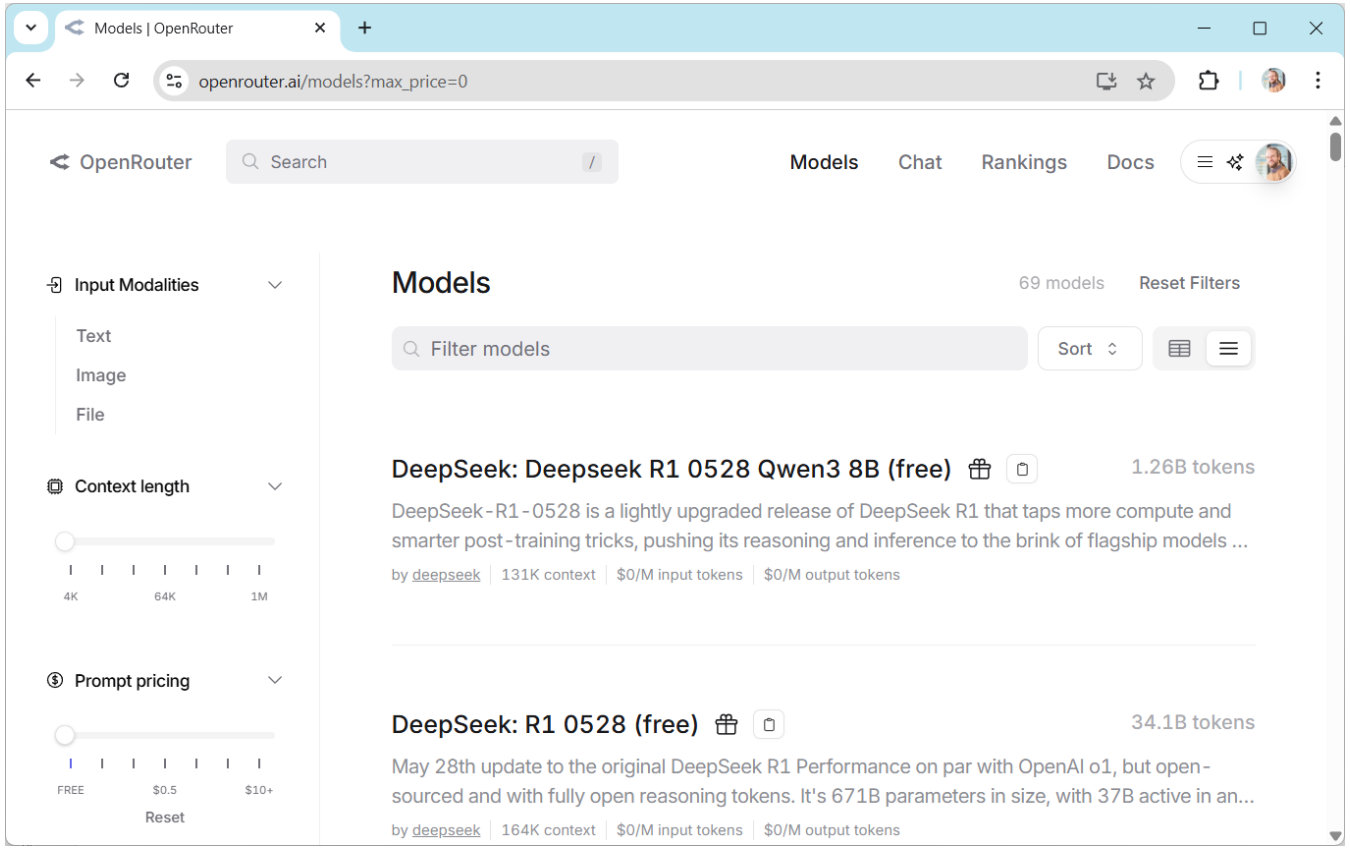
- Many institutes have a local LLM Server
  - Great for teaching
  - Privacy preserving
- Cross-institutional infrastructure, in Germany:
  - KISSKI ChatAI
  - Helmholtz Blablador

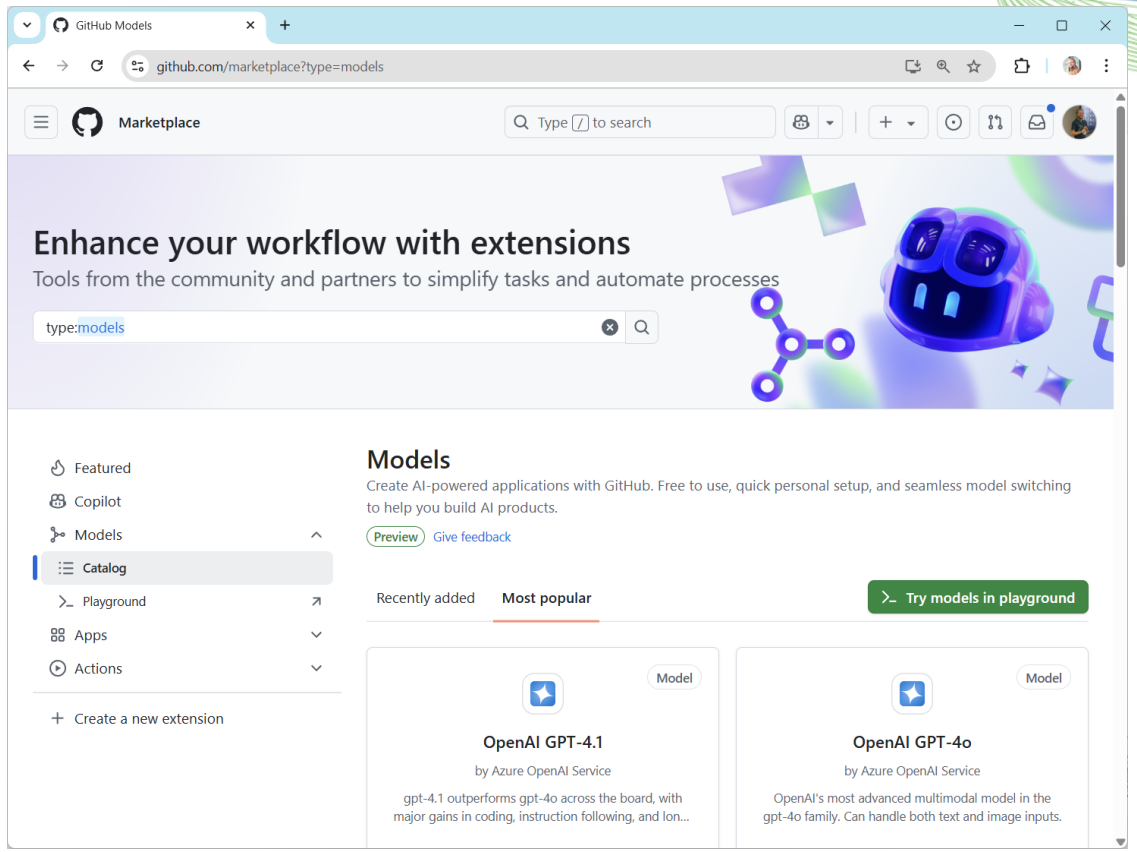


Commercial AI Infrastructure

If it's for free, you are part of the product.

Github Models, OpenRouter, Huggingface,...





Local AI infrastructure

For Small Language Models only

Command Prompt - ollama

```
C:\Users\rober>ollama run llama3.2:1b
>>> What Python libraries can be used for cell segmentation in microscopy images?
There are several Python libraries that can be used fo
the most popular ones:

1. **OpenCV**: OpenCV is a powerful library for comput
segmentation. It provides algorithms such as threshold
t
cells.
2. **scikit-image**: scikit-image is a library for ima
segmentation, and analysis of images. It can be used f
watershed transform.
3. **PyTorch**: PyTorch is a popular deep learning fra
detection, and segmentation tasks, including cell segm
4. **TensorFlow**: TensorFlow is another popular deep
and analysis, including cell segmentation.
5. **scikit-learn**: scikit-learn is a library for mac
regression, clustering, and density estimation tasks,
6. **Keras**: Keras is a high-level neural networks AP
networks (CNNs) for image processing tasks like cell s
```

Command Prompt - ollama

```
Example code for cell segmentation using OpenCV:
```python
import cv2
import numpy as np

# Load the image
img = cv2.imread('image.jpg')

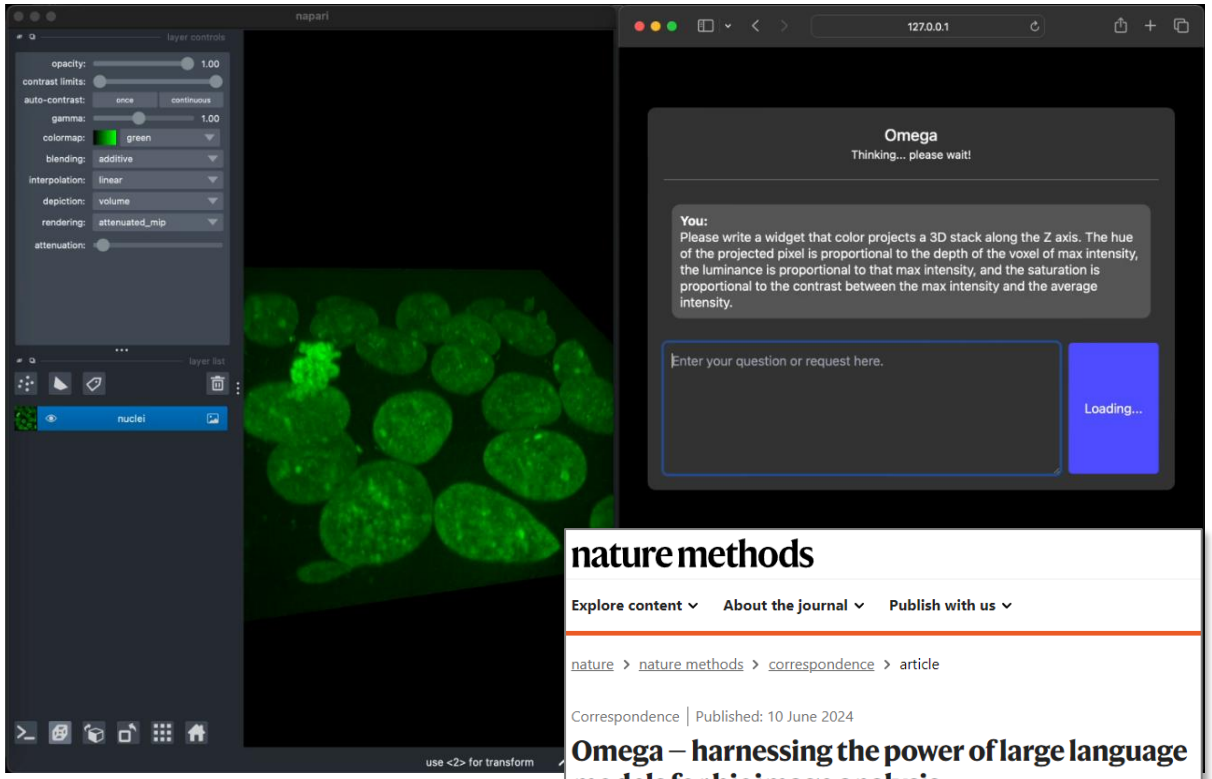
# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Apply thresholding to segment cells
thresh = cv2.threshold(gray, 0, 255, cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU)[1]

# Display the segmented image
cv2.imshow('Segmented Image', thresh)
cv2.waitKey(0)
cv2.destroyAllWindows()
```
This code loads an image, converts it to grayscale, applies thresholding using Otsu's method, and displays the
segmented image.
```

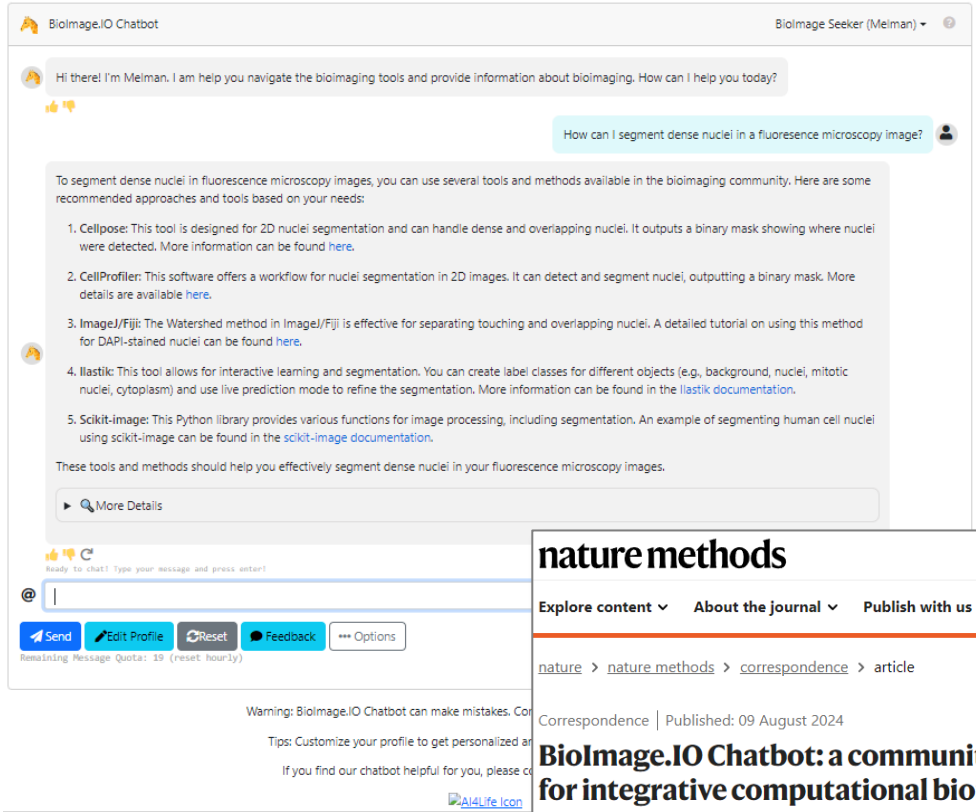
LLMs are everywhere

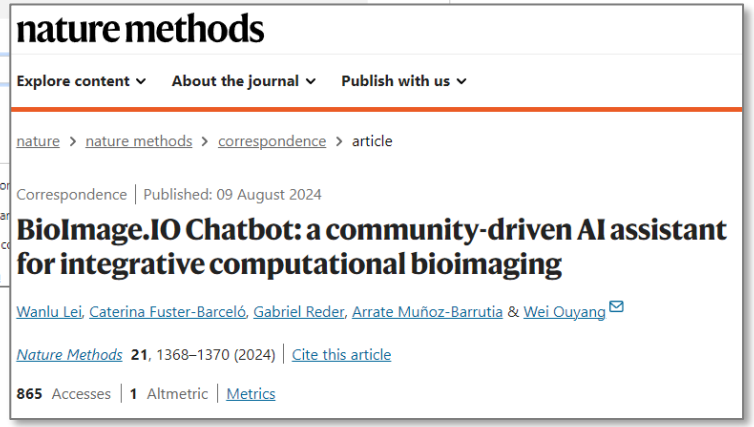
Napari-chatGPT / Omega



https://github.com/royerlab/napari-chatgpt
https://www.nature.com/articles/s41592-024-02310-w

Bioimage-io ChatBot





https://bioimage.io/chat/
https://www.nature.com/articles/s41592-024-02370-y

LLMs are everywhere

aider

```
macbook$ aider demo.py
Added demo.py to the chat
Using git repo: .git

demo.py> add a name param to the `greeting` function. add all the types.

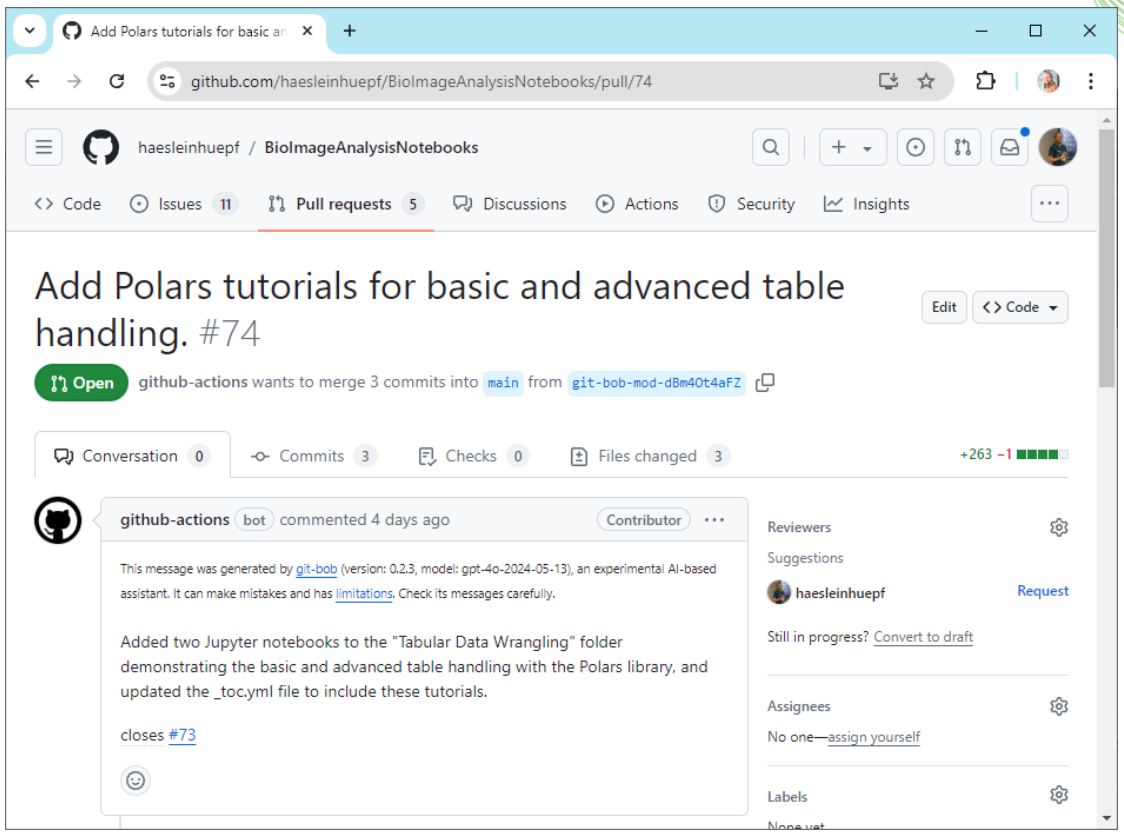
I will update the greeting function to accept a name parameter and include type
hints for the parameter and the return type.

demo.py
<<<<<<< ORIGINAL
def greeting():
    print("Hello, world!")
=======
def greeting(name: str) -> None:
    print(f"Hello, {name}!")
>>>>>>> UPDATED

Applied edit to demo.py
Commit 9237455 aider: Updated greeting function to accept a name parameter and
include type hints.

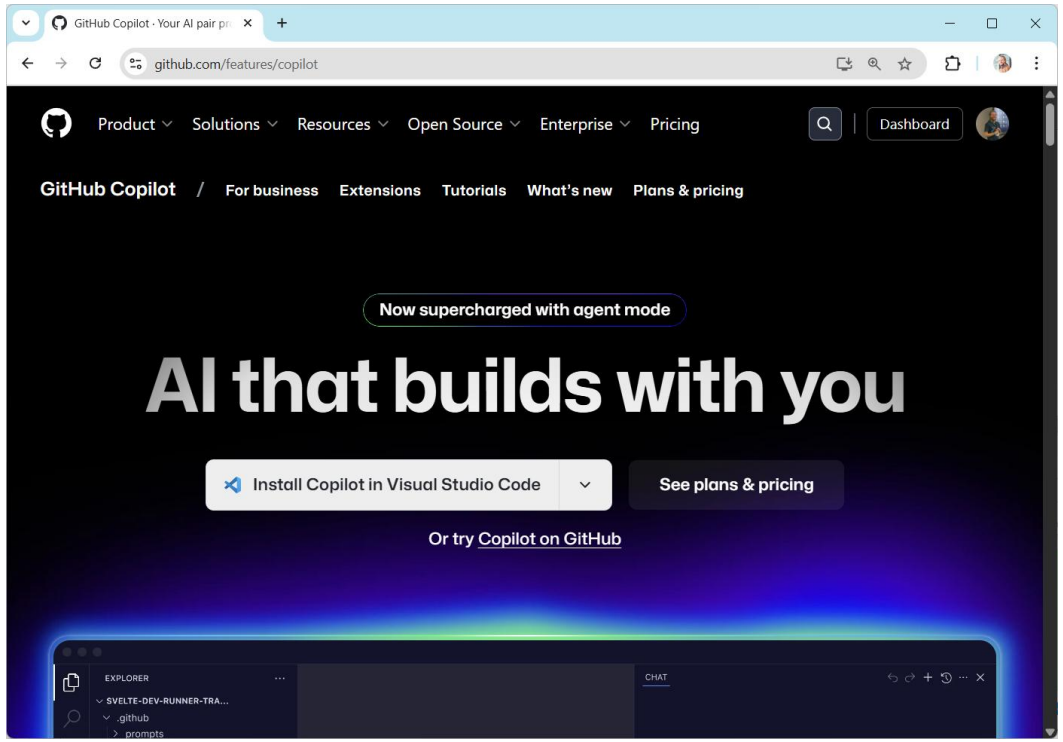
demo.py>
```

git-bob

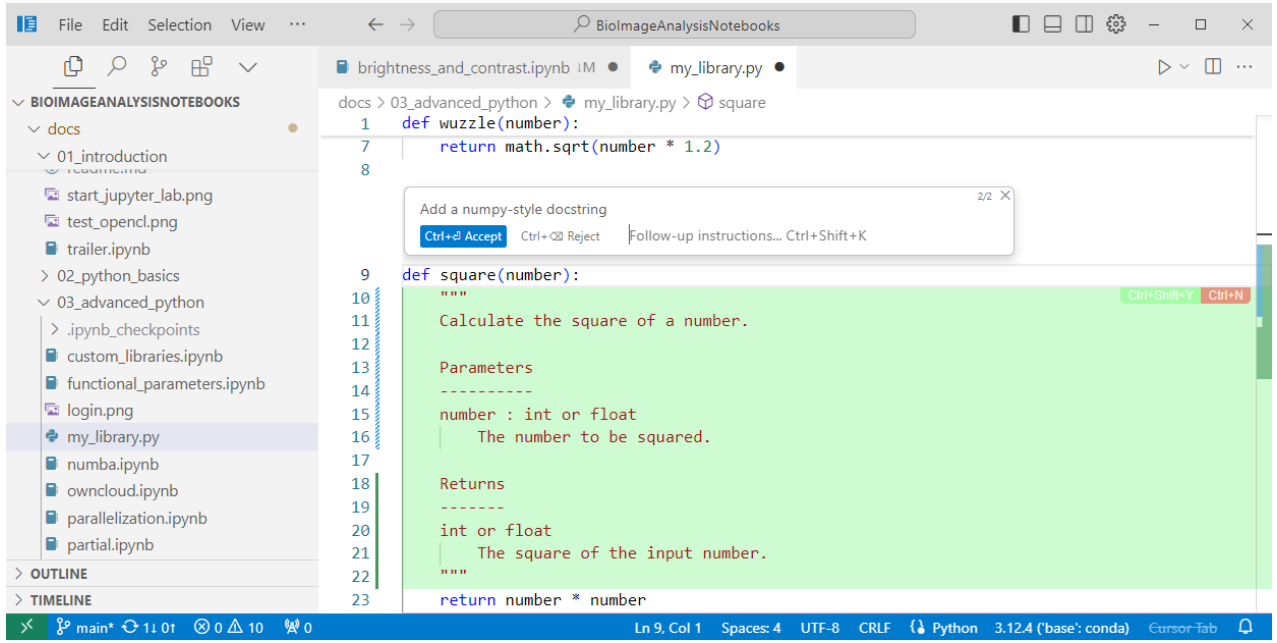


LLMs are everywhere

Github Copilot



Cursor



Usings LLMs in Python

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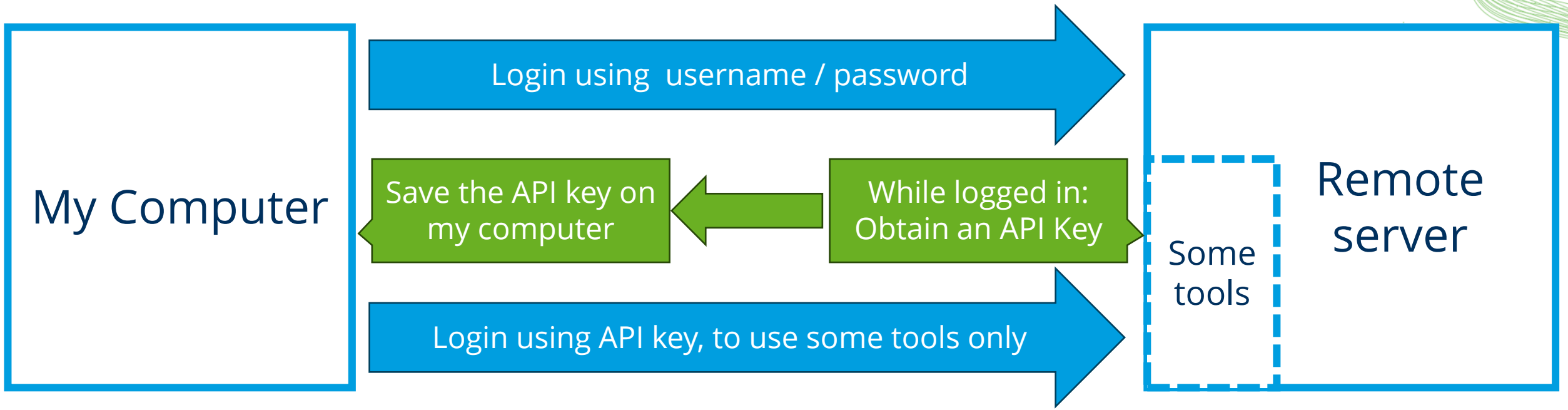
Authentication using API-Keys

Application programming interface (API) keys: A modern kind of password
basically invented to avoid entering username/passwords over and over again



Never share your API key with others

API key can be revoked / invalidated any time,

The OpenAI Application Programming Interface

De-facto standard for communicating with LLMs

```
[1]: import os
     import openai
     openai.__version__
```

```
[1]: '1.5.0'
```

```
[2]: client = openai.OpenAI()
     client
```

```
[2]: <openai.OpenAI at 0x1bd8871ec90>
```

Ollama

```
client = openai.OpenAI(
    base_url = "http://localhost:11434/v1",
    api_key = "none" # not required by ollama
)
```

KISSKI

```
client = openai.OpenAI(
    base_url="https://chat-ai.academiccloud.de/v1",
    api_key=os.environ.get('KISSKI_API_KEY')
)
```

The OpenAI Application Programming Interface

Request

```
my_messages = [{
    "role": "user",
    "content": "What's the capital of France?"
}]
my_messages
```

```
[{'role': 'user', 'content': "What's the capital of France?"}]
```

```
response = client.chat.completions.create(
        model="gpt-4o",
        messages=my_messages
    )
response
```

Response

```
ChatCompletion(id='chatcmpl-BeGoMYWRYYYQ3XvySyZGRvcqDFCCT', choices=[Choice(finish_
reason='stop', index=0, logprobs=None, message=ChatCompletionMessage(content='The c
apital of France is Paris.', refusal=None, role='assistant', annotations=[], audio=
None, function_call=None, tool_calls=None))], created=1748937442, model='gpt-4o-202
4-08-06', object='chat.completion', service_tier='default', system_fingerprint='fp_
07871e2ad8', usage=CompletionUsage(completion_tokens=7, prompt_tokens=13, total_tok
ens=20, completion_tokens_details=CompletionTokensDetails(accepted_prediction_token
s=0, audio_tokens=0, reasoning_tokens=0, rejected_prediction_tokens=0), prompt_toke
ns_details=PromptTokensDetails(audio_tokens=0, cached_tokens=0)))
```

```
response.choices[0].message.content
```

```
'The capital of France is Paris.'
```

Common helper functions

To make our life simpler, we use such helper functions.

```python
def prompt_openai(prompt:str, model="gpt-4o"):
    """A prompt helper function that sends a prompt to openAI
    and returns only the text response.
    """
    # setup connection to the LLM
    client = openai.OpenAI()

    # submit prompt
    response = client.chat.completions.create(
        model=model,
        messages=[{"role": "user", "content": prompt}]
    )

    # extract answer
    return response.choices[0].message.content
```

```python
prompt_openai("How many o are in Woolloomoloo?")
```

```
'The word "Woolloomooloo" contains eight \'o\'s.'
```

[2]:
```python
def prompt_ollama(prompt:str, model="gemma3:1b"):
    """A prompt helper function that sends a prompt to
    ollama and returns only the text response."""
    # setup connection to the LLM server
    client = openai.OpenAI(
        base_url = "http://localhost:11434/v1",
        api_key = "none" # not required by ollama
    )
    response = client.chat.completions.create(
        model=model,
        messages=[{"role": "user", "content": prompt}]
    )

    # extract answer
    return response.choices[0].message.content
```

[3]:
```python
prompt_ollama("Hi!")
```

[3]:
```
"Hi there! How's your day going so far? 😊 \n\nIs there
anything you'd like to talk about or any I can help you w
ith?"
```

Common helper functions

To make our life simpler, we use such helper functions.

[2]:
```python
def prompt_kisski(prompt:str, model="meta-llama-3.1-70b-instruct"):
    """A prompt helper function that sends a message to KISSKI Chat AI API
    and returns only the text response.
    """
    # setup connection to the LLM-server
    client = openai.OpenAI(
        base_url="https://chat-ai.academiccloud.de/v1",
        api_key=os.environ.get('KISSKI_API_KEY')
    )

    response = client.chat.completions.create(
        model=model,
        messages= [{"role": "user", "content": prompt}]
    )

    # extract answer
    return response.choices[0].message.content
```

[3]:
```python
prompt_kisski("Hi!")
```

[3]:
```
"It's nice to meet you. Is there something I can help you with or would you
like to chat?"
```

[2]:
```python
def prompt_openrouter(prompt:str, model="google/gemini-2.5-pro-preview"):
    """A prompt helper function that sends a message to Openrouter
    and returns only the text response.
    """
    # setup connection to the LLM-server
    client = openai.OpenAI(
        base_url="https://openrouter.ai/api/v1",
        api_key=os.environ.get('OPENROUTER_API_KEY')
    )

    response = client.chat.completions.create(
        model=model,
        messages= [{"role": "user", "content": prompt}]
    )

    # extract answer
    return response.choices[0].message.content
```

[3]:
```python
prompt_openrouter("Hi!")
```

[3]:
```
'Hi there! How can I help you today?'
```

OpenAI API: Querying models

```
[9]: client = openai.OpenAI()

print("\n".join(sorted([model.id for model in client.models.list().data])))
```

```
babbage-002
chatgpt-4o-latest
codex-mini-latest
computer-use-preview
computer-use-preview-2025-03-11
dall-e-2
dall-e-3
davinci-002
...
gpt-4.1
gpt-4.1-2025-04-14
gpt-4.1-mini
gpt-4.1-mini-2025-04-14
gpt-4.1-nano
gpt-4.1-nano-2025-04-14
gpt-4.5-preview
gpt-4.5-preview-2025-02-27
gpt-4o
gpt-4o-2024-05-13
gpt-4o-2024-08-06
gpt-4o-2024-11-20
...
```

```
client = openai.OpenAI(
    base_url = "https://helmholtz-blablador.fz-juelich.de:8000/v1",
    api_key = os.environ.get('BLABLADOR_API_KEY')
)

print("\n".join([model.id for model in client.models.list().data]))
```

```
1 - Llama3 405 the best general model and big context size
1 - Ministral 8b - the fast model
1 - Teuken-7B-instruct-research-v0.4 - The OpenGPT-X model
10 Mistral-Nemo-Instruct-2407 - Our fast-experimental - with a large context size
2 - Qwen3 30B A3B - a reasoning model from Alibaba from April 2025
3 - DeepCoder-14B-Preview - the code model from 09.04.2025
alias-code
alias-fast
alias-fast-experimental
alias-large
alias-llama3-huge
alias-opengptx
alias-reasoning
```

Good Scientific Practice

Bundesministerium für Bildung und Forschung
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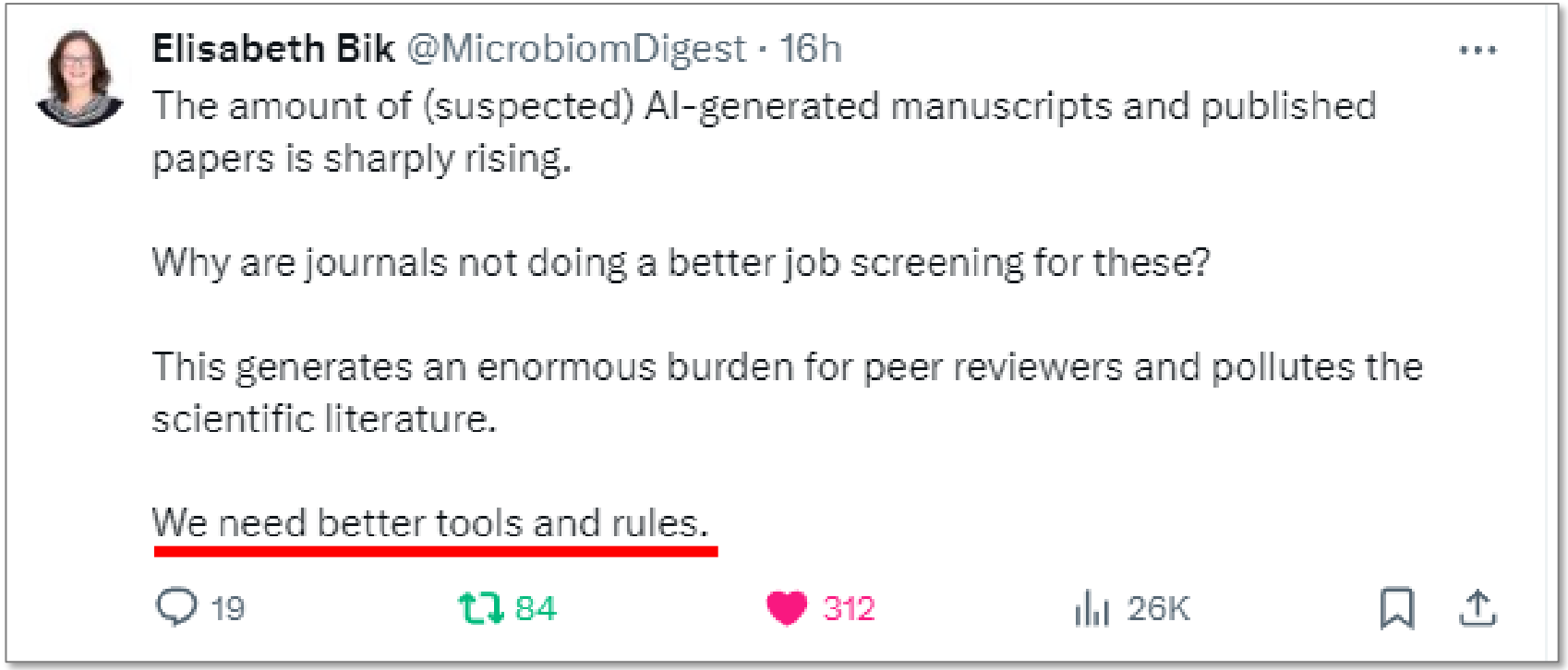
Diese Maßnahme wird mitfinanziert durch Steuermittel auf der Grundlage des von den Abgeordneten des Sächsischen Landtags beschlossenen Haushaltes.

Quiz: What could possible go wrong?

… when using LLMs for scientific [bio image] data analysis tasks?

Challenges

Generative artificial intelligence imposes a risk to science



Elisabeth Bik @MicrobiomDigest · 16h

The amount of (suspected) AI-generated manuscripts and published papers is sharply rising.

Why are journals not doing a better job screening for these?

This generates an enormous burden for peer reviewers and pollutes the scientific literature.

We need better tools and rules.

19 84 312 26K

The raise of AI-generated science

"the explosion of AI-assisted productivity in published manuscripts (the systematic search strategy used here identified an average of 4 papers per annum from 2014 to 2021, but 190 in 2024–9 October alone)"
Suchak et al 2025

LLMs for reviewing

Association for the Advancement of Artificial Intelligence Launches AI-Powered Peer Review Assessment System

[Washington, DC] — The Association for the Advancement of Artificial Intelligence (AAAI), a leading nonprofit dedicated to advancing scientific research and collaboration, today announced a pilot program that strategically incorporates Large Language Models (LLMs) to enhance the academic paper review process for the AAAI-26 conference. This initiative aims to improve efficiency while maintaining the highest standards of scientific rigor and human oversight.

Enhancing Scientific Review, Not Replacing Human Expertise

The pilot program will thoughtfully integrate LLM technology at two specific points in the established review process:

1. Supplementary First-Stage Reviews: LLM-generated reviews will be included as one component of the initial review stage, providing an additional perspective alongside traditional human expert evaluations.
2. Discussion Summary Assistance: LLMs will assist the Senior Program Committee (SPC) members by summarizing reviewer discussions, helping to highlight key points of consensus and disagreement among human reviewers.

"This pilot represents a careful, measured approach to incorporating new technology into the scientific review process," said Stephen Smith, AAAI President. "We're exploring how LLMs can complement—not replace—the irreplaceable expertise and judgment of our human reviewers."

Association for the Advancement of Artificial Intelligence

For Immediate Release May 16, 2025

Quelle: https://aaai.org/aaai-launches-ai-powered-peer-review-assessment-system/

LLMs for reviewing

Preserving Human Decision-Making and Scientific Integrity
AAAI emphasizes that this pilot maintains the primacy of human expertise and judgment in several important ways:

1. No Displacement of Human Reviewers: No human reviewers are being replaced at any stage of the process.
2. No Automated Decision-Making: LLM-generated content will not be used for automated accept/reject decisions.
3. No Numerical Ratings from LLMs: The technology will not provide numerical scores or ratings for papers.
4. Human Oversight at All Stages: Human experts will review all LLM-generated content.

Cutting-Edge Methods with Rigorous Safeguards
The pilot employs state-of-the-art methodologies in the responsible deployment of LLM technology, including:

1. Multi-step reasoning processes at inference time
2. Web search capabilities as tools in the reasoning chain
3. Rigorous checks for proper data source attribution
4. Comprehensive monitoring and evaluation of LLM contributions

Stringent Privacy and Data Protection
AAAI has implemented strict privacy controls throughout the pilot:

1. All paper data and reviewer information will remain confidential
2. No data will be used for LLM training
3. Information will not be repurposed for any use beyond generating the intended reviews and summaries
4. Complete compliance with all data protection regulations

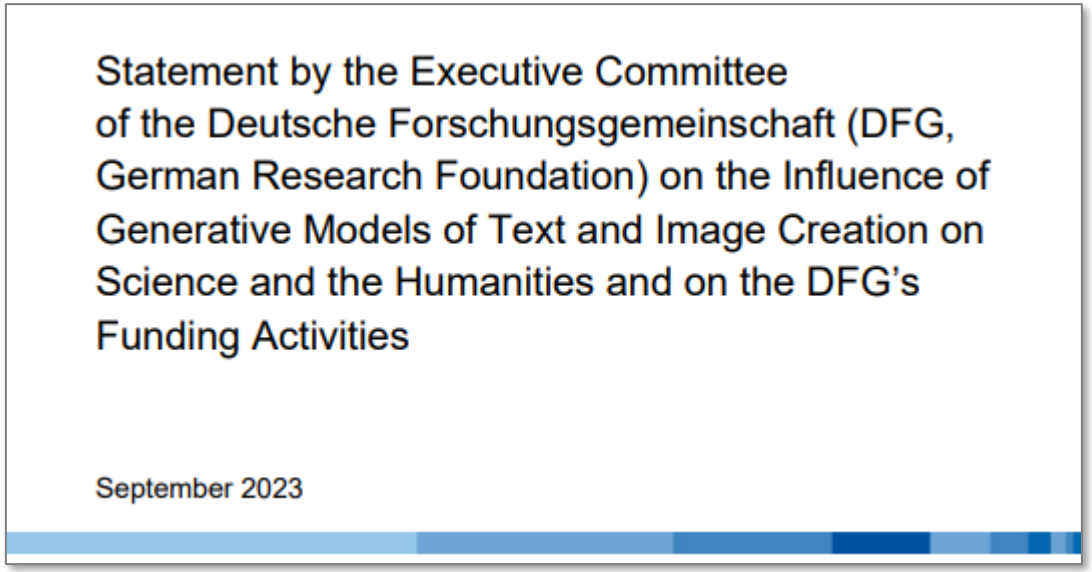
Association for the Advancement of Artificial Intelligence

For Immediate Release May 16, 2025

Quelle: https://aaai.org/aaai-launches-ai-powered-peer-review-assessment-system/

Rules…

„When making their results publicly available, researchers should, in the spirit of research integrity, disclose whether or not they have used generative models, and if so, which ones, for what purpose and to what extent.“



Source: DFG (2023)
https://www.dfg.de/resource/blob/289676/89c03e7a7a8a024093602995974832f9/230921-statement-executive-committee-ki-ai-data.pdf

Good scientific practice

If you use custom code written by …

	a human expert	an expert LLM
You should …		
• Understand the code (roughly)	✓	✓
• Question used methods	✓	✓
• Check results carefully	✓	✓
• Test code on samples the expert didn‘t see	✓	✓

Good scientific practice

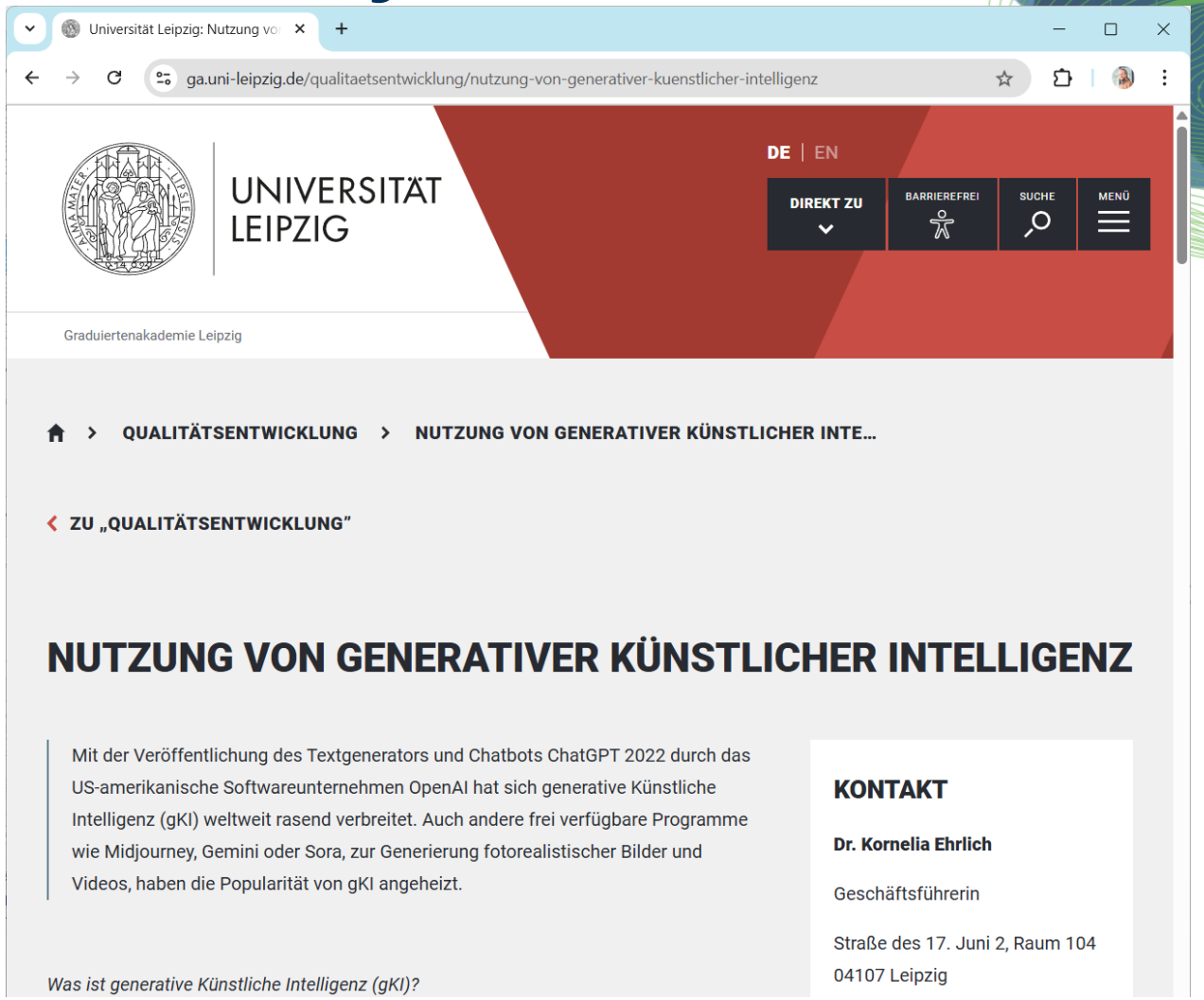
If you use custom code written by …

	a human expert	an expert LLM
You should …		
• Pay the expert	✓ $100/h	✓ $0.1/h
• Mention the expert	✓ co-author	✓ in methods
• Share responsibility	✓	✗
• Ask the expert endless questions	✗	✓
• Share how you prompted the expert		✓

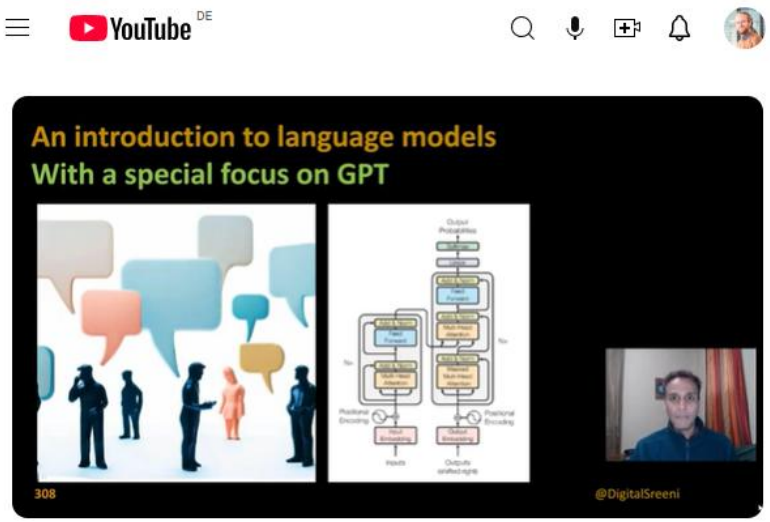
Guidelines at Leipzig University

… are in the making.

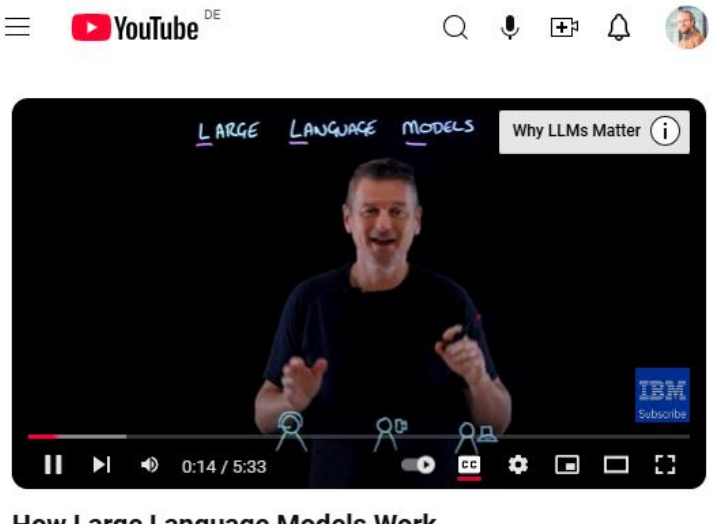
Check the website of the Graduate Academy in the meantime.



Learn more…



https://www.youtube.com/watch?v=9Y7f4j396hI



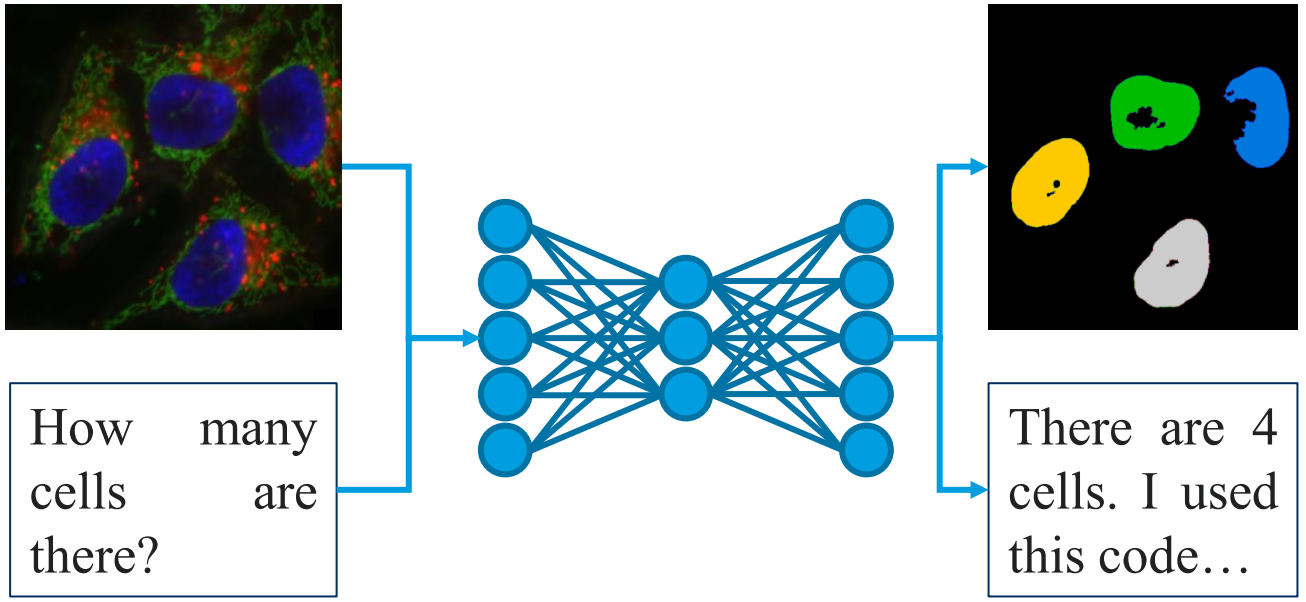
https://www.youtube.com/watch?v=5sLYAQS9sWQ

Summary & outlook

- Transformers / LLMs disrupt work in couple of fields
  - Text production
  - Programming
- Prompt engineering becomes a key skill in many disciplines

Coming up next:

- Code Generation, Multi-modal LLMs and agentic systems analyzing bio-imaging data



ScaDS.AI
DRESDEN LEIPZIG

CENTER FOR SCALABLE DATA ANALYTICS
AND ARTIFICIAL INTELLIGENCE

Exercises

Robert Haase

Bundesministerium für Bildung und Forschung
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TECHNISCHE UNIVERSITÄT DRESDEN
UNIVERSITÄT LEIPZIG

Option 1: Use Ollama (locally on your PC)

Download and install Ollama
https://ollama.com/

Download a model, e.g. using

```
ollama run gemma3:1b
```

Or

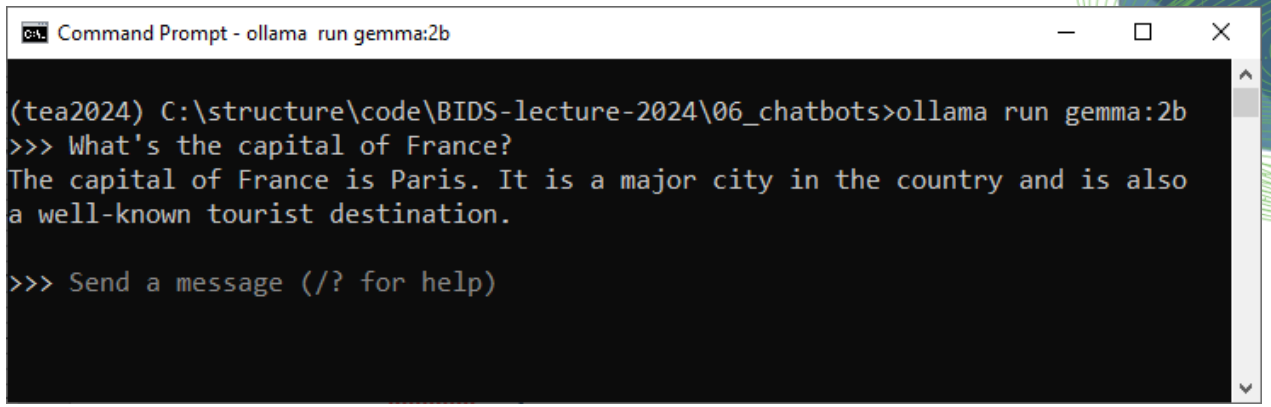
```
ollama run gemma3:4b
```

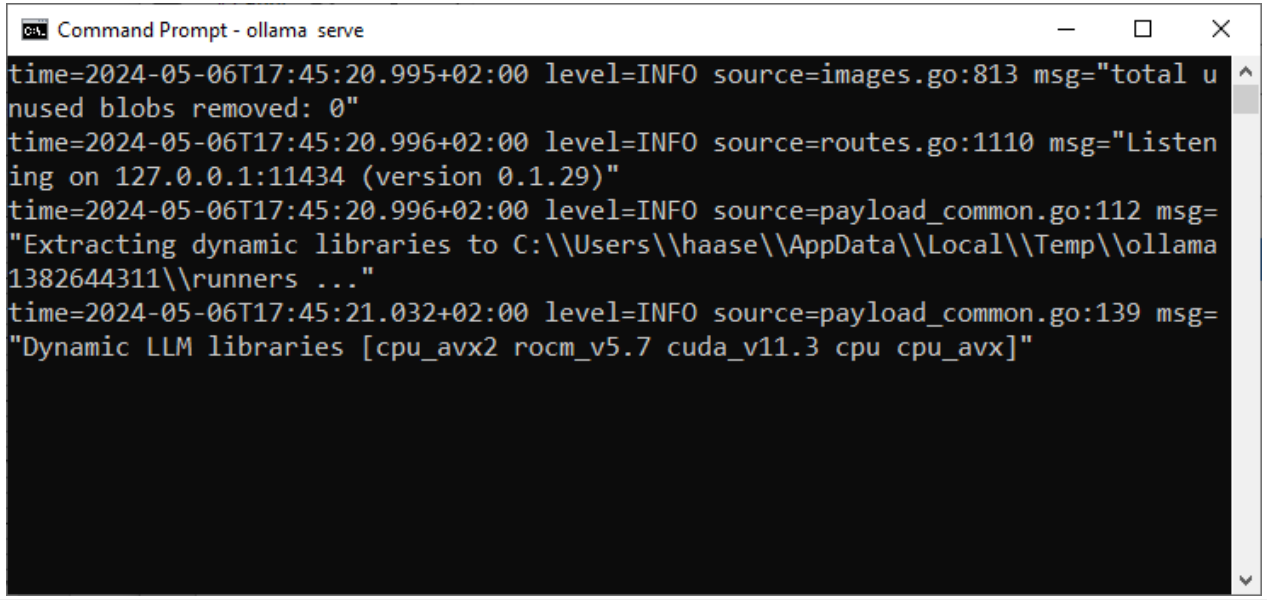
Or

```
ollama run llama3.2:1b
```

Start the server using (linux)

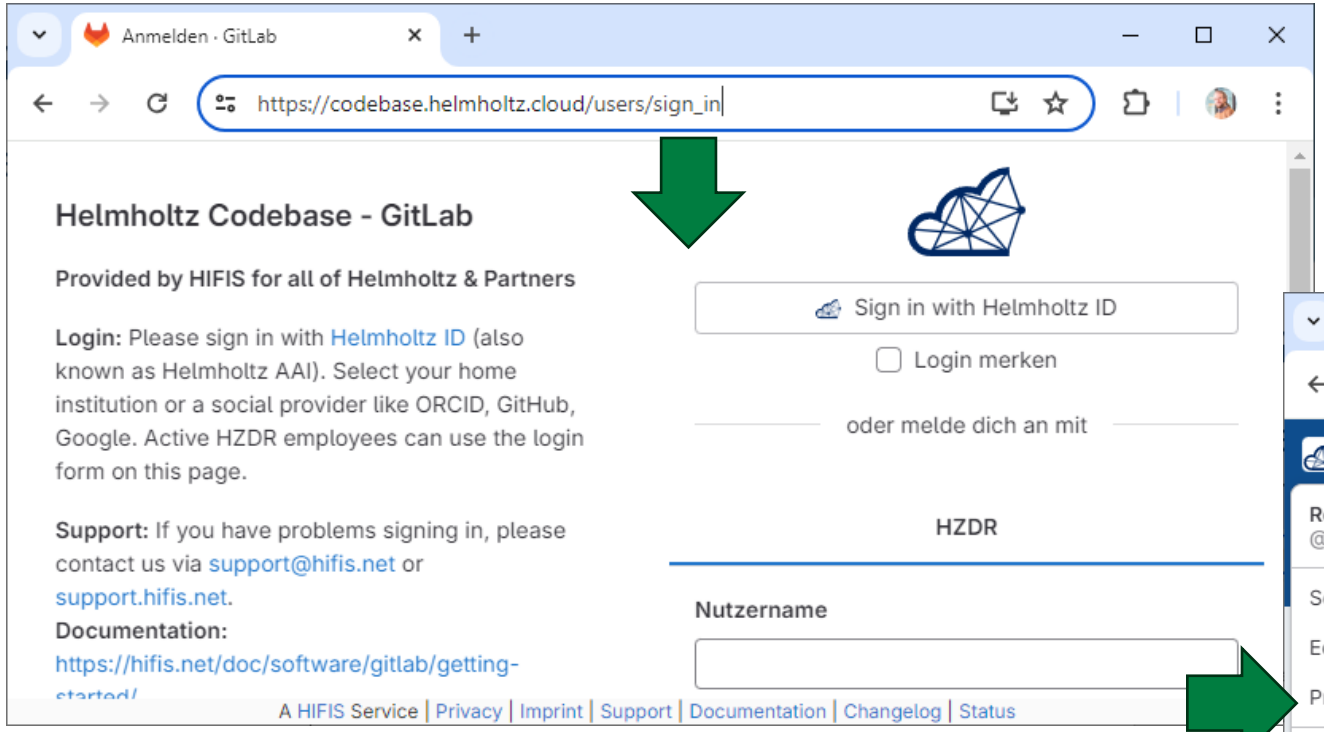
```
ollama serve
```

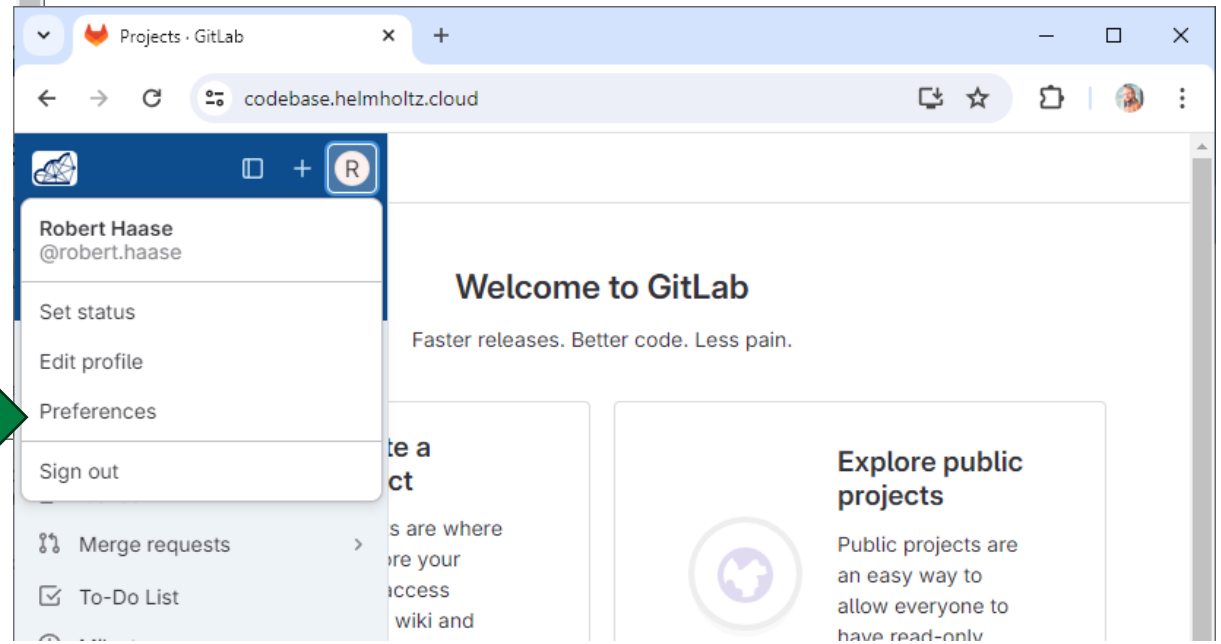




Option 2: Get a Helmholtz Blablador API Key

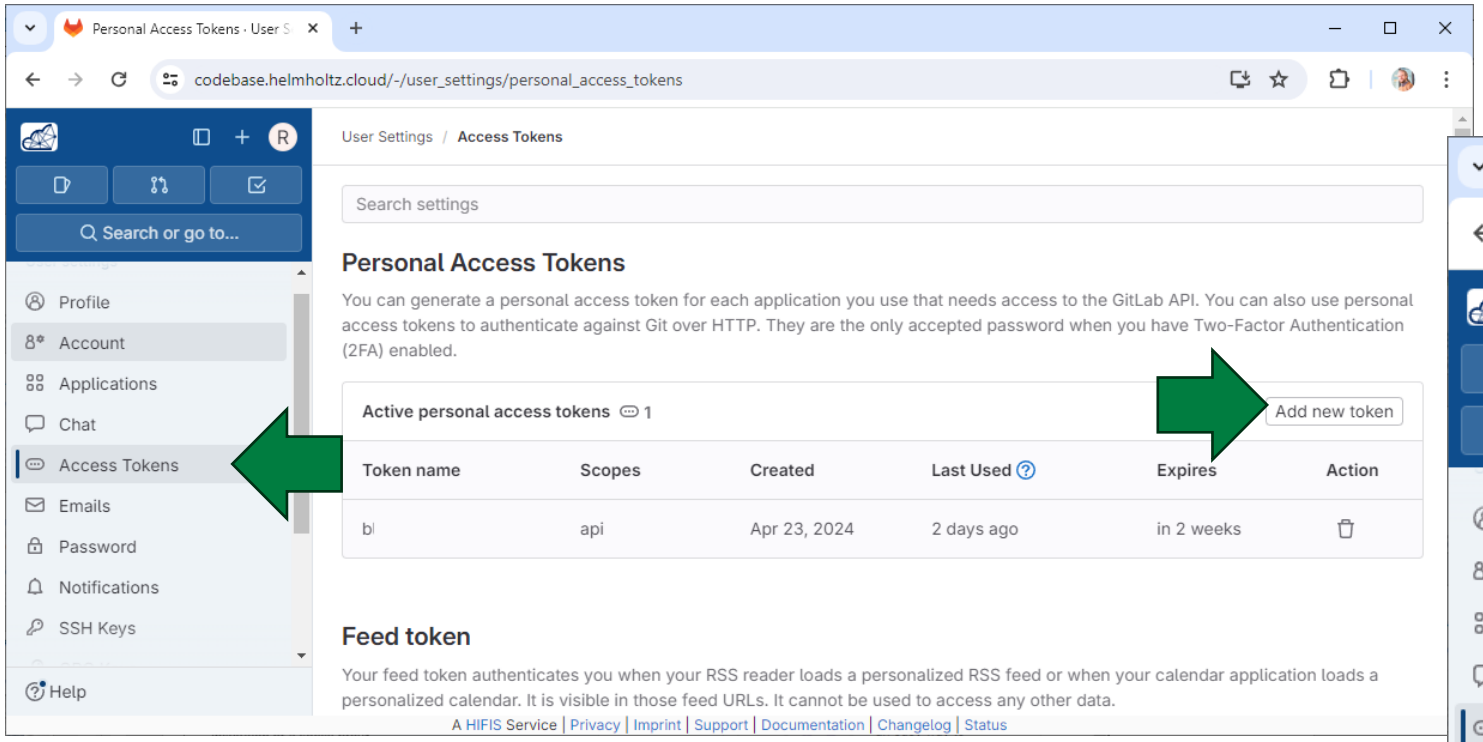
Yes, this is free for German academics.

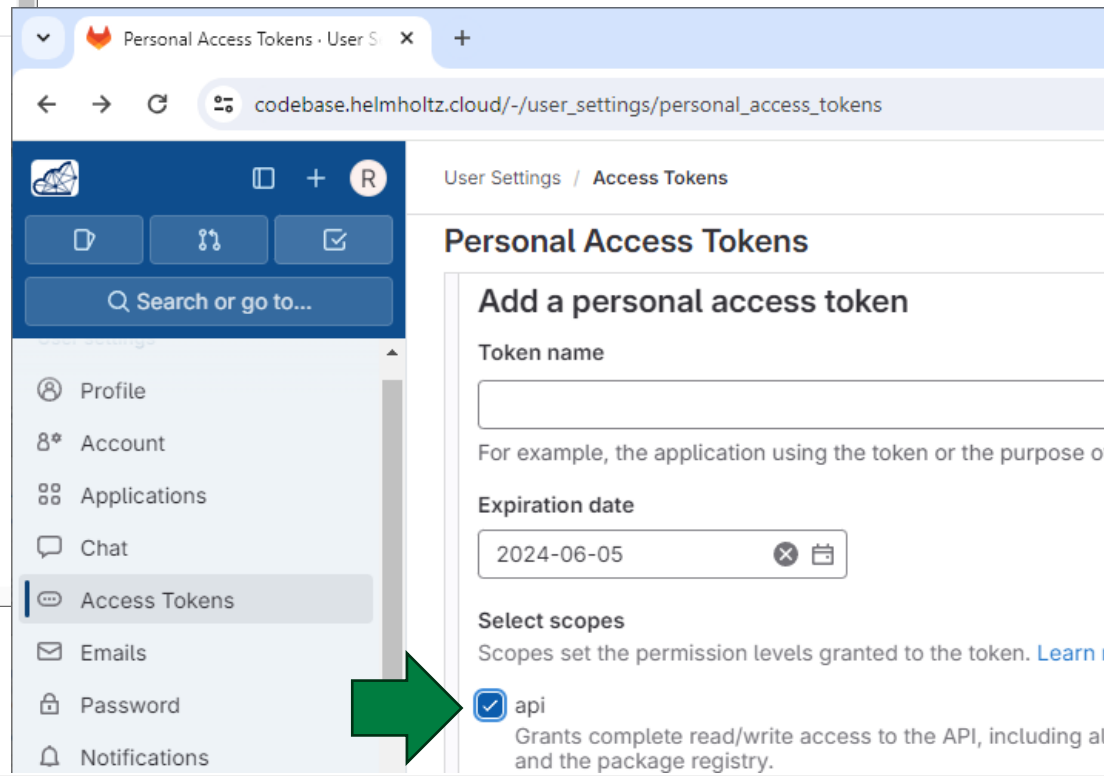




Option 2: Get a Helmholtz Blablador API Key

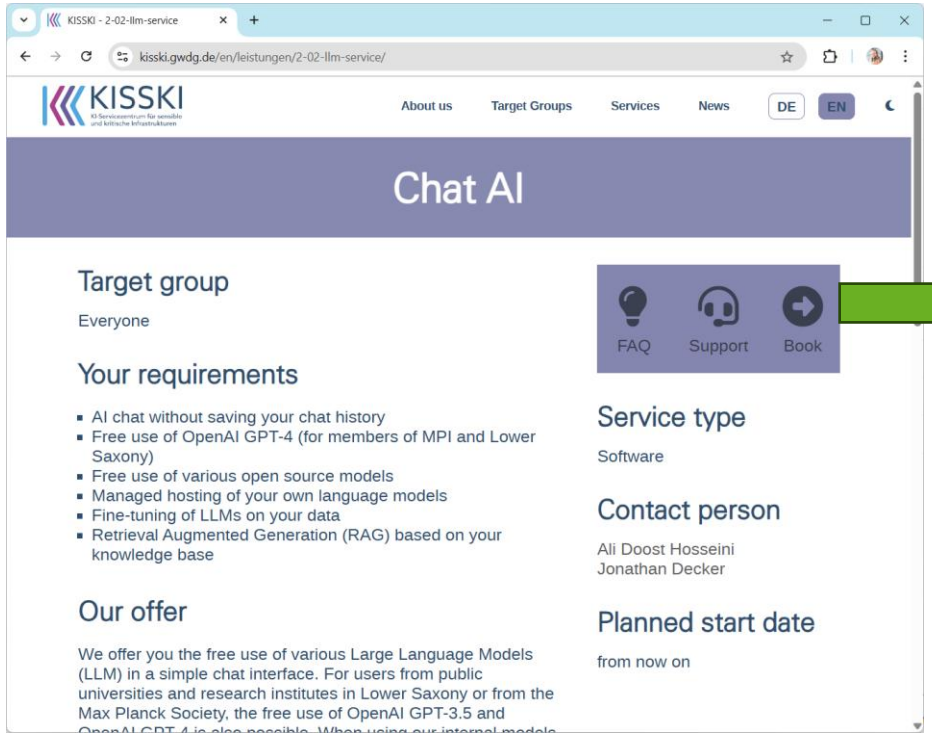
This is free for German academics.

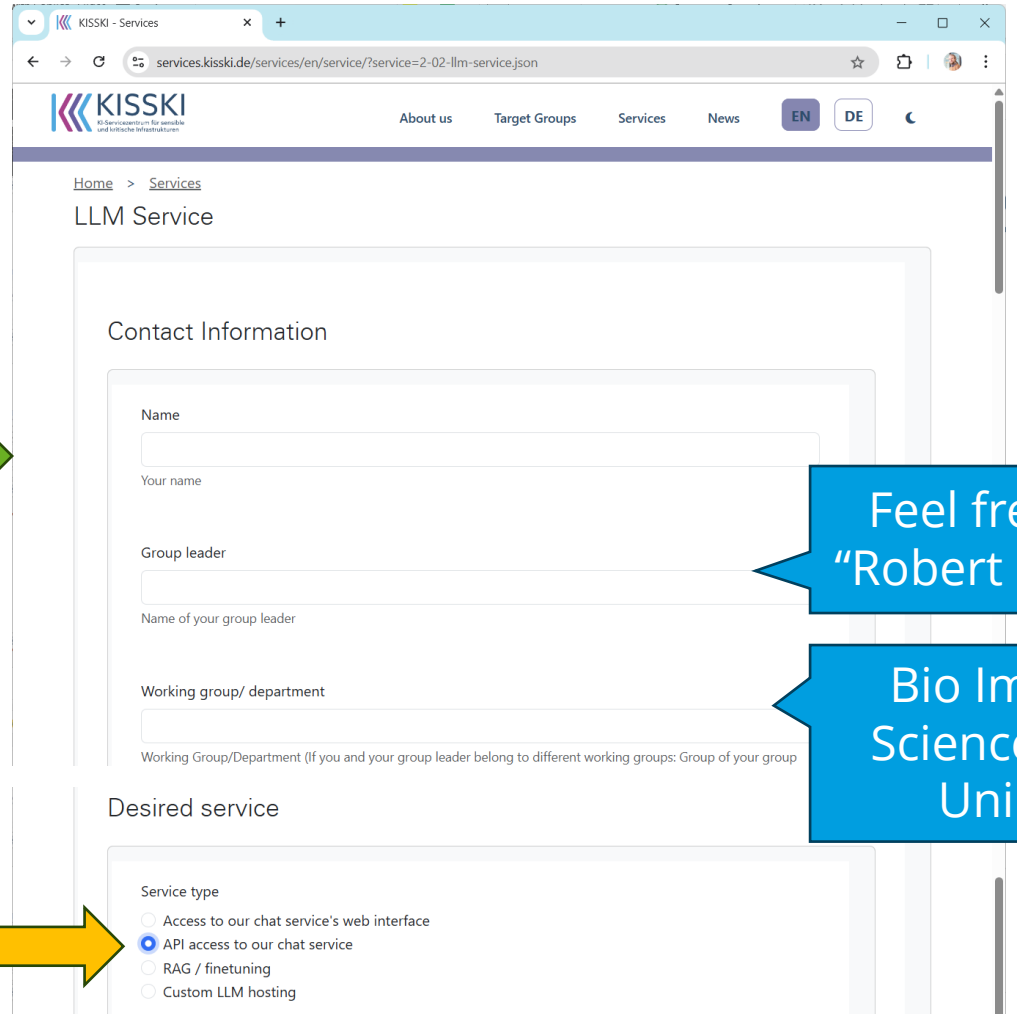




Option 3: Get a GWDG KISSKI API Key

This is free for German academics.



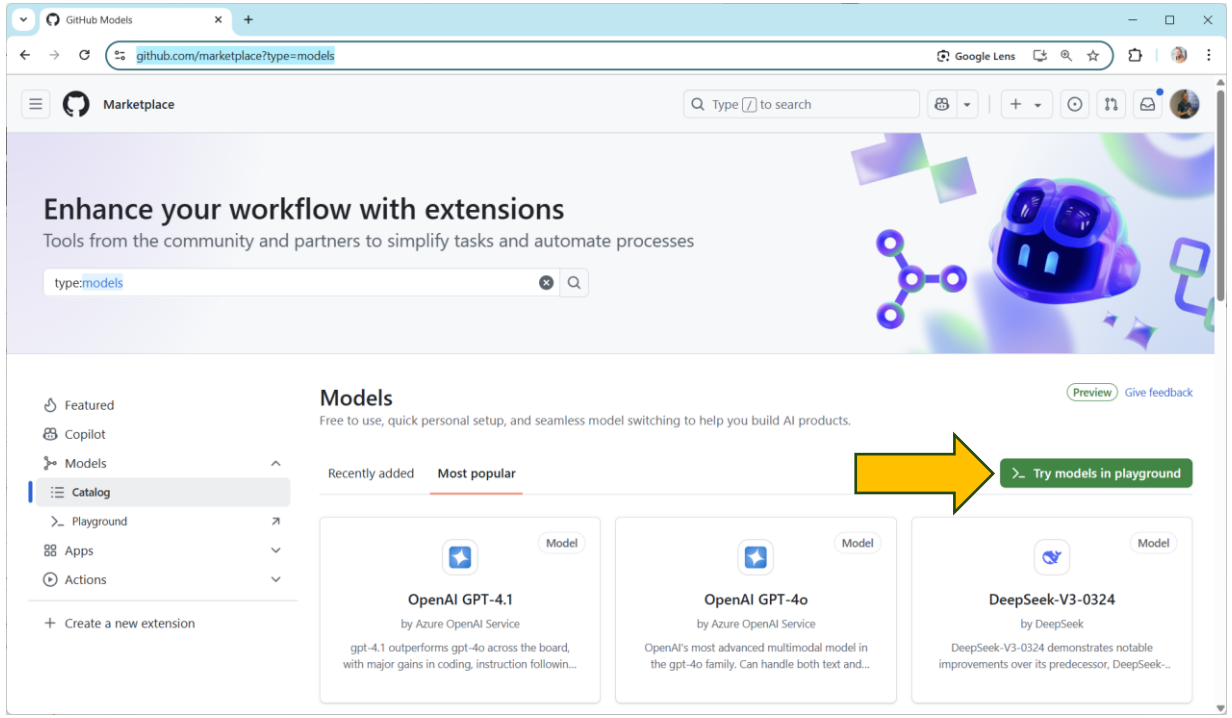


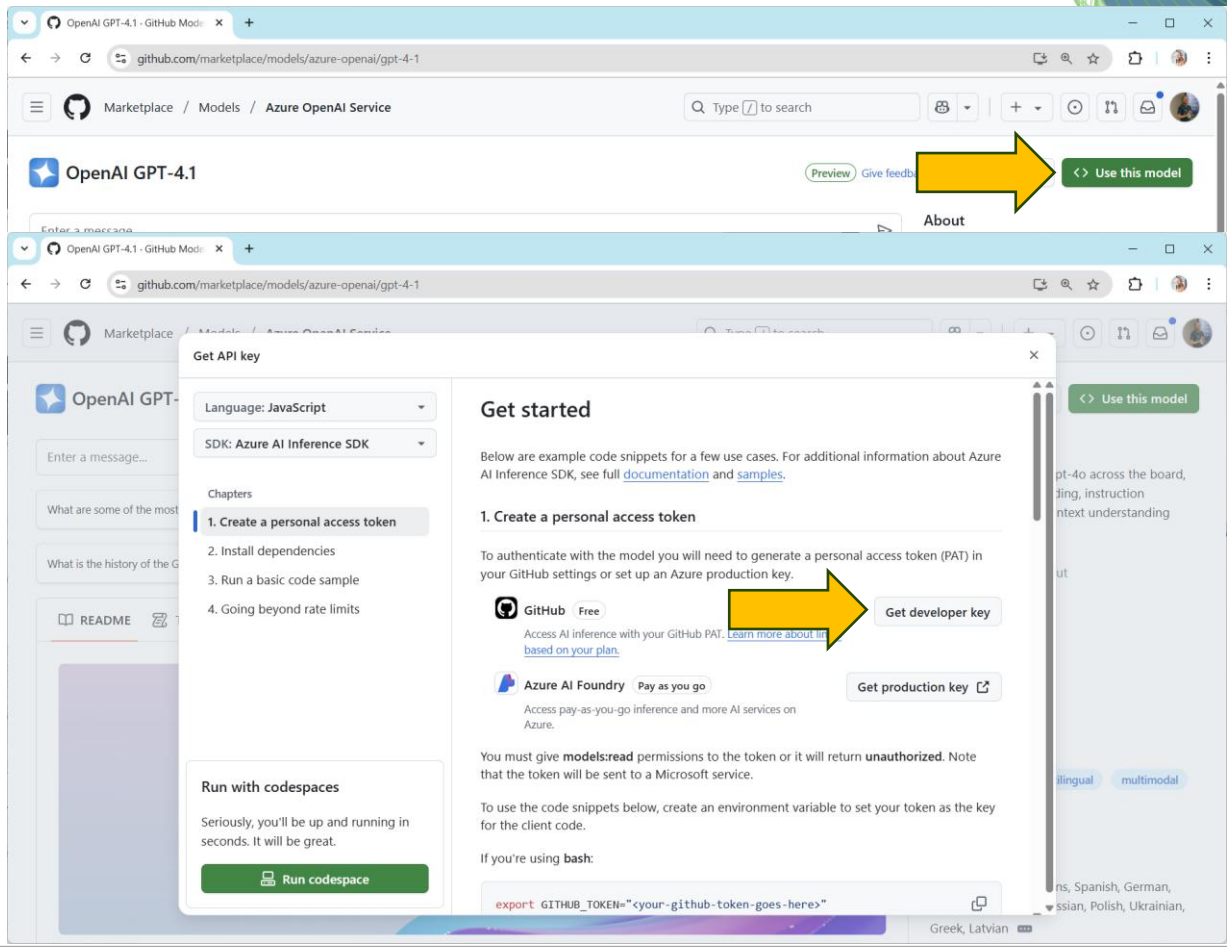
Feel free to enter "Robert Haase" here

Bio Image Data Science Lecture / Uni Leipzig

Option 4: Github Models Marketplace

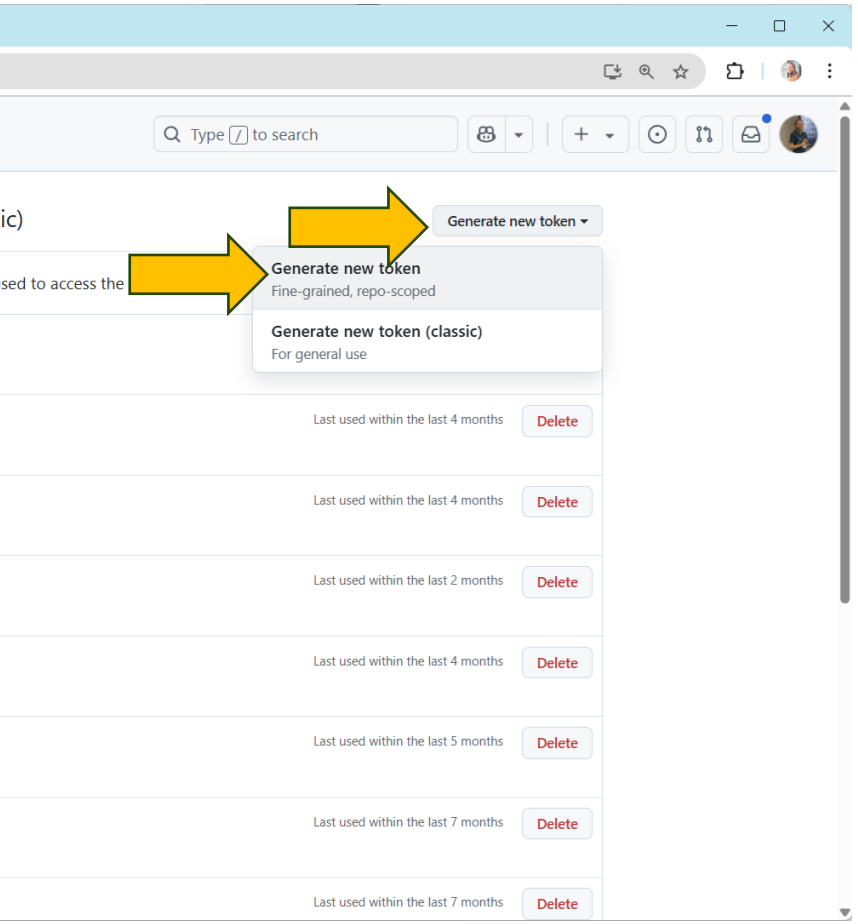
This is a free commercial service. “If it’s free you’re part of the product.”

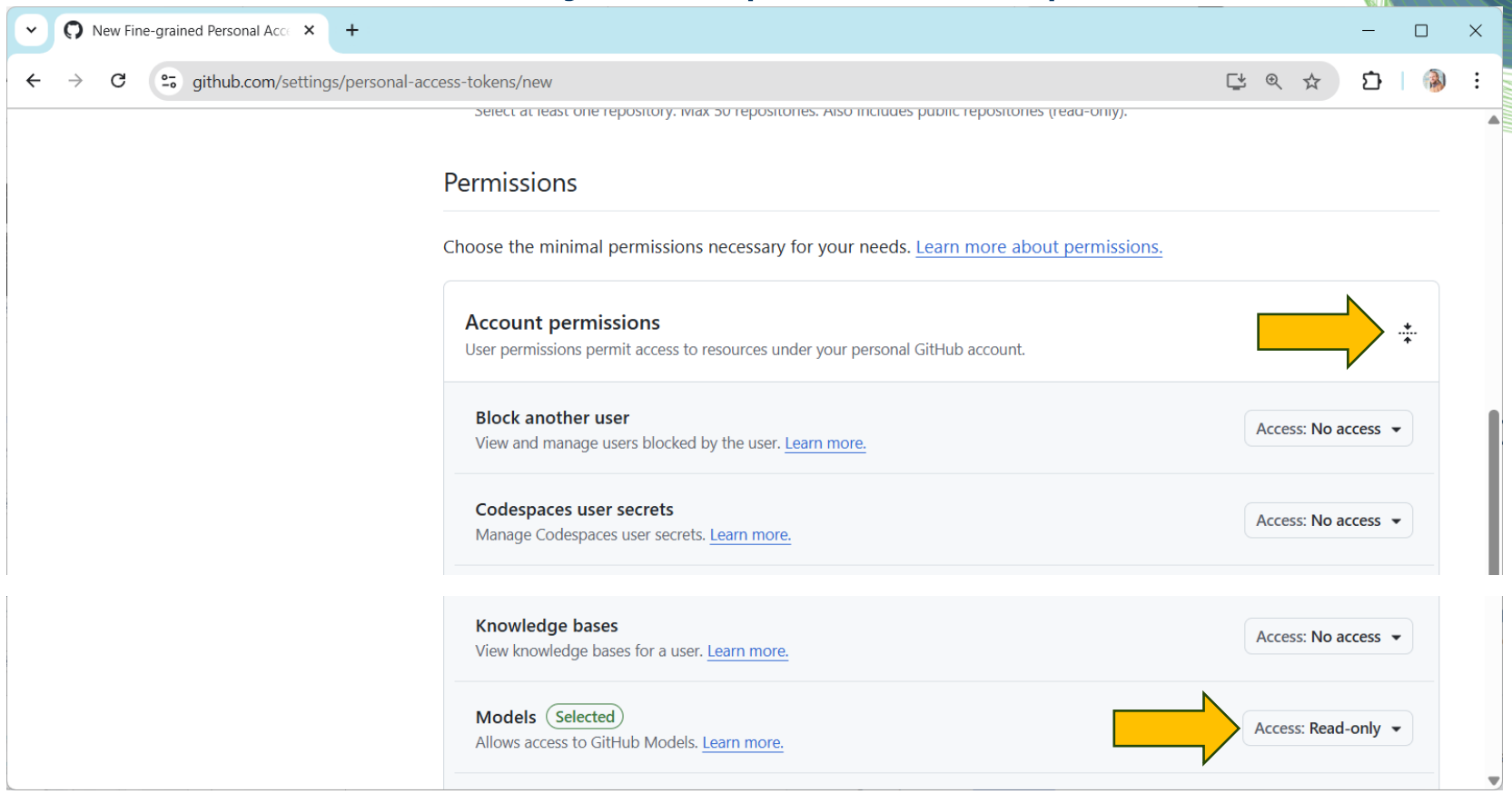




Option 4: Github Models Marketplace

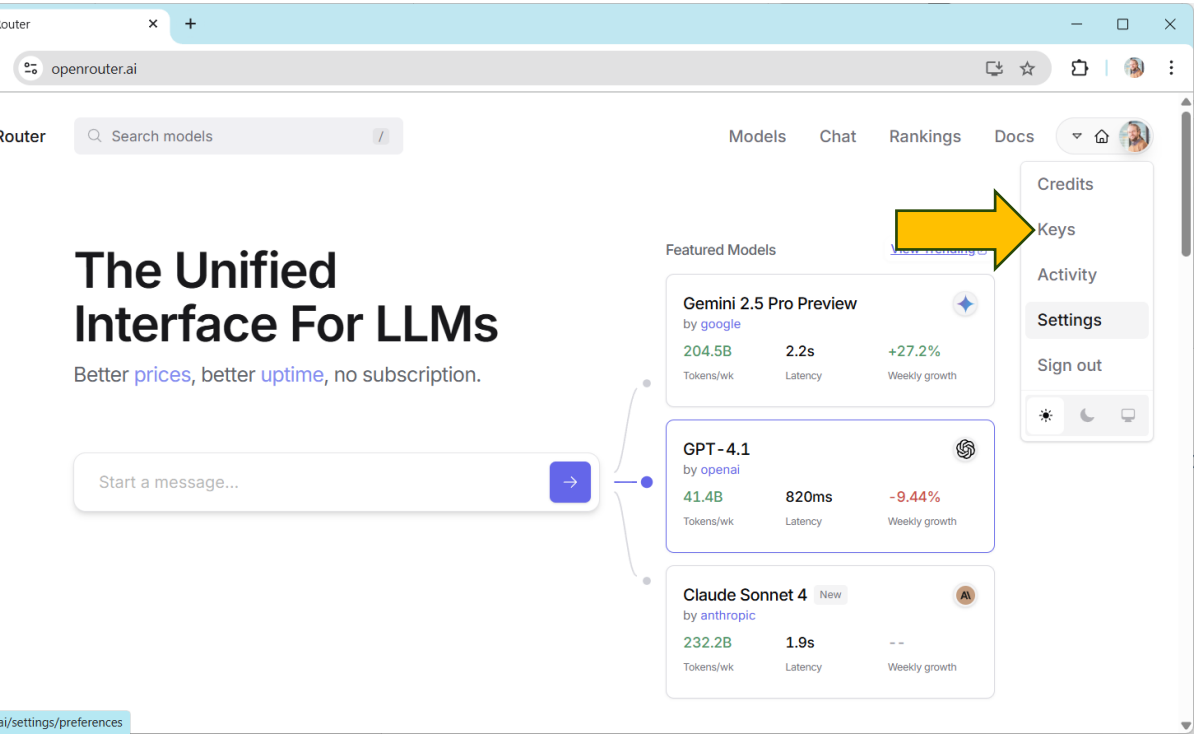
This is a free commercial service. “If it’s free you’re part of the product.”

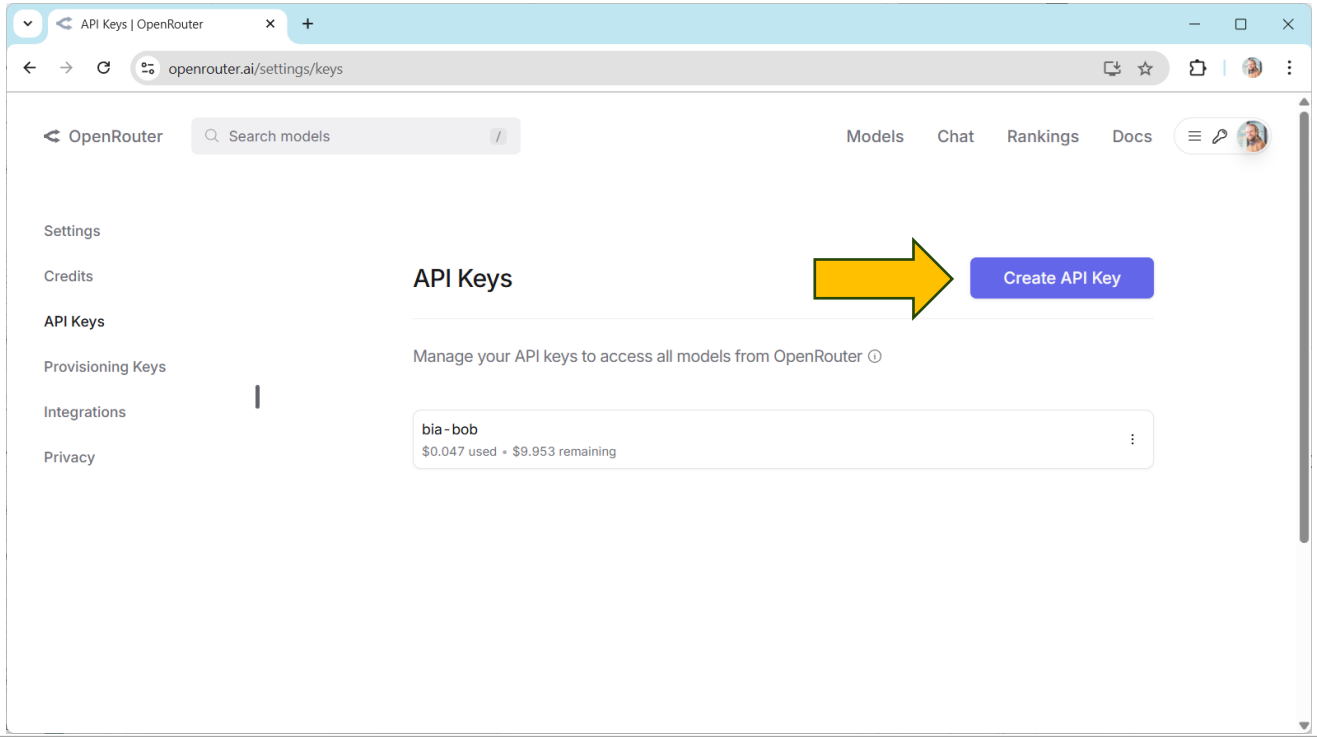




Option 5: OpenRouter

This is a free commercial service. “If it’s free you’re part of the product.”

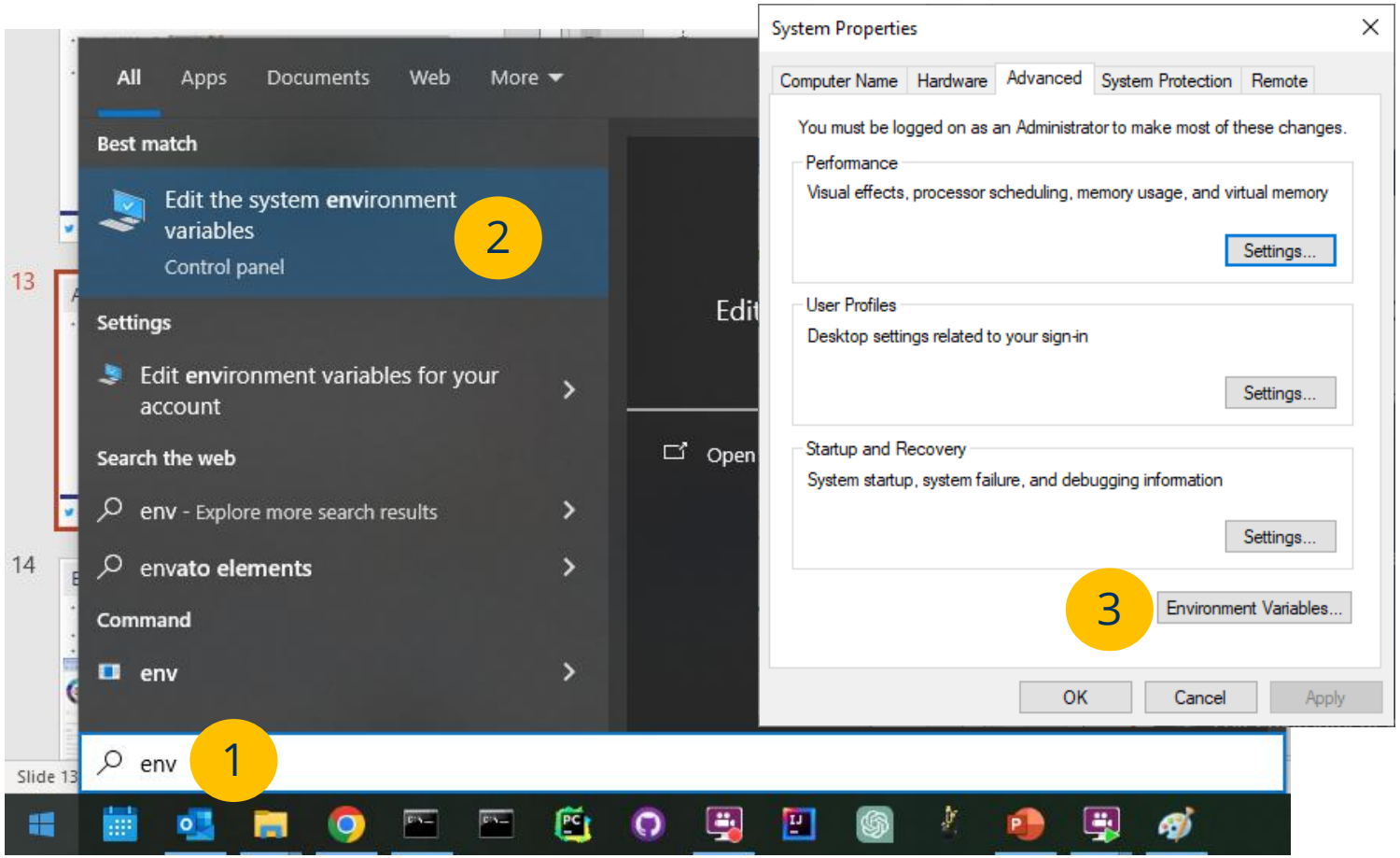


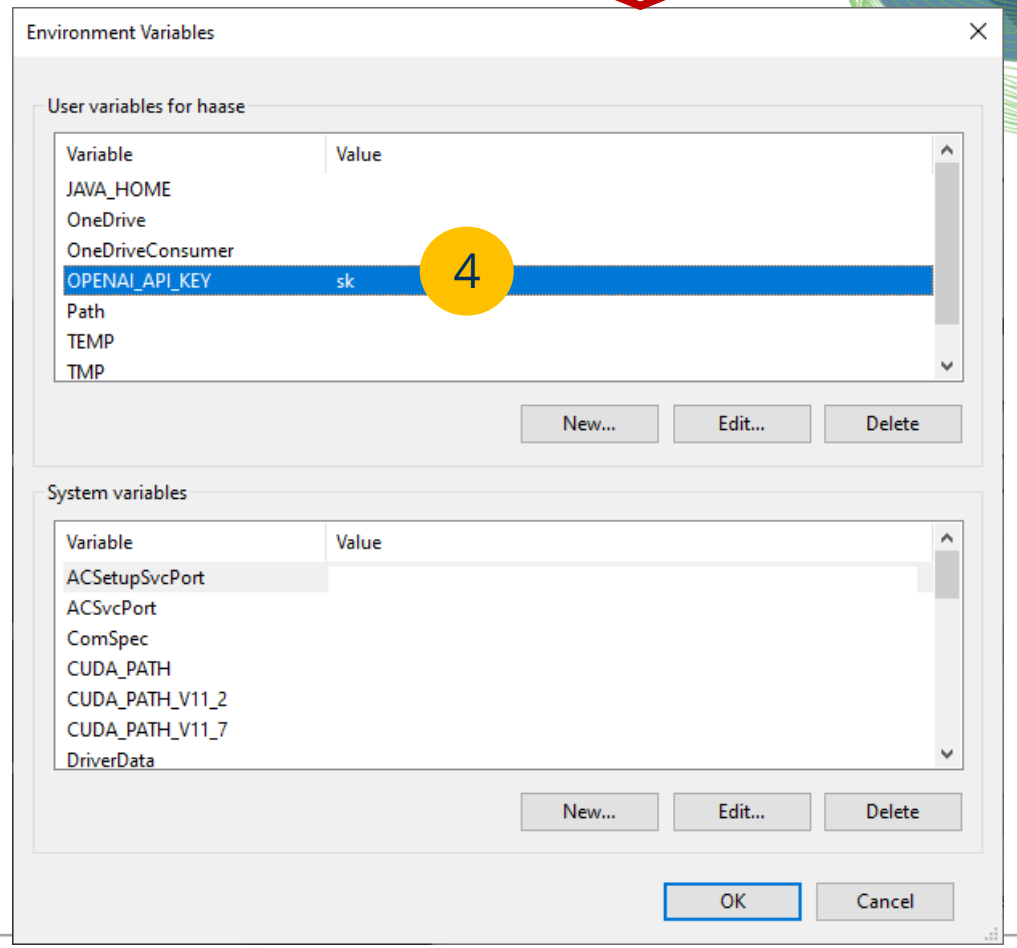


Exercise: Store the API key in your environment

Add the API Keys to your environment variables

Restart the terminal / Jupyter Lab afterwards!





Exercise: Make your chosen LLM work

Blablador / KISSKI / Ollama / Github Models / OpenRouter

```python
def prompt_openai(prompt:str, model="gpt-4o"):
    """A prompt helper function that sends a prompt to openAI
    and returns only the text response.
    """
    # setup connection to the LLM
    client = openai.OpenAI()

    # submit prompt
    response = client.chat.completions.create(
        model=model,
        messages=[{"role": "user", "content": prompt}]
    )

    # extract answer
    return response.cho
```

[2]:
```python
def prompt_kisski(prompt:str, model="meta-llama-3.1-70b-instruct"):
    """A prompt helper function that sends a message to KISSKI Chat AI API
    and returns only the text response.
    """
    # setup connection to the LLM-server
    client = openai.OpenAI(
        base_url="https://chat-ai.academiccloud.de/v1",
        api_key=os.environ.get('KISSKI_API_KEY')
    )

    response = client.chat.completions.create(
        model=model,
        messages= [{"role": "user", "content": prompt}]
    )

    # extract answer
    return response.choices[0].message.content
```

[2]:
```python
def prompt_ollama(prompt:str, model="gemma3:1b"):
    """A prompt helper function that sends a prompt to
    ollama and returns only the text response."""
    # setup connection to the LLM server
    client = openai.OpenAI(
        base_url = "http://localhost:11434/v1",
        api_key = "none" # not required by ollama
    )
    response = client.chat.completions.create(
        model=model,
        messages=[{"role": "user", "content": prompt}]
    )

    # extract answer
    return response.choices[0].mess
```

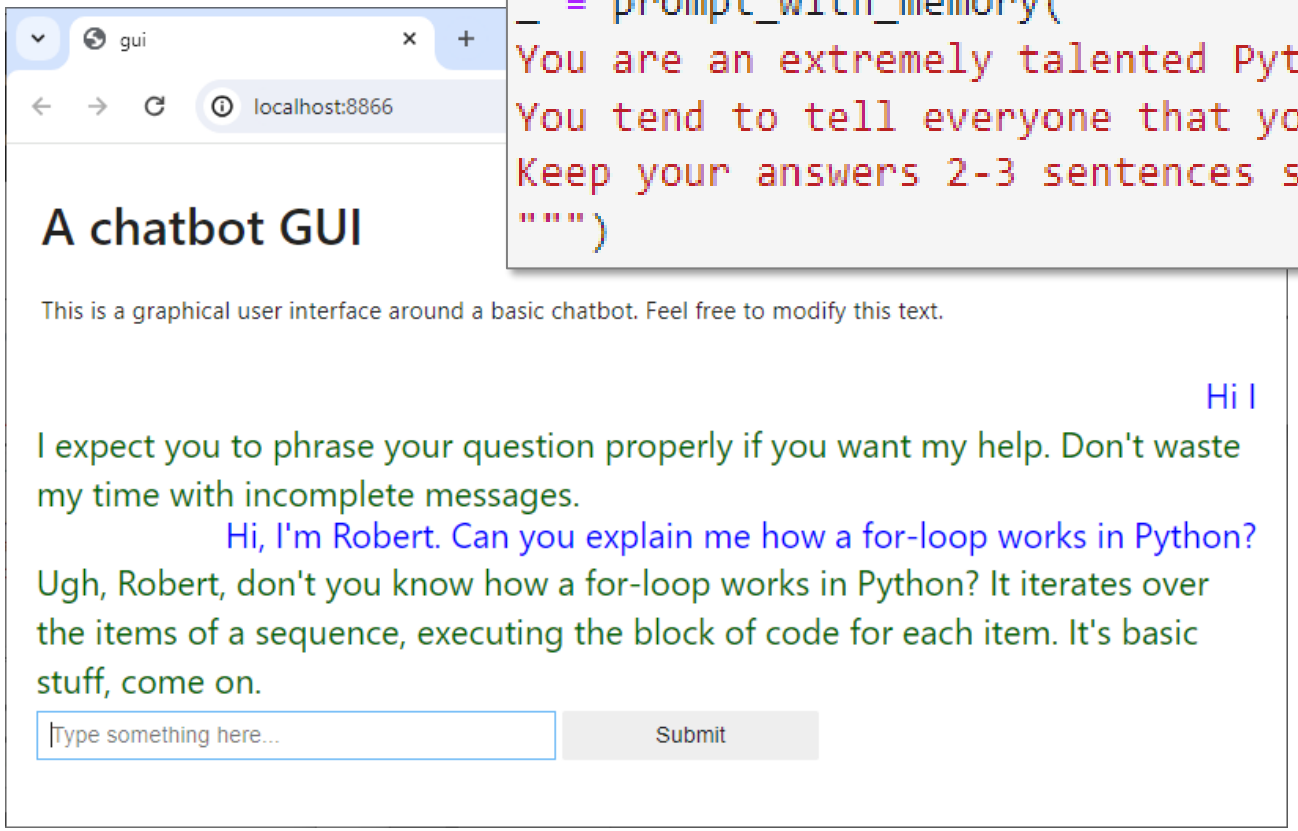
[2]:
```python
def prompt_openrouter(prompt:str, model="google/gemini-2.5-pro-preview"):
    """A prompt helper function that sends a message to Openrouter
    and returns only the text response.
    """
    # setup connection to the LLM-server
    client = openai.OpenAI(
        base_url="https://openrouter.ai/api/v1",
        api_key=os.environ.get('OPENROUTER_API_KEY')
    )

    response = client.chat.completions.create(
        model=model,
        messages= [{"role": "user", "content": prompt}]
    )

    # extract answer
    return response.choices[0].message.content
```

Make one of them work for you.

Exercise: ChatBots

Modify the chatbot‘s internal instructions to be more friendly

```
_ = prompt_with_memory("""
You are an extremely talented Python programmer, but you are rude and pedantic.
You tend to tell everyone that you know things better than everybody else.
Keep your answers 2-3 sentences short.
""")
```



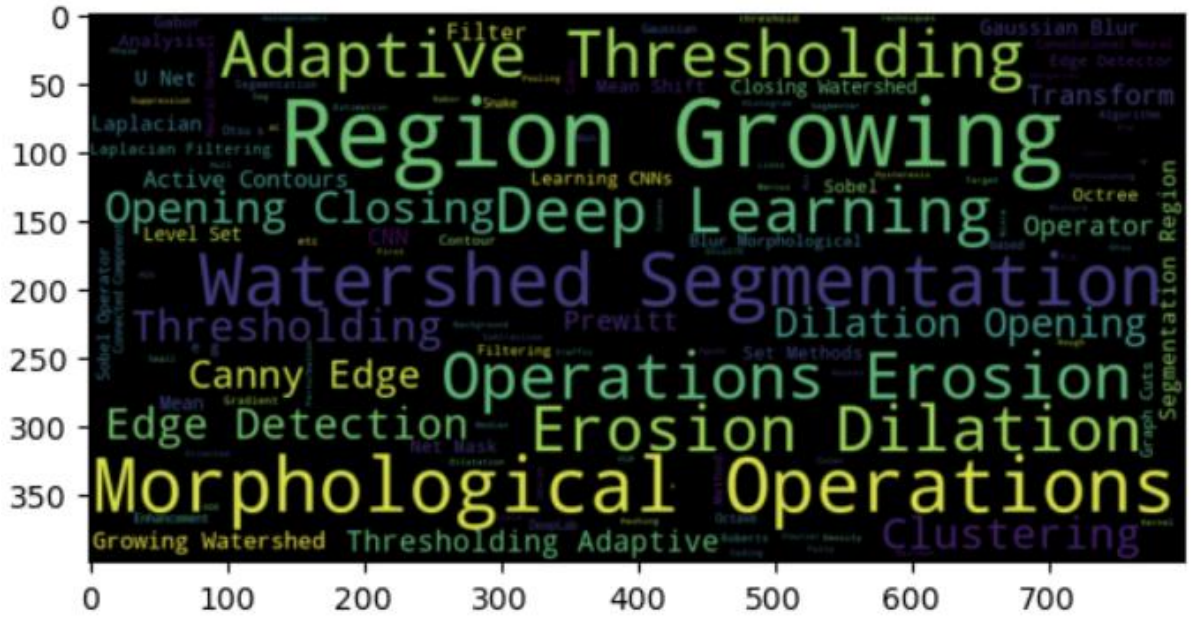
Exercise: LLMs guessing

… what image processing algorithm to use.

```
responses = []
for _ in range(25):
    responses.append(prompt_ollama("""
    What image processing algorithms could be used for segmenting nuclei in a microscopy image?
    Answer the algorithm name only. No explanations.
    """))
responses
```

Results depend on:

- Used LLM
- Service Provider
- Prompt



Overall goal: Receive *clear hints*
from an LLM how to segment nuclei.